

Biomolecules

Question1

Identify the correct statements.

- A. Arginine and Tryptophan are essential amino acids.
- B. Histidine does not contain heterocyclic ring in its structure.
- C. Proline is a six membered cyclic ring amino acid.
- D. Glycine does not have chiral centre.
- E. Cysteine has characteristic feature of side chain as $\text{MeS} - \text{CH}_2 - \text{CH}_2 -$.

Choose the correct answer from the options given below :

JEE Main 2026 (Online) 21st January Morning Shift

Options:

A.

C and D Only

B.

A and D Only

C.

C and E Only

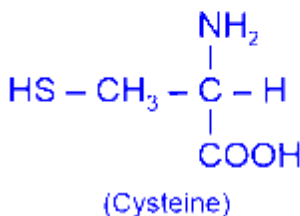
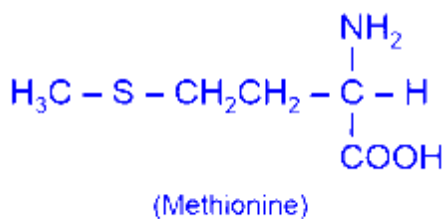
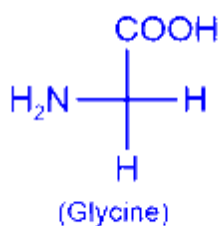
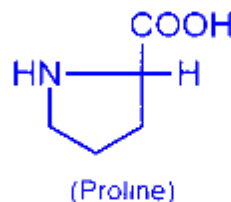
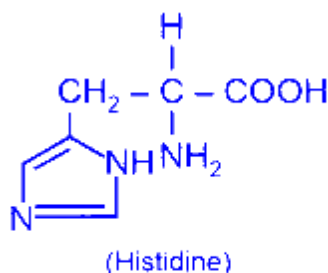
D.

B and E Only

Answer: B

Solution:

Arginine and Tryptophan are essential amino acids.



Question2

The correct statements are :

A. Activation energy for enzyme catalysed hydrolysis of sucrose is lower than that of acid catalysed hydrolysis.

B. During denaturation, secondary and tertiary structures of a protein are destroyed but primary structure remains intact.

C. Nucleotides are joined together by glycosidic linkage between C₁ and C₄ carbons of the pentose sugar.

D. Quaternary structure of proteins represents overall folding of the polypeptide chain.

Choose the correct answer from the options given below :

JEE Main 2026 (Online) 21st January Evening Shift

Options:

A. A, B and D Only

B. A, C and D Only

C. B and C Only

D. A and B Only

Answer: D

Solution:

A. Activation energy for enzyme catalysed hydrolysis of sucrose is lower than that of acid catalysed hydrolysis.

- Enzymes are biological catalysts that increase the rate of reactions by lowering their activation energy.
- Acid catalysis also provides an alternative reaction pathway, but enzymes are typically much more efficient at lowering activation energy under physiological conditions.
- For the hydrolysis of sucrose, the enzyme sucrase (invertase) significantly reduces the activation energy compared to acid-catalyzed hydrolysis, making the reaction much faster.
- Therefore, statement A is correct.

B. During denaturation, secondary and tertiary structures of a protein are destroyed but primary structure remains intact.

- Denaturation is the process by which proteins lose their native three-dimensional structure (secondary, tertiary, and if present, quaternary structures) due to disruptions of weak interactions (like hydrogen bonds, hydrophobic interactions, ionic bonds).
- The primary structure, which is the sequence of amino acids linked by strong covalent peptide bonds, is generally not broken during denaturation. Breaking peptide bonds would require harsh conditions like strong acid/base hydrolysis or specific proteases.
- Therefore, statement B is correct.

C. Nucleotides are joined together by glycosidic linkage between C1 and C4 carbons of the pentose sugar.

- Nucleotides are linked together in nucleic acids (DNA and RNA) by phosphodiester bonds.
- A phosphodiester bond forms between the phosphate group attached to the 5' carbon of one nucleotide and the hydroxyl group on the 3' carbon of the pentose sugar of the next nucleotide.
- A glycosidic linkage (specifically, an N-glycosidic bond) connects the nitrogenous base to the C1 carbon of the pentose sugar *within* a single nucleotide. It does not link nucleotides together, nor does it

involve C1 and C4 carbons *between* different pentose sugars.

- Therefore, statement C is incorrect.

D. Quaternary structure of proteins represents overall folding of the polypeptide chain.

- The **tertiary structure** describes the overall three-dimensional folding of a *single polypeptide chain*.
- The **quaternary structure** refers to the arrangement of *multiple polypeptide subunits* (each having its own tertiary structure) to form a functional protein complex. It describes how these individual chains associate with each other.
- Therefore, the statement describes tertiary structure, not quaternary structure.
- Therefore, statement D is incorrect.

Conclusion:

Statements A and B are correct. Statements C and D are incorrect.

The correct option should include only A and B.

The final answer is

A and B Only

Question3

Given below are two statements :

Statement I : Sucrose is dextrorotatory. However, sucrose upon hydrolysis gives a solution having mixture of products. This solution shows laevorotation.

Statement II : Hydrolysis of sucrose gives glucose and fructose. Since the laevorotation of glucose is more than the dextrorotation of fructose, the resulting solution becomes laevorotatory.

In the light of the above statements, choose the correct answer from the options given below :

JEE Main 2026 (Online) 22nd January Morning Shift

Options:

A.

Both Statement I and Statement II are false

B.

Both Statement I and Statement II are true

C.

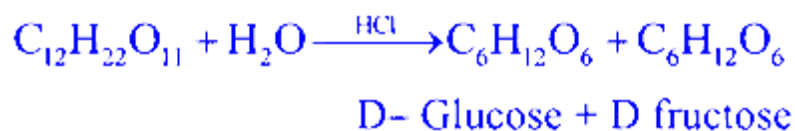
Statement I is true but Statement II is false

D.

Statement I is false but Statement II is true

Answer: C

Solution:



$$[\alpha]_{D\text{-sucrose}} = +66.5^\circ, \quad [\alpha]_{D\text{-Glucose}} = +52.5^\circ$$
$$[\alpha]_{D\text{-Fructose}} = -92.4^\circ$$

⇒ Sucrose is dextrorotatory and hydrolysed product is laevorotatory.

Question4

Match List - I with List - II.

List - I Reaction of Glucose with		List - II Product formed	
A.	Hydroxylamine	I.	Gluconic acid
B.	Br ₂ water	II.	Glucose pentacetate
C.	Excess acetic anhydride	III.	Saccharic acid
D.	Concentrated HNO ₃	IV.	Glucoxime

Choose the correct answer from the options given below :

JEE Main 2026 (Online) 22nd January Evening Shift

Options:

A.

A-IV, B-III, C-II, D-I

B.

A-I, B-III, C-IV, D-II

C.

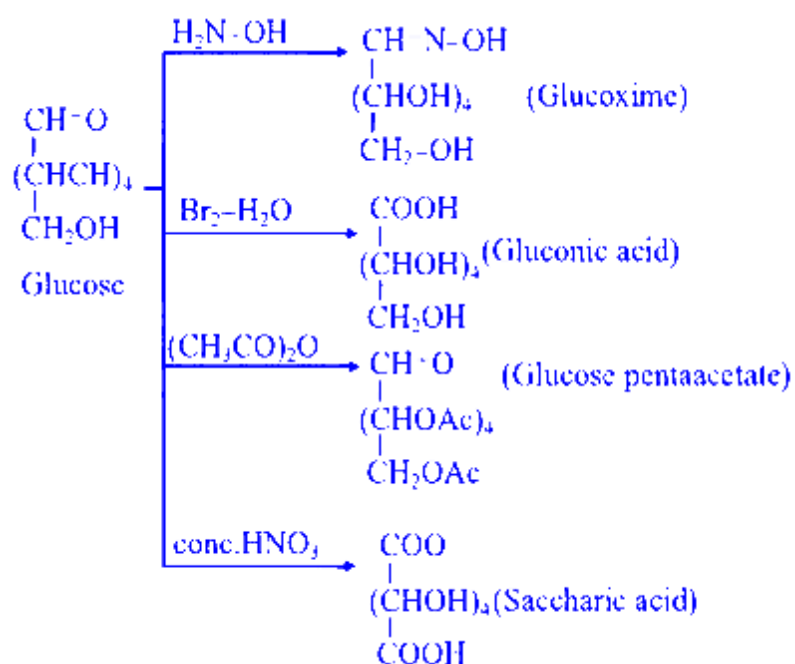
A-III, B-I, C-IV, D-II

D.

A-IV, B-I, C-II, D-III

Answer: D

Solution:



Question5

Both human DNA and RNA are chiral molecules. The chirality in DNA and RNA arises due to the presence of

JEE Main 2026 (Online) 23rd January Evening Shift

Options:

A.

L-sugar component

B.

D-sugar component

C.

Chiral phosphate unit

D.

Base unit

Answer: B

Solution:

DNA and RNA show chirality because they contain a chiral sugar in their structure.

In DNA, the sugar present is chiral **2-deoxy-Ribose**.

In RNA, the sugar present is chiral **Ribose**.

Question6

The number of possible tripeptides formed involving alanine (ala), glycine (gly) and valine (val), where no amino acid has been used more than once is:

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Options:

A.

8

B.

3

C.

4

D.

6

Answer: D

Solution:

Gly ala val

Gly val ala

Val gly ala

Val ala gly

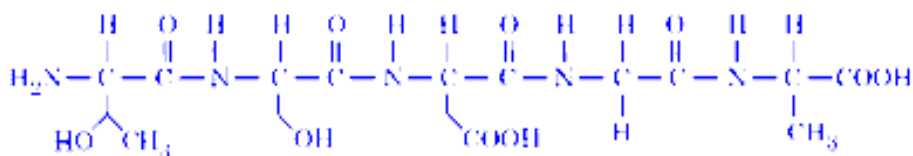
Ala val gly

Ala gly val

Total tri peptides = 6

Question7

In the given pentapeptide, find out an essential amino acid (Y) and the sequence present in the pentapeptide :



Choose the correct answer from the options given below :

JEE Main 2026 (Online) 28th January Morning Shift

Options:

A.

(Y)	(Sequence)
Serine	Thr-Ser-Asp-Ala-Gly

B.

(Y)	(Sequence)
Serine	Ser-Asp-Thr-Ala-Gly

C.

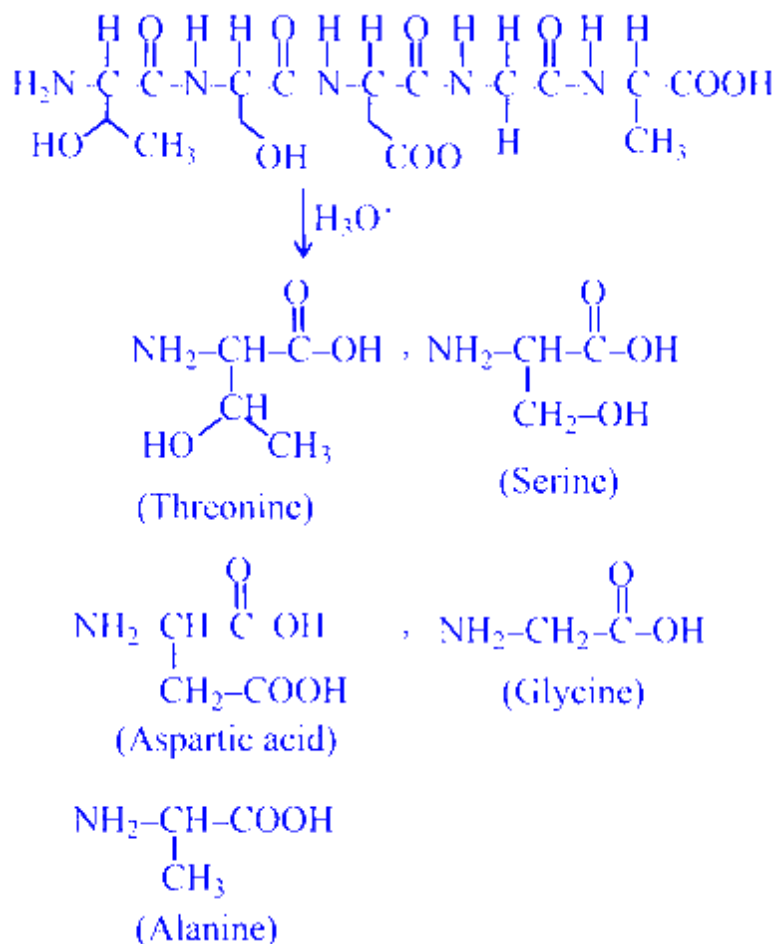
(Y)	(Sequence)
Threonine	Ser-Thr-Asp-Gly-Ala

D.

(Y)	(Sequence)
Threonine	Thr-Ser-Asp-Gly-Ala

Answer: D

Solution:



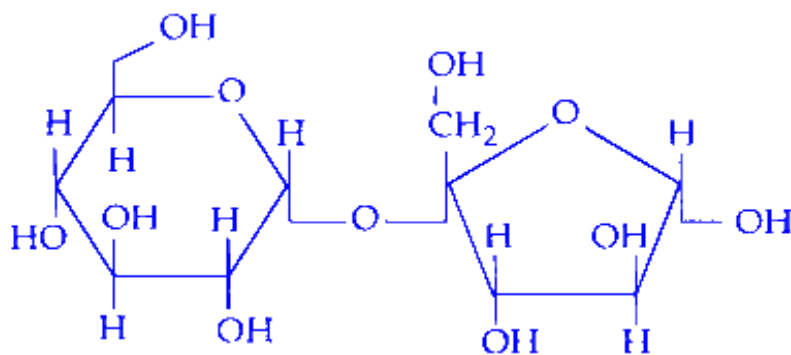
Question8

Structures of four disaccharides are given below. Among the given disaccharides, the non-reducing sugar is :

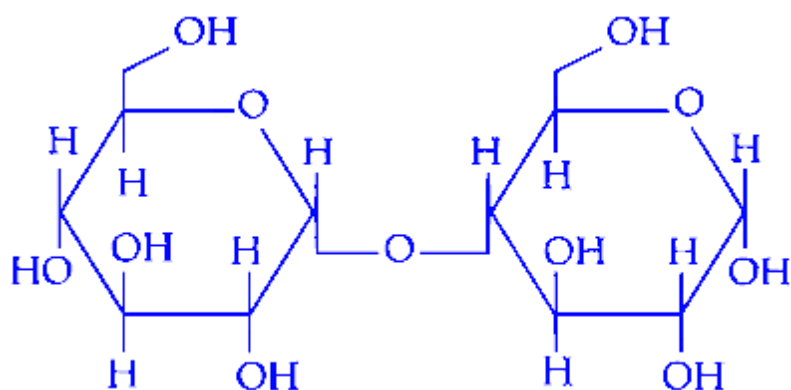
JEE Main 2026 (Online) 28th January Evening Shift

Options:

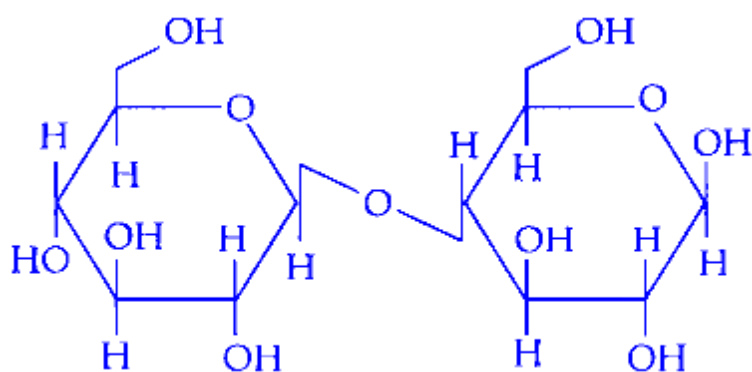
A.



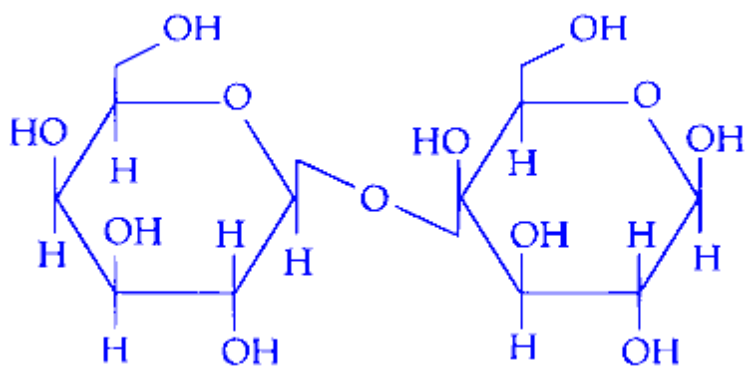
B.



C.



D.

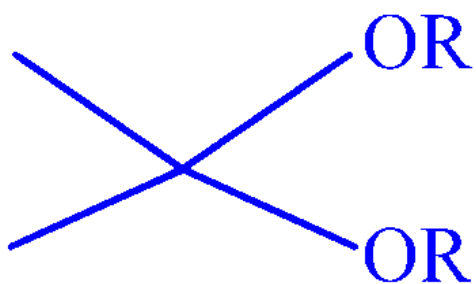


Answer: A

Solution:

Structure (1) given is of sucrose which is non reducing.

For non reducing sugar compound should have acetal linkage not hemi acetal linkage.



Question9

Match List - I with List - II.

	List - I Vitamin		List - II Name
A	Vitamin B ₁	I	Pyridoxine
B	Vitamin B ₂	II	Ascorbic acid
C	Vitamin B ₆	III	Thiamine
D	Vitamin C	IV	Riboflavin

Choose the correct answer from the options given below:

JEE Main 2026 (Online) 2nd April Morning Shift

Options:

A.

A-II, B-I, C-III, D-IV

B.

A-IV, B-III, C-II, D-I

C.

A-III, B-IV, C-I, D-II

D.

A-I, B-III, C-II, D-IV

Answer: C

Solution:

1. **Vitamin B₁**: The chemical name for Vitamin B₁ is Thiamine. Its deficiency in the human body can lead to a disease known as Beri-Beri.

Therefore, A matches with III.

2. **Vitamin B₂**: The chemical name for Vitamin B₂ is Riboflavin. A deficiency in this vitamin leads to cheilosis, which is the fissuring at the corners of the mouth and lips.

Therefore, B matches with IV.

3. **Vitamin B₆**: The chemical name for Vitamin B₆ is Pyridoxine. Its deficiency can cause convulsions.

Therefore, C matches with I.

4. **Vitamin C**: The chemical name for Vitamin C is Ascorbic acid. It is a water-soluble vitamin found in citrus fruits, and its deficiency causes Scurvy (bleeding gums).

Therefore, D matches with II.

Now, let us combine our findings:

A → III

B → IV

C → I

D → II

Looking at the options provided, the sequence A-III, B-IV, C-I, D-II perfectly matches Option C.

Option C is the correct answer.

Question10

Identify the correct pair having amino acid (A) and the hormone (B) that is iodinated derivative of the amino acid (A).

(T and Y represent one letter code for amino acids)

JEE Main 2026 (Online) 2nd April Evening Shift

Options:

A.

Amino acid (A) - T Hormone (B) - Insulin

B.

Amino acid (A) - T Hormone (B) - Thyroxine

C.

Amino acid (A) - Y Hormone (B) - Thyroxine

D.

Amino acid (A) - Y Hormone (B) - Insulin

Answer: C

Solution:

The correct pair is:

Amino acid (A) = Y, Hormone (B) = Thyroxine

So, the correct option is:

Option C

Reason:

- The one-letter code *Y* stands for **tyrosine**.
- **Thyroxine** is an **iodinated derivative of tyrosine**.
- **Insulin** is a **protein hormone**, not an iodinated derivative of an amino acid.

Hence,

$Y \rightarrow \text{Thyroxine}$

Question11

Match the List-I with List-II

List-I Name of amino acid	List-II One letter symbol/type
A. Arginine	I. D/Non-essential
B. Aspartic acid	II. R/Essential
C. Lysine	III. E/Non-essential
D. Glutamic acid	IV. K/Essential

Choose the correct answer from the options given below :

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Options:

A.

A-II, B-I, C-IV, D-III

B.

A-IV, B-III, C-II, D-I

C.

A-III, B-IV, C-I, D-II

D.

A-II, B-IV, C-I, D-III

Answer: A

Solution:

Match each amino acid with its correct one-letter symbol and whether it is essential or non-essential:

- **Arginine** → one-letter symbol is *R*

In standard school-level classification here, it is matched as **Essential**

- So, $A \rightarrow II$

Aspartic acid $\rightarrow$ one-letter symbol is D

It is **Non-essential**

So, $B \rightarrow I$

- **Lysine** $\rightarrow$ one-letter symbol is K

It is **Essential**

So, $C \rightarrow IV$

- **Glutamic acid** $\rightarrow$ one-letter symbol is E

It is **Non-essential**

So, $D \rightarrow III$

Thus the correct matching is:

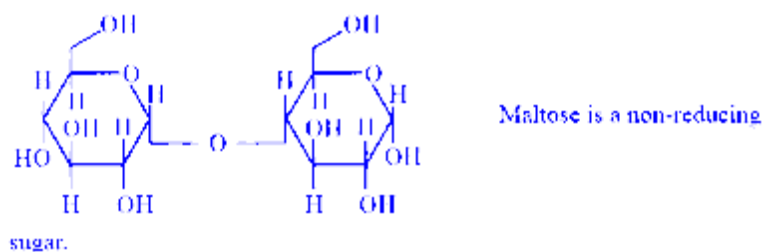
$A - II$, $B - I$, $C - IV$, $D - III$

So, the correct option is **Option A**.

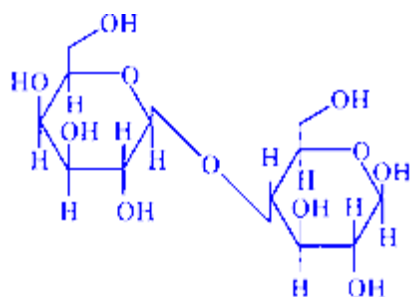
Question12

Given below are two statements :

Statement I : The structure of Maltose is given below:



Statement II : The structure of Lactose is given below :



Lactose is a reducing sugar.

In the light of the above statements, choose the correct answer from the options given below :

JEE Main 2026 (Online) 4th April Morning Shift

Options:

A.

Both Statement I and Statement II are true

B.

Both Statement I and Statement II are false

C.

Statement I is true but Statement II is false

D.

Statement I is false but Statement II is true

Answer: D

Solution:

Maltose and lactose are both **disaccharides**.

In each of these structures, one anomeric carbon is used to form the **glycosidic linkage**, but the other anomeric carbon remains **free** as a **hemiacetal (–OH at anomeric carbon)**.

If a disaccharide has a **free hemiacetal group**, it can open up to the open-chain form and can reduce mild oxidising agents (like Tollens' reagent or Fehling's solution).

Therefore, **both maltose and lactose are reducing sugars** because both contain a **hemiacetal linkage (free anomeric carbon)**.

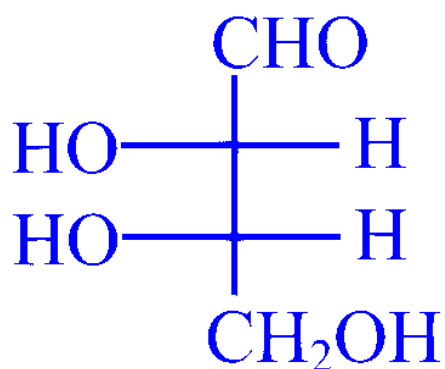
Question13

A D-aldotetrose on oxidation with concentrated HNO_3 resulted in optically inactive dicarboxylic acid. The structure of the D-aldotetrose is :

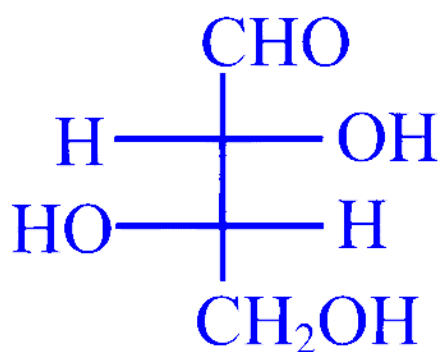
JEE Main 2026 (Online) 4th April Evening Shift

Options:

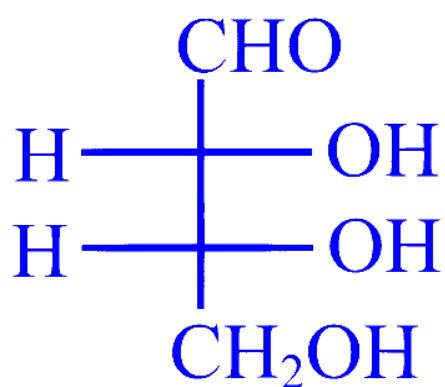
A.



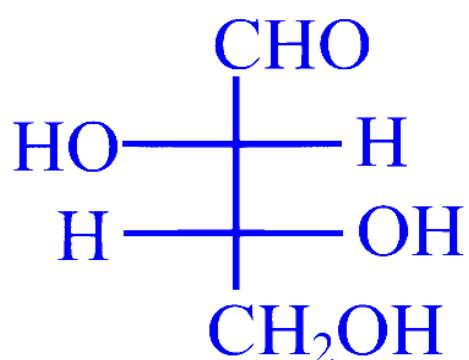
B.



C.

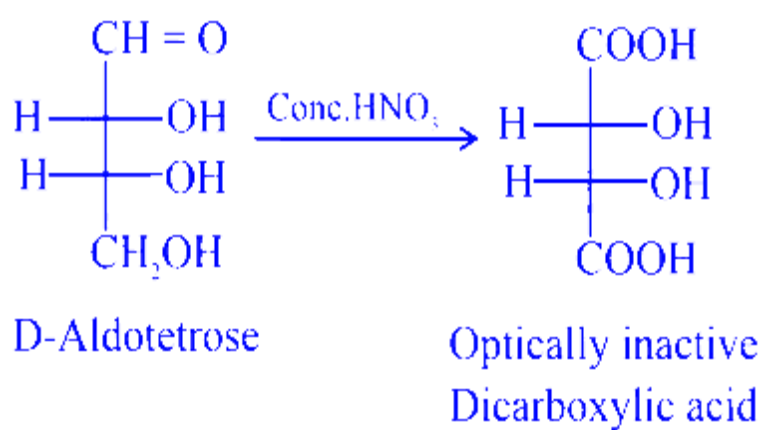


D.



Answer: C

Solution:



Question14

Identify the correct statements.

A. Glucose exists in two anomeric forms.

B. Anomers of glucose differ in configuration at C – 1 in cyclic hemiacetal structure.

C. Melting point of α - anomer of glucose is greater than β - anomer.

D. Specific rotation of α - anomer is $+19^\circ$ while for β - anomer is $+112^\circ$

E. α and β - anomers of glucose are prepared by crystallization of saturated glucose solution at 303 K and 371 K respectively.

Choose the correct answer from the options given below :

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Options:

A.

A and B Only

B.

B and C Only

C.

A, B and D Only

D.

A, B and E Only

Answer: D

Solution:

Statement A: Glucose exists in two anomeric forms.

This statement is correct. Glucose exists in two different crystalline forms, which are named as the α -form and the β -form. These are known as anomers.

Statement B: Anomers of glucose differ in configuration at C – 1 in cyclic hemiacetal structure.

This statement is correct. The α and β forms of glucose are cyclic hemiacetals that differ only in the spatial arrangement of the hydroxyl group (-OH) at the first carbon atom (C – 1). This specific carbon is referred to as the anomeric carbon.

Statement C: Melting point of α - anomer of glucose is greater than β - anomer.

This statement is incorrect. According to experimental data, the melting point of the α -form of glucose is 419 K, whereas the melting point of the β -form is slightly higher, at 423 K. Therefore, the melting point of the β -anomer is greater than that of the α -anomer.

Statement D: Specific rotation of α - anomer is $+19^\circ$ while for β - anomer is $+112^\circ$

This statement is incorrect because the values are interchanged. The specific rotation of the pure α -anomer is $+112.2^\circ$, and the specific rotation of the pure β -anomer is $+19.2^\circ$. Over time, both undergo mutarotation to reach an equilibrium specific rotation of $+52.5^\circ$.

Statement E: α and β - anomers of glucose are prepared by crystallization of saturated glucose solution at 303 K and 371 K respectively.

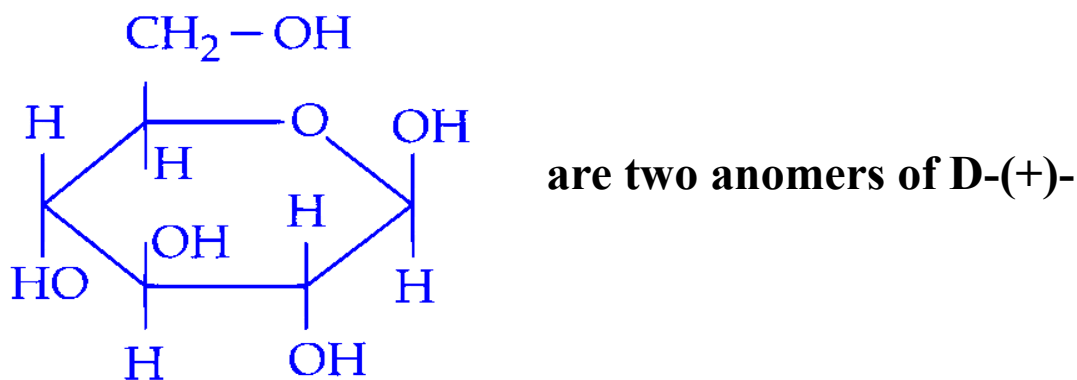
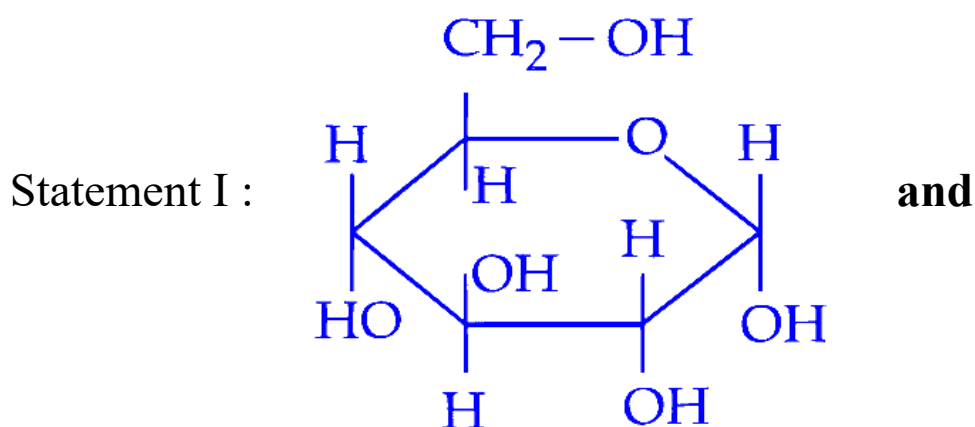
This statement is correct. The α -form is typically obtained by the crystallization from a concentrated aqueous solution of glucose at room temperature, around 303 K. On the other hand, the β -form is obtained by crystallizing it from a hot and saturated aqueous solution at 371 K.

From the analysis above, the mathematically and factually correct statements are A, B, and E.

Therefore, the correct choice is **Option D**.

Question15

Given below are two statements :



glucose.

Statement II : The open chain forms of D-glucose and D-fructose contain three similar chiral carbons at C_3 , C_4 and C_5 .

In the light of the above statements, choose the correct answer from the options given below :

JEE Main 2026 (Online) 5th April Evening Shift

Options:

A.

Both Statement I and Statement II are true

B.

Both Statement I and Statement II are false

C.

Statement I is true but Statement II is false

D.

Statement I is false but Statement II is true

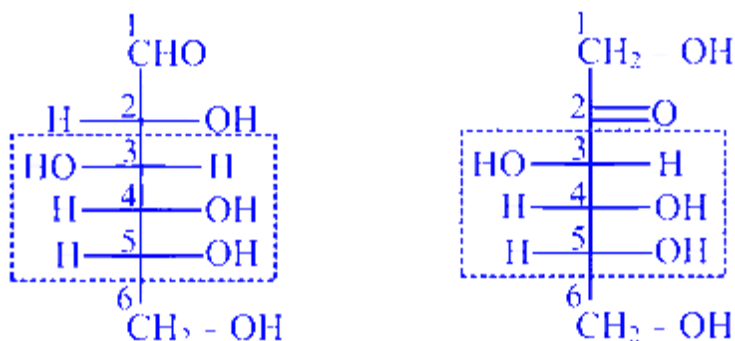
Answer: A

Solution:

Statement I: The two given cyclic structures are the α and β forms (anomers) of D-(+)-glucose.

They are called anomers because they differ only in the position (orientation) of the $-OH$ group at the anomeric carbon, i.e. C_1 in glucose.

Statement II:



In the open-chain forms of D-glucose and D-fructose, the chiral carbons C_3 , C_4 , and C_5 have the same arrangement of groups (same configuration) in both molecules.

$-OH$ position is same at C_3 , C_4 and C_5 .

Question16

Identify the incorrect statement about tertiary structure of proteins.

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Options:

A.

They can be fibrous or globular in structure.

B.

The main forces that stabilize the structure are hydrogen bonding, disulphide links, van der Waals and electrostatic forces of attraction.

C.

The structure remains intact when exposed to pH changes.

D.

A linear polypeptide chain will convert to a secondary structure and then further folding of the secondary structure will convert to tertiary structure.

Answer: C

Solution:

When the pH changes, the tertiary structure of a protein gets disturbed or broken. However, the amino acids in the protein remain connected to each other through peptide bonds.

Question17

Match the List-I with List-II

List-I Amino acid		List-II Positive reaction/Test for functional group present in side chain of amino acid	
A.	Glutamine	I.	Hinsberg's test
B.	Lysine	II.	Neutral FeCl_3 test
C.	Tyrosine	III.	Ceric ammonium nitrate test
D.	Serine	IV.	Hoffman bromamide degradation

Choose the correct answer from the options given below :

JEE Main 2026 (Online) 6th April Morning Shift

Options:

A.

A-IV, B-II, C-I, D-III

B.

A-IV, B-I, C-II, D-III

C.

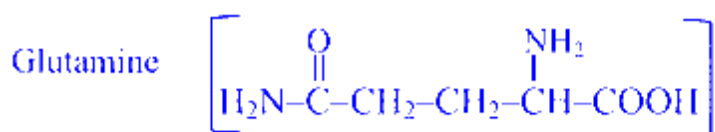
A-III, B-II, C-I, D-IV

D.

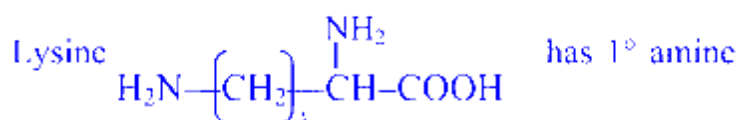
A-IV, B-I, C-III, D-II

Answer: B

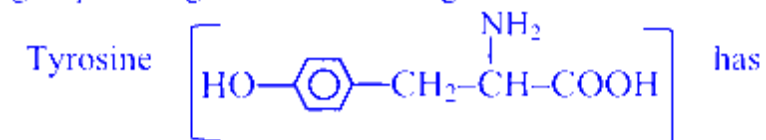
Solution:



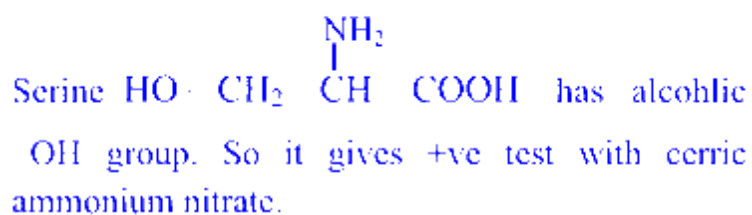
has amide functional group inside chain. So it gives Hoffmann bromamide reaction.



group. So it gives +ve Hinsberg test.



phenolic OH. So it gives +ve test with neutral FeCl_3 .



Question18

Match the List-I with List-II

List-I Deficiency Disease		List-II Vitamin	
A.	Scurvy	I.	Pyridoxine
B.	Convulsions	II.	Vitamin A
C.	Cheilosis	III.	Ascorbic Acid
D.	Xerophthalmia	IV.	Riboflavin

Choose the correct answer from the options given below :

JEE Main 2026 (Online) 6th April Morning Shift

Options:

A.

A-I, B-III, C-II, D-IV

B.

A-I, B-III, C-IV, D-II

C.

A-III, B-I, C-IV, D-II

D.

A-III, B-I, C-II, D-IV

Answer: C

Solution:

Let us recall the vitamin deficiencies and the corresponding diseases:

1. **Scurvy** – caused due to deficiency of **Vitamin C (Ascorbic acid)**.

→ So, $A \rightarrow III$

2. **Convulsions** – occur due to deficiency of **Vitamin B₆ (Pyridoxine)**.

→ So, $B \rightarrow I$

3. **Cheilosis** – characterized by cracks at the corners of the mouth; caused by deficiency of **Vitamin B₂ (Riboflavin)**.

→ So, $C \rightarrow IV$

4. **Xerophthalmia** – dryness of the cornea and conjunctiva; caused by deficiency of **Vitamin A**.

→ So, $D \rightarrow II$

Hence, the correct matching is:

$A - III, B - I, C - IV, D - II$

Therefore, the correct answer is **Option C**.

Question19

The incorrect statement from the following with respect to carbohydrates is :

JEE Main 2026 (Online) 8th April Evening Shift

Options:

A.

All monosaccharides are reducing sugars.

B.

The monosaccharide units obtained from hydrolysis of oligosaccharides are always the same.

C.

Starch and cellulose are typical examples of polysaccharides, which are very high molecular weight compounds of more than ten monosaccharide units.

D.

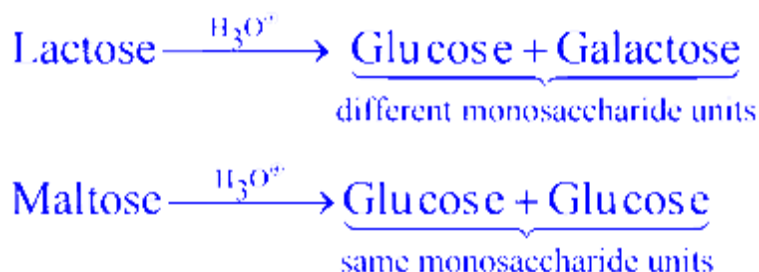
Open chain and cyclic structures co-exist at equilibrium that are responsible for certain properties as in the case of D – (+)-glucose.

Answer: B

Solution:

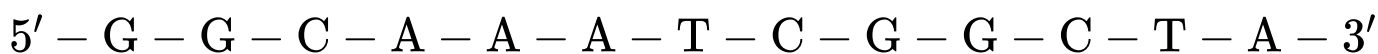
The monosaccharide units obtained from hydrolysis of oligosaccharides may be or may not be same.

Example :



Question 20

The total number of hydrogen bonds of a DNA-double Helix strand whose one strand has the following sequence of bases is _____.



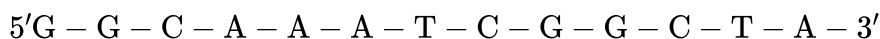
JEE Main 2025 (Online) 4th April Morning Shift

Answer: 33

Solution:

Two nucleic acid chains are wound about each other and held together by H bonds between pair of bases.

Adenine forms two hydrogen bonds with thymine and Guanine forms three hydrogen bonds with cytosine.



In given DNA strand total seven guanine and cytosine bases which form total 21 H-bonds and six adenine and thymine base which will form total 12 H-bonds with other DNA strand.

$$\text{Total no. of H bonds} = 7 \times 3 + 6 \times 2 = 33$$

Ans. 33

Question21

Which of the following acids is a vitamin ?

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Options:

- A. Saccharic acid
- B. Ascorbic acid
- C. Aspartic acid
- D. Adipic acid

Answer: B

Solution:

Correct Answer: Ascorbic Acid

Ascorbic acid is commonly known as Vitamin C. It is a water-soluble vitamin essential for various bodily functions, including:

Antioxidant Function: It helps protect cells from damage caused by free radicals.

Collagen Synthesis: It is crucial for the formation of collagen, a protein necessary for skin, cartilage, tendons, ligaments, and blood vessels.

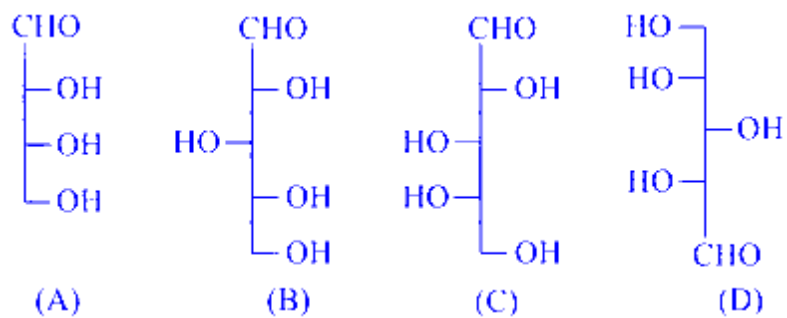
Immune Support: It supports the immune system by enhancing the body's natural defenses.

Iron Absorption: It improves the absorption of non-heme iron from plant-based foods.

Among the given options, only ascorbic acid qualifies as a vitamin.

Question22

Identify the number of structure/s from the following which can be correlated to D-glyceraldehyde.



JEE Main 2025 (Online) 22nd January Evening Shift

Options:

A. three

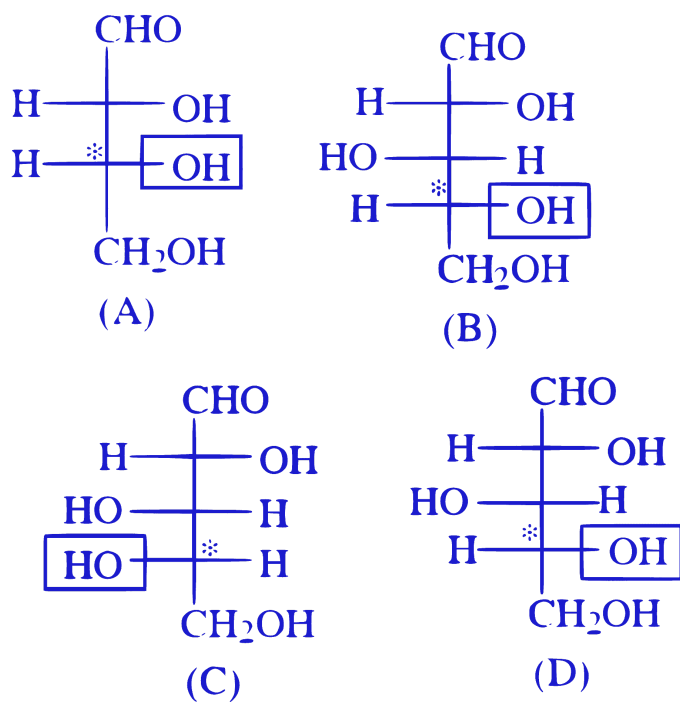
B. two

C. one

D. four

Answer: A

Solution:



In A, B, D — OH group in right hand side then D-configuration is assigned.

Question23

Given below are two statements:

Statement I: Fructose does not contain an aldehydic group but still reduces Tollen's reagent

Statement II: In the presence of base, fructose undergoes rearrangement to give glucose.

In the light of the above statements, choose the correct answer from the options given below

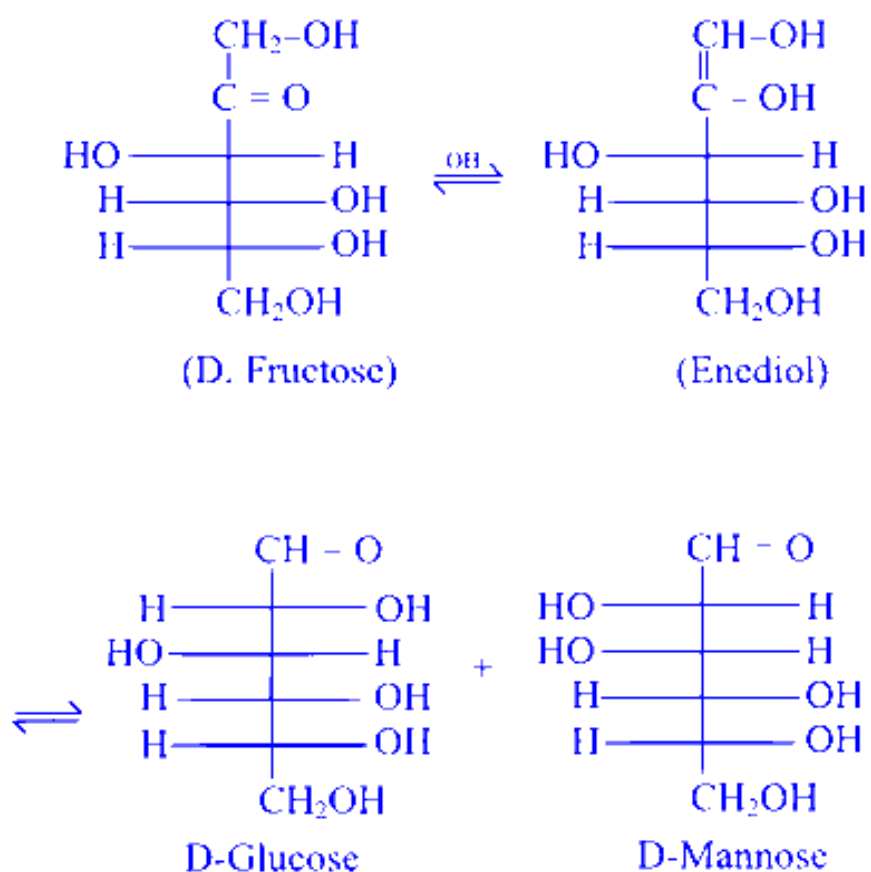
JEE Main 2025 (Online) 23rd January Morning Shift

Options:

- A. Statement I is true but Statement II is false
- B. Statement I is false but Statement II is true
- C. Both Statement I and Statement II are true
- D. Both Statement I and Statement II are false

Answer: C

Solution:



Question24

The α -Helix and β - Pleated sheet structures of protein are associated with its :

JEE Main 2025 (Online) 23rd January Evening Shift

Options:

- A. tertiary structure
- B. quaternary structure
- C. primary structure
- D. secondary structure

Answer: D

Solution:

α -helix and β -pleated sheet belongs to secondary structure of protein, which have hydrogen bonds.

Question25

The carbohydrate "Ribose" present in DNA, is

- A. A pentose sugar**
- B. present in pyranose form**
- C. in "D" configuration**
- D. a reducing sugar, when free**
- E. in α -anomeric form**

Choose the correct answer from the options given below:

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Options:

A. A, D and E Only

B. B, D and E Only

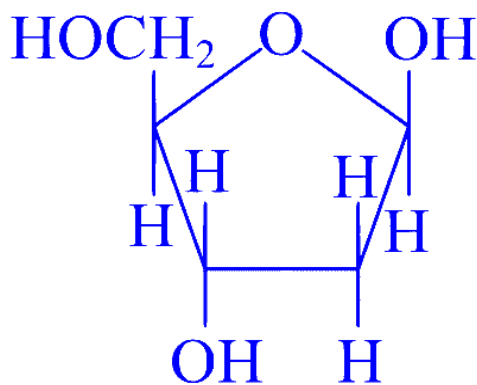
C. A, C and D Only

D. A, B and E Only

Answer: C

Solution:

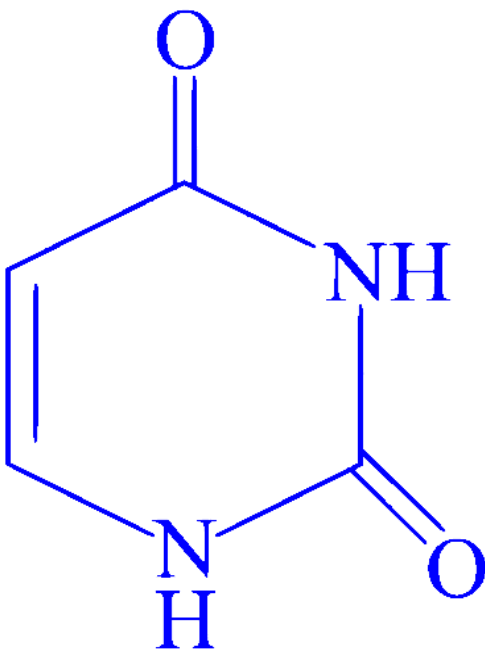
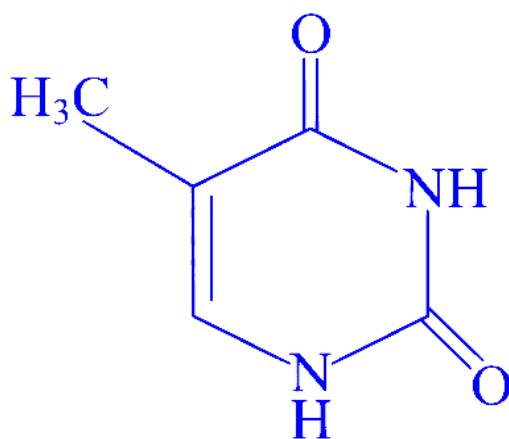
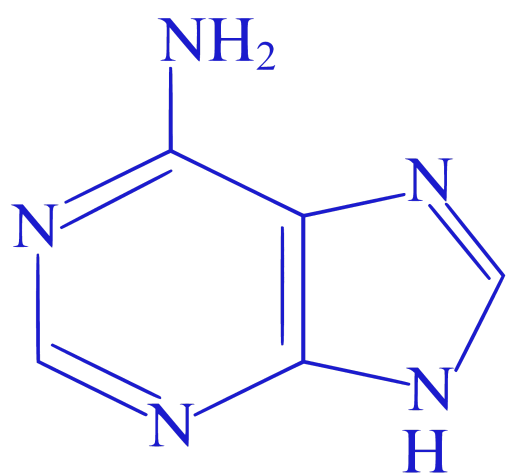
In Ribose carbohydrate present in DNA is β - 2- Deoxy-D-Ribose whose structure is

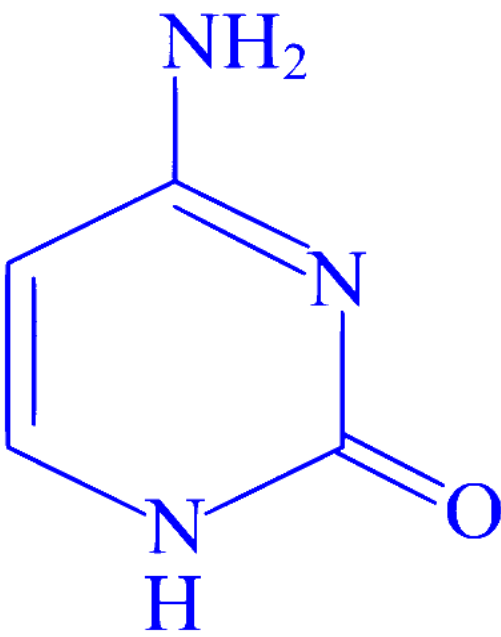


which is a reducing *D*-sugar in β anomeric form & it is a pentose sugar.

Question26

Match List - I with List - II.

	List - I		List - II
(A)	Adenine	(I)	 The chemical structure of Adenine is a purine base, consisting of a fused pyrimidine and imidazole ring system. It features two carbonyl groups (C=O) and two NH groups. The structure is drawn in blue.
(B)	Cytosine	(II)	 The chemical structure of Cytosine is a pyrimidine base, consisting of a single six-membered ring. It features two carbonyl groups (C=O) and two NH groups, with a methyl group (H₃C) attached to the ring. The structure is drawn in blue.
(C)	Thymine	(III)	 The chemical structure of Thymine is a pyrimidine base, consisting of a single six-membered ring. It features two carbonyl groups (C=O) and two NH groups, with a methyl group (H₃C) attached to the ring. The structure is drawn in blue.

	List - I		List - II
(D)	Uracil	(IV)	

Choose the correct answer from the options given below :

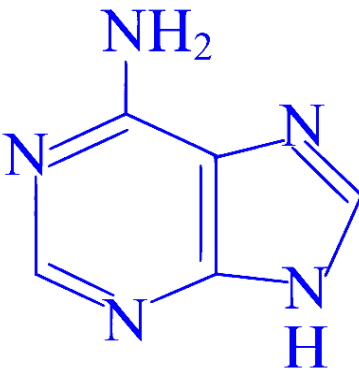
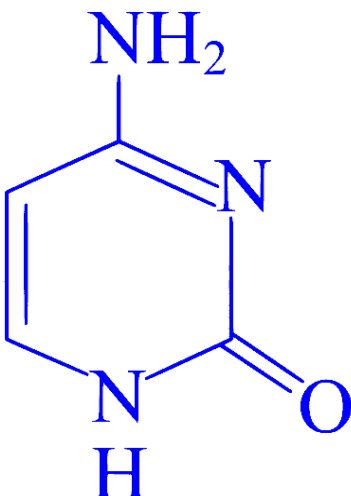
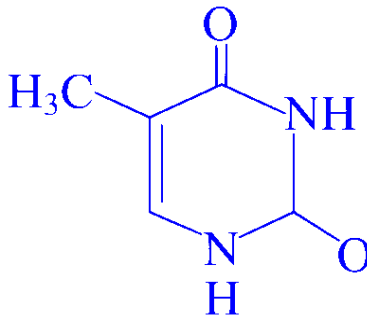
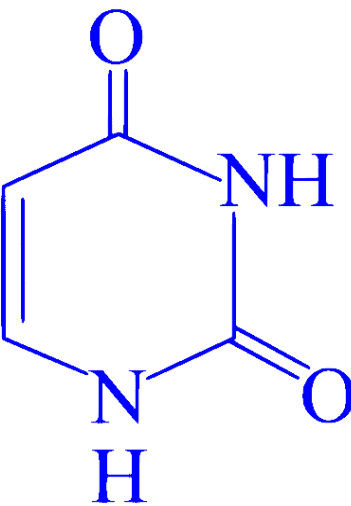
JEE Main 2025 (Online) 24th January Evening Shift

Options:

- A. (A)-(IV), (B)-(III), (C)-(II), (D)-(I)
- B. (A)-(III), (B)-(IV), (C)-(II), (D)-(I)
- C. (A)-(III), (B)-(I), (C)-(IV), (D)-(II)
- D. (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

Answer: B

Solution:

(A)	Adenine	(III)	
(B)	Cytosine	(IV)	
(C)	Thymine	(II)	
(D)	Uracil	(I)	

Question 27

Given below are two statements:

Statement I : D-glucose pentaacetate reacts with 2,4 - dinitrophenylhydrazine

Statement II : Starch, on heating with concentrated sulfuric acid at 100°C and 2 – 3 atmosphere pressure produces glucose.

In the light of the above statements, choose the correct answer from the options given below

JEE Main 2025 (Online) 28th January Morning Shift

Options:

- A. Statement I is true but Statement II is false
- B. Both Statement I and Statement II are true
- C. Statement I is false but Statement II is true
- D. Both Statement I and Statement II are false

Answer: C

Solution:

Statement I: D-glucose pentaacetate does not react with 2,4-dinitrophenylhydrazine (DNPH). This is because D-glucose pentaacetate is a derivative where all the hydroxyl groups of D-glucose have been acetylated, eliminating the presence of a free aldehyde group. Since DNPH reacts specifically with free aldehyde or ketone groups to form hydrazones, D-glucose pentaacetate does not participate in this reaction.

Statement II: Starch, when heated with concentrated sulfuric acid at 100°C and under a pressure of 2-3 atmospheres, undergoes hydrolysis. This process breaks the glycosidic linkages in starch, resulting in the production of glucose. This statement is accurate as starch is a polysaccharide composed of glucose units and can be broken down into glucose under these conditions.

The correct information based on the statements is:

Statement I is false, as D-glucose pentaacetate does not react with 2,4-dinitrophenylhydrazine due to the absence of a free aldehyde group.

Statement II is true, as the described method effectively converts starch into glucose.

Question 28

Identify correct conversion during acidic hydrolysis from the following :

(A) starch gives galactose.

(B) cane sugar gives equal amount of glucose and fructose.

(C) milk sugar gives glucose and galactose.

(D) amylopectin gives glucose and fructose.

(E) amylose gives only glucose.

Choose the correct answer from the options given below :

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Options:

A.

(A), (B) and (C) only

B.

(B), (C) and (D) only

C.

(B), (C) and (E) only

D.

(C), (D) and (E) only

Answer: C

Solution:

(A) Starch $\xrightarrow{\text{H}^+/\text{H}_2\text{O}}$ Glucose

(B) Cane sugar $\xrightarrow{\text{H}^+/\text{H}_2\text{O}}$ glu cose + fructose
(Sucrose) 50% 50%

(C) Milk sugar $\xrightarrow{\text{H}^+/\text{H}_2\text{O}}$ glu cose + galactose
(Lactose)

(D) Amylopectin $\xrightarrow{\text{H}^+/\text{H}_2\text{O}}$ Glucose

(E) Amylose $\xrightarrow{\text{H}^+/\text{H}_2\text{O}}$ Glucose

So, correct options are B, C and E only

Question29

Match List-I with List-II.

List - I (Saccharides)	List - II (Glycosidic-linkages found)
(A) Sucrose	(I) α 1 – 4
(B) Maltose	(II) α 1 – 4 and α 1 – 6
(C) Lactose	(III) α 1 – β 2
(D) Amylopectin	(IV) β 1 – 4

Choose the correct answer from the options given below :

JEE Main 2025 (Online) 28th January Evening Shift

Options:

A.

(A)-(III), (B)-(I), (C)-(IV), (D)-(II)

B.

(A)-(I), (B)-(II), (C)-(III), (D)-(IV)

C.

(A)-(II), (B)-(IV), (C)-(III), (D)-(I)

D.

(A)-(IV), (B)-(II), (C)-(I), (D)-(III)

Answer: A

Solution:

(A) Sucrose $\rightarrow \alpha_1 - \beta_2$ Glycosidic linkage

(B) Maltose $\rightarrow \alpha$ 1-4 Glycosidic linkage

(C) Lactose $\rightarrow \beta$ 1-4 Glycosidic linkage

(D) Amylopectin $\rightarrow \alpha$ 1-4 and α 1-6 Glycosidic linkage

A – III, B – I, C – IV, D – II

Question30

Match List - I with List - II.

List - I (Carbohydrate)	List - II (Linkage Source)
(A) Amylose	(I) β - $C_1 - C_4$, plant
(B) Cellulose	(II) α - $C_1 - C_4$, animal
(C) Glycogen	(III) α - $C_1 - C_4$, α - $C_1 - C_6$, plant
(D) Amylopectin	(IV) α - $C_1 - C_4$, plant

Choose the correct answer from the options given below :

JEE Main 2025 (Online) 29th January Morning Shift

Options:

A.

(A)-(II), (B)-(III), (C)-(I), (D)-(IV)

B.

(A)-(III), (B)-(III), (C)-(I), (D)-(IV)

C.

(A)-(IV), (B)-(I), (C)-(III), (D)-(II)

D.

(A)-(IV), (B)-(I), (C)-(II), (D)-(III)

Answer: D

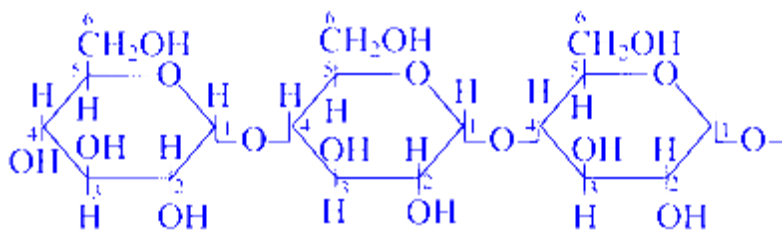
Solution:

A) Amylose - Polysaccharide of glucose units with glycosidic linkage $\alpha - 1, 4$.

The source is plant.

So, the linkage is $\alpha - c_1 - C_4$.

Glucose units in amylose are linked by alpha bonds between the 1st and 4th carbon atoms.

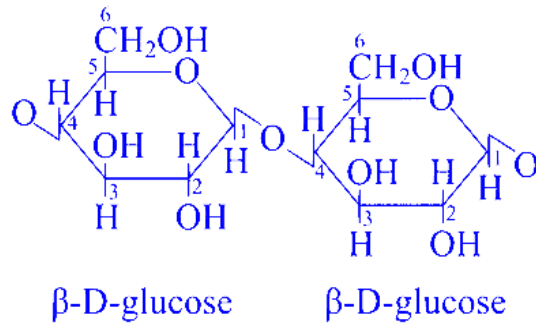


Amylose match with $\alpha - c_1 - c_4$, plant

(A) with (IV)

B) Cellulose - The glycosidic linkage in cellulose is $\beta - 1, 4$. Glucose monomers are joined by linkage between the c_1 of the one glucose ring and the C_4 of the next ring.

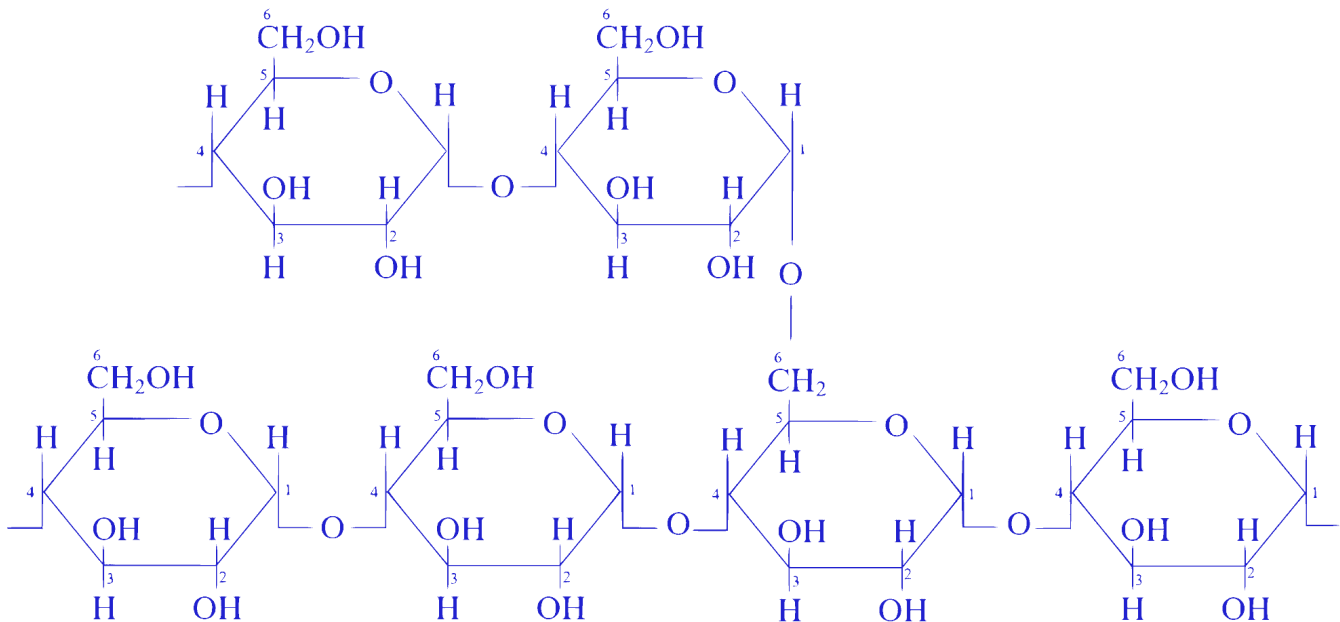
Source is plant



Cellulose match with $\beta - C_1 - C_4$, plant.

(B) with (I)

c) Glycogen - Glycogen is a polymer of glucose that is stored in animals as an energy reserve. It is used as carbohydrate storage in animals. It is a polymer of α -glucose with $\alpha - 1, 4$ -glycosidic and $\alpha - 1, 6$ -glycosidic links.



$\alpha - 1, 4$ -linkage forms during linear polymer.

$\alpha - 1, 6$ - linkage forms during the formation of branching.

Glycogen match with $\alpha - C_1 - C_4$, animal

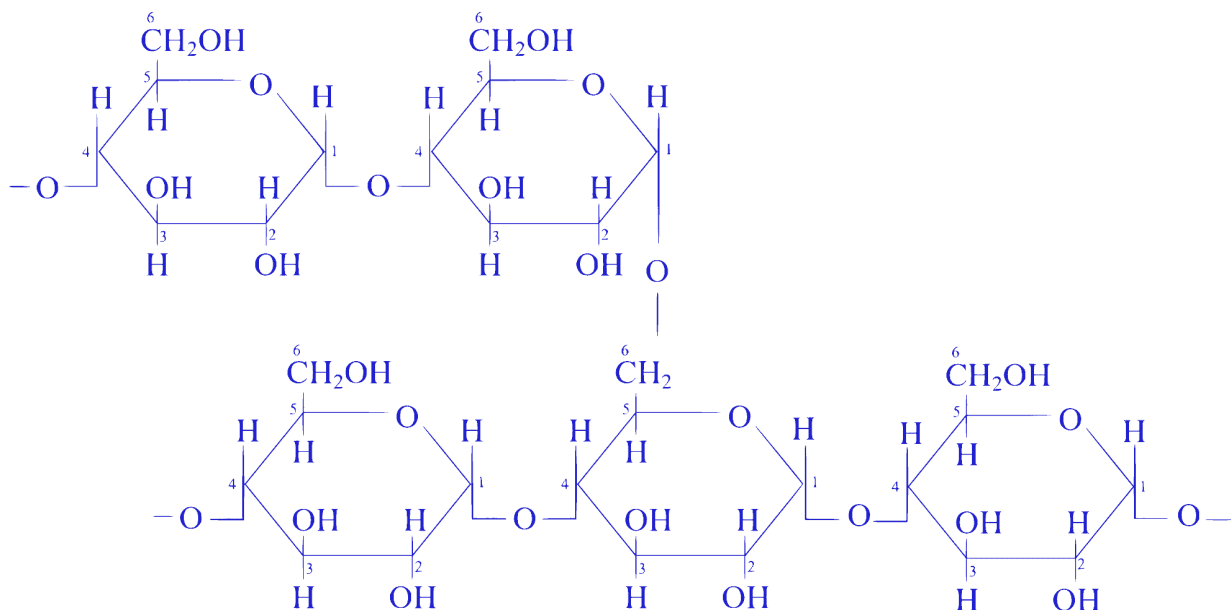
(C) with (II)

D) Amylopectin - It is the polymer of glucose units linked by $\alpha - 1, 4$ and $\alpha - 1, 6$ glycosidic bonds.

$\alpha - 1, 4$ - glucose units linked in a linear way.

$\alpha - 1, 6$ - These bonds branch out from the linear chains of glucose units.

They are stored in large complexes called starch granules in plants.



Amylopectin match with $\alpha - c_1, c_4$, $\alpha - c_1 - c_6$, plant (D) with (III).

Answer: (A) – (IV), (B) – (I), (C) – (II), (D) – (III). Option (3).

Question31

Identify the essential amino acids from below:

(A) Valine

(B) Proline

(C) Lysine

(D) Threonine

(E) Tyrosine

Choose the correct answer from the options given below:

JEE Main 2025 (Online) 29th January Evening Shift

Options:

A.

(A), (C) and (D) only

B.

(C), (D) and (E) only

C.

(A), (C) and (E) only

D.

(B), (C) and (E) only

Answer: A

Solution:

The nine essential amino acids that must be obtained from the diet are:

Histidine

Isoleucine

Leucine

Lysine

Methionine

Phenylalanine

Threonine

Tryptophan

Valine

From the provided list:

Valine (A) is essential.

Proline (B) is non-essential.

Lysine (C) is essential.

Threonine (D) is essential.

Tyrosine (E) is non-essential (it is considered conditionally essential, but under normal conditions it is non-essential).

Thus, the essential amino acids among those listed are Valine, Lysine, and Threonine.

So the correct answer is:

Option A: (A), (C) and (D) only

Question 32

Identify the correct statement among the following :

JEE Main 2025 (Online) 2nd April Morning Shift

Options:

- A. All naturally occurring amino acids except glycine contain one chiral centre.
- B. Amino acid, cysteine can easily undergo dimerisation due to the presence of free SH group.
- C. Glutamic acid is the only amino acid that contains a —COOH group at the side chain.
- D. All naturally occurring amino acids are optically active.

Answer: B

Solution:

To clarify the statements regarding amino acids:

Chirality in Amino Acids: Most naturally occurring amino acids contain at least one chiral center, but there are exceptions. For example, Isoleucine has two chiral centers, and Glycine is an exception as it is optically inactive due to lacking a chiral center.

Cysteine and Dimerization: The amino acid Cysteine can easily undergo dimerization. This is due to its free sulfhydryl (SH) group, which can form disulfide bonds, leading to the formation of a dimer.

Carboxyl Groups in Amino Acids: Glutamic acid is not the only amino acid with a carboxyl group in its side chain. Aspartic acid also contains a carboxyl group in its side chain.

Optical Activity: While many naturally occurring amino acids are indeed optically active, Glycine is an exception as it does not possess a chiral center and is hence optically inactive.

In summary:

Not all naturally occurring amino acids are optically active (e.g., Glycine).

Not all naturally occurring amino acids contain only one chiral center (e.g., Isoleucine has two).

Both Glutamic acid and Aspartic acid contain side chain carboxyl groups.

Cysteine can dimerize due to its SH group.

Question33

A tetrapeptide, " x " on complete hydrolysis produced glycine (Gly), alanine (Ala), valine (Val), leucine (Leu) in equimolar proportion each. The number of tetrapeptides (sequences) possible involving each of these amino acids is :

JEE Main 2025 (Online) 2nd April Evening Shift

Options:

A. 32

B. 24

C. 8

D. 16

Answer: B

Solution:

To determine the number of tetrapeptide sequences possible using the amino acids glycine (Gly), alanine (Ala), valine (Val), and leucine (Leu), we consider the concept of permutations. Since each tetrapeptide sequence consists of four different amino acids, we need to calculate the number of ways to arrange these four different elements.

The formula for the number of permutations of a set of n distinct elements is given by $n!$ (n factorial), where $n! = n \times (n - 1) \times (n - 2) \times \cdots \times 1$.

For our case:

$$n = 4$$

Hence, the number of possible sequences (or permutations) is:

$$4! = 4 \times 3 \times 2 \times 1 = 24$$

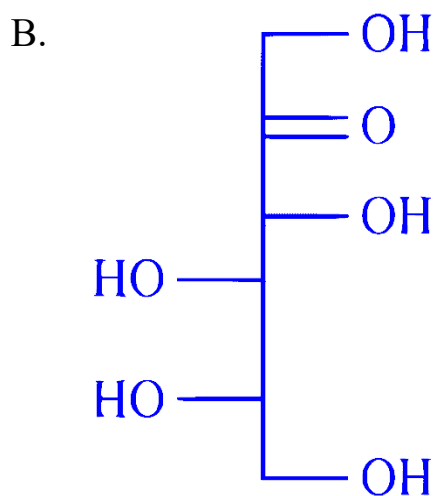
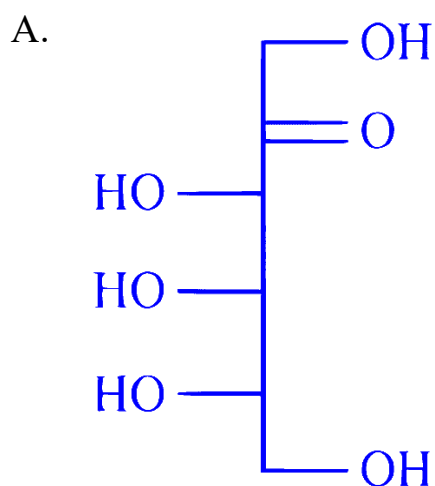
Thus, there are 24 different tetrapeptide sequences possible using each of the four amino acids exactly once.

Question34

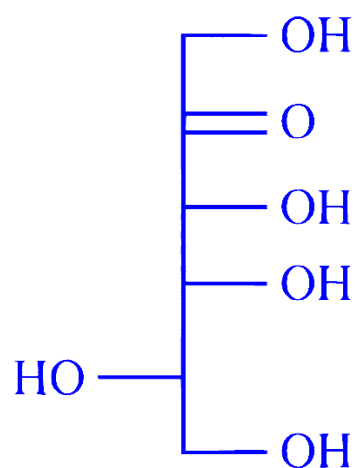
Which of the following is the correct structure of L-Fructose?

JEE Main 2025 (Online) 3rd April Morning Shift

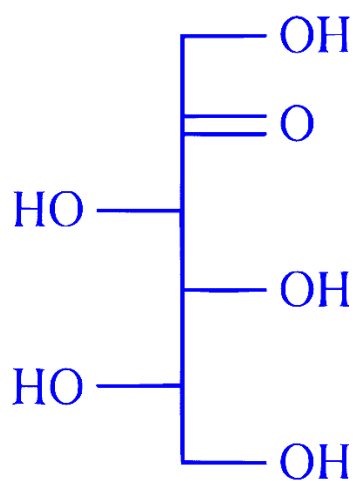
Options:



C.

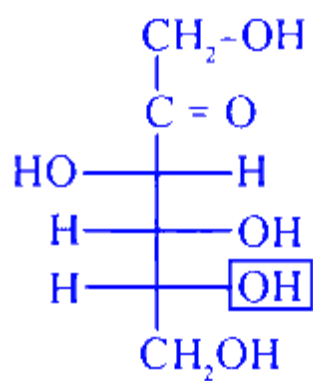


D.

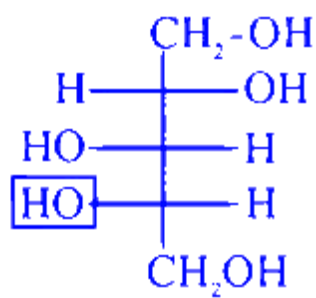


Answer: B

Solution:



D - Fructose



L - Fructose

Question35

Fat soluble vitamins are:

- A. Vitamin B₁**
- B. Vitamin C**
- C. Vitamin E**
- D. Vitamin B 12**
- E. Vitamin K**

Choose the correct answer from the options given below:

JEE Main 2025 (Online) 3rd April Evening Shift

Options:

- A. B & C Only
- B. C & D Only
- C. C & E Only
- D. A & B Only

Answer: C

Solution:

Fat-soluble vitamins are A, D, E and K. From the list:

A: Vitamin B₁ – water-soluble

B: Vitamin C – water-soluble

C: Vitamin E – fat-soluble

D: Vitamin B₁₂ – water-soluble

E: Vitamin K – fat-soluble

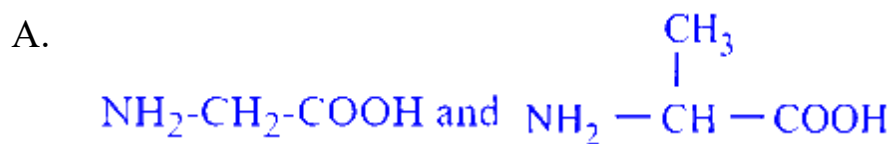
So the fat-soluble ones are C (E) and E (K), which is Option C.

Question36

Identify the pair of reactants that upon reaction, with elimination of HCl will give rise to the dipeptide Gly-Ala.

JEE Main 2025 (Online) 4th April Morning Shift

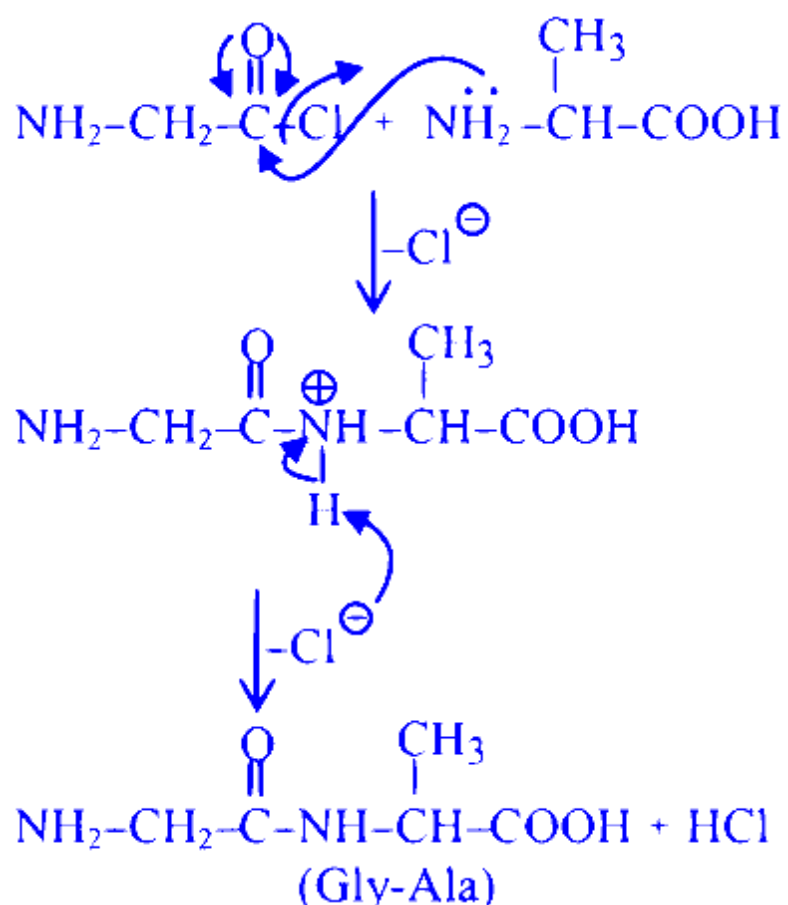
Options:





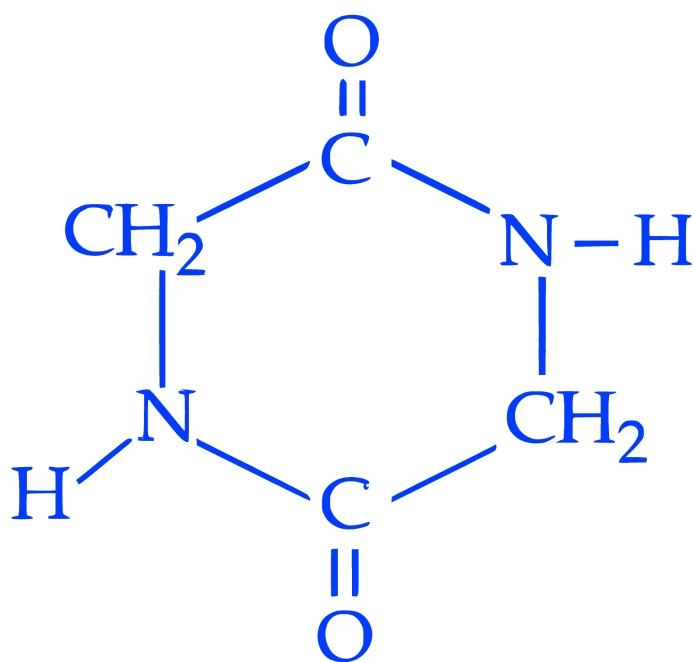
Answer: C

Solution:



Question37

A dipeptide, " x " on complete hydrolysis gives " y " and " z ". " y " on treatment with aq. HNO_2 produces lactic acid. On the other hand " z " on heating gives the following cyclic molecule.



Based on the information given, the dipeptide X is :

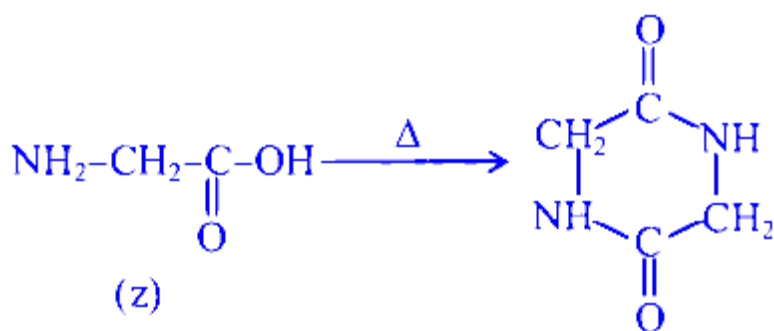
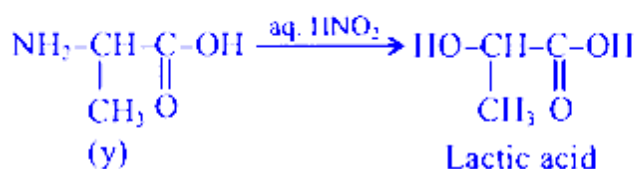
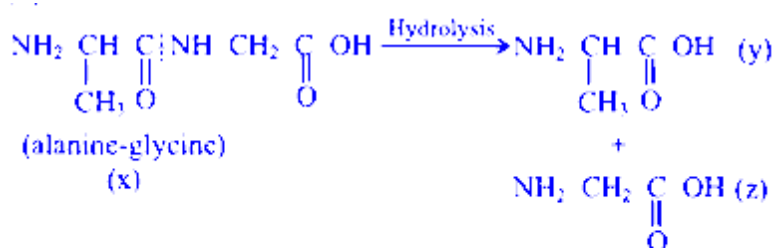
JEE Main 2025 (Online) 4th April Evening Shift

Options:

- A. alanine-alanine
- B. valine-glycine
- C. valine-leucine
- D. alanine-glycine

Answer: D

Solution:



Question38

Given below are two statements:

Statement I: D-(+)-glucose + D-(+) fructose $\xrightarrow{-\text{H}_2\text{O}}$ Sucrose

sucrose $\xrightarrow{\text{hydrolysis}}$ D – (+) glucose + D – (+) fructose

Statement II: Invert sugar is formed during sucrose hydrolysis

In the light of the above statements, choose the correct answer from the options given below

JEE Main 2025 (Online) 7th April Morning Shift

Options:

- A. Both Statement I and Statement II are False
- B. Statement I is false but Statement II are true
- C. Both Statement I and Statement II are true
- D. Statement I is true but Statement II is false

Answer: B

Solution:

In Statement I, the reaction shown is incorrect. During the hydrolysis of sucrose, D-(+)-glucose and D-(-)-fructose are produced. However, the statement incorrectly specifies D-(+)-fructose, making Statement I inaccurate.

Statement II is accurate. When sucrose undergoes hydrolysis, it results in the formation of invert sugar, a mixture of glucose and fructose. This change involves the inversion of optical rotation, which is where the term "invert sugar" comes from.

Question39

Given below are two statements :

Statement (I) : On hydrolysis, oligo peptides give rise to fewer number of α -amino acids while proteins give rise to a large number of β -amino acids.

Statement (II) : Natural proteins are denatured by acids which convert the water soluble form of fibrous proteins to their water insoluble form.

In the light of the above statements, choose the most appropriate answer from the options given below :

JEE Main 2025 (Online) 7th April Evening Shift

Options:

A.

Statement I is incorrect but **Statement II** is correct

B.

Both **Statement I** and **Statement II** are incorrect

C.

Both **Statement I** and **Statement II** are correct

D.

Statement I is correct but **Statement II** is incorrect

Answer: B

Solution:

When oligopeptides undergo hydrolysis, they yield a small number of α -amino acids. In contrast, proteins do not produce β -amino acids upon hydrolysis. Thus, Statement I is incorrect.

Furthermore, natural proteins, when subjected to denaturation by acids, do not convert fibrous proteins from a water-soluble to a water-insoluble form, as fibrous proteins are inherently not water-soluble. Therefore, Statement II is also incorrect.

In conclusion, both Statement I and Statement II are incorrect.

Question40

**Two nucleotides are joined together by a linkage known as :
[27-Jan-2024 Shift 1]**

Options:

A. Phosphodiester linkage

B. Glycosidic linkage

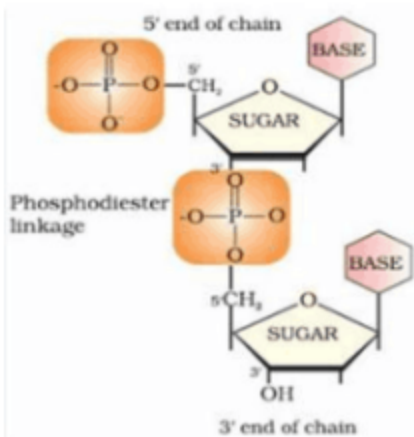
C. Disulphide linkage

D. Peptide linkage

Answer: A

Solution:

Phosphodiester linkage



Question41

Two nucleotides are joined together by a linkage known as :
[27-Jan-2024 Shift 1]

Options:

- A. Phosphodiester linkage
- B. Glycosidic linkage
- C. Disulphide linkage
- D. Peptide linkage

Answer: A

Question42

Which structure of protein remains intact after coagulation of egg white on boiling?
[27-Jan-2024 Shift 2]

Options:

- A. Primary
- B. Tertiary
- C. Secondary
- D. Quaternary

Answer: A

Solution:

Boiling an egg causes denaturation of its protein resulting in loss of its quaternary, tertiary and secondary structures.

Question43

**Type of amino acids obtained by hydrolysis of proteins is :
[29-Jan-2024 Shift 1]**

Options:

- A. β
- B. α
- C. δ
- D. γ

Answer: B

Solution:

Proteins are natural polymers composed of α -amino acids which are connected by peptide linkages. Hence proteins upon acidic hydrolysis produce α -amino acids.

Question44

Match List I with List II

LIST-I (Substances)	LIST-II (Element Present)
A.Ziegler catalyst	I.Rhodium
B.Blood Pigment	II. Cobalt
C.Wilkinson catalyst	III.Iron
D.Vitamin B ₁₂	IV.Titanium

Choose the correct answer from the options given below:
[29-Jan-2024 Shift 1]

Options:

- A. A-II, B-IV, C-I, D-III
- B. A-II, B-III, C-IV, D-I
- C. A-III, B-II, C-IV, D-I
- D. A-IV, B-III, C-I, D-II

Answer: D

Solution:

Ziegler catalyst → Titanium
Blood pigment → Iron
Wilkinson catalyst → Rhodium
Vitamin B₁₂ → Cobalt

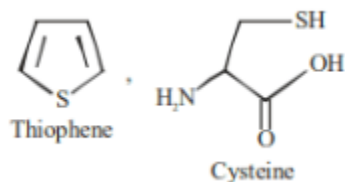
Question45

Number of compounds among the following which contain sulphur as heteroatom is

Furan, Thiophene, Pyridine, Pyrrole, Cysteine, Tyrosine
[29-Jan-2024 Shift 1]

Answer: 2

Solution:



Question46

Match List I with List II

	List I (Bio Polymer)		List II (Monomer)
A.	Starch	I.	nucleotide
B.	Cellulose	II.	α -glucose
C.	Nucleic acid	III.	β -glucose
D.	Protein	IV.	α -amino acid

**Choose the correct answer from the options given below :-
[29-Jan-2024 Shift 2]**

Options:

- A. A-II, B-I, C-III, D-IV
- B. A-IV, B-II, C-I, D-III
- C. A-I, B-III, C-IV, D-II
- D. A-II, B-III, C-I, D-IV

Answer: D

Solution:

Question47

Sugar which does not give reddish brown precipitate with Fehling's reagent is:

[30-Jan-2024 Shift 1]

Options:

- A. Sucrose
- B. Lactose
- C. Glucose
- D. Maltose

Answer: A

Solution:

Sucrose do not contain hemiacetal group.
Hence it does not give test with Fehling solution.
While all other give positive test with Fehling solution

Question48

The total number of correct statements, regarding the nucleic acids is

- A. RNA is regarded as the reserve of genetic information.**
- B. DNA molecule self-duplicates during cell division**
- C. DNA synthesizes proteins in the cell.**
- D. The message for the synthesis of particular proteins is present in DNA**
- E. Identical DNA strands are transferred to daughter cells.**

[30-Jan-2024 Shift 2]

Answer: 3

Solution:

- A. RNA is regarded as the reserve of genetic information. (False)
B. DNA molecule self-duplicates during cell division. (True)
C. DNA synthesizes proteins in the cell. (False)
D. The message for the synthesis of particular proteins is present in DNA. (True)
E. Identical DNA strands are transferred to daughter cells. (True)
-

Question49

Match List I with List II

	LIST-I		LIST-II
A.	Glucose/ NaHCO_3/Δ	I.	Gluconic acid
B.	Glucose/ HNO_3	II.	No reaction
C.	Glucose/ HI/Δ	III.	n-hexane
D.	Glucose/Bromine water	IV.	Saccharic acid

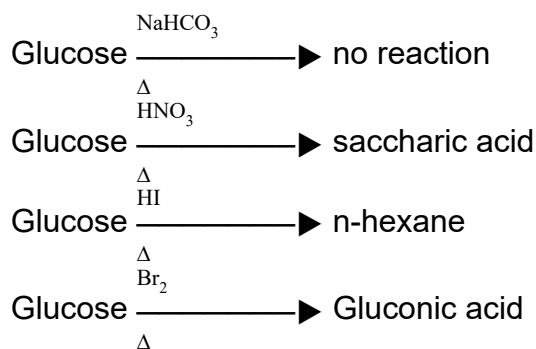
**Choose the correct answer from the options given below:
[31-Jan-2024 Shift 1]**

Options:

- A. A-IV, B-I, C-III, D-II
B. A-II, B-IV, C-III, D-I
C. A-III, B-II, C-I, D-IV
D. A-I, B-IV, C-III, D-II

Answer: B

Solution:



Question50

From the vitamins A, B₁, B₆, B₁₂, C, D, E and K, the number vitamins that can be stored in our body is _____
[31-Jan-2024 Shift 2]

Answer: 5

Solution:

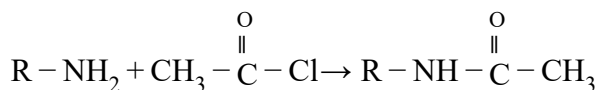
Vitamins A, D, E, K and B₁₂ are stored in liver and adipose tissue.

Question51

A compound (x) with molar mass 108gmol^{-1} undergoes acetylation to give product with molar mass 192gmol^{-1} . The number of amino groups in the compound (x) is _____
[31-Jan-2024 Shift 2]

Answer: 2

Solution:



Gain in molecular weight after acylation with one $-NH_2$ group is 42 .

Total increase in molecular weight = 84

$$\therefore \text{Number of amino group in } x = \frac{84}{42} = 2$$

Question52

If one strand of a DNA has the sequence ATGCTTCA, sequence of the bases in complementary strand is:
[1-Feb-2024 Shift 1]

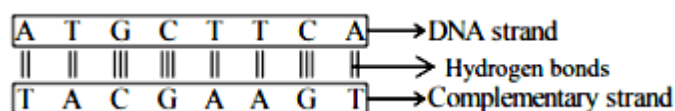
Options:

- A. CATTAGCT
- B. TACGAAGT
- C. GTACTTAC
- D. ATGCGACT

Answer: B

Solution:

Adenine base pairs with thymine with 2 hydrogen bonds and cytosine base pairs with guanine with 3 hydrogen bonds.



Question53

The number of tripeptides formed by three different amino acids using each amino acid once is _____
[1-Feb-2024 Shift 2]

Answer: 6

Solution:

Let 3 different amino acid are A, B, C then following combination of tripeptides can be formed-
ABC, ACB, BAC, BCA, CAB, CBA

Question54

**A short peptide on complete hydrolysis produces 3 moles of glycine (G), two moles of leucine (L) and two moles of valine (V) per mole of peptide. The number of peptide linkages in it are _____.
[30-Jan-2023 Shift 2]**

Answer: 6

Solution:

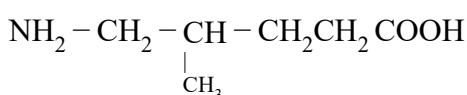
Number of peptide linkage = (amino acid – 1)
= 7 – 1 = 6

Question55

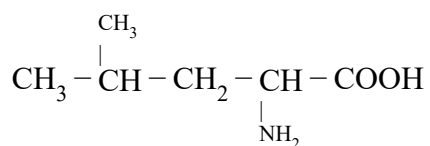
**A protein ' X ' with molecular weight of 70, 000u, on hydrolysis gives amino acids. One of these amino acid is
[31-Jan-2023 Shift 1]**

Options:

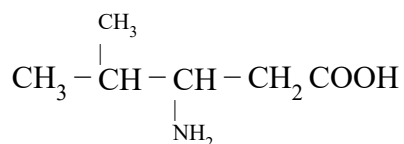
A.



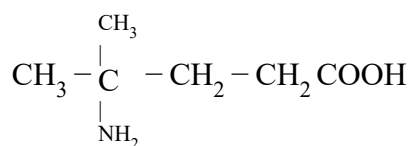
B.



C.



D.



Answer: B

Solution:

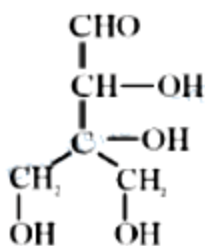
Only in option (2) α -Amino acid is given all the other options are not α -Amino acids.

Question56

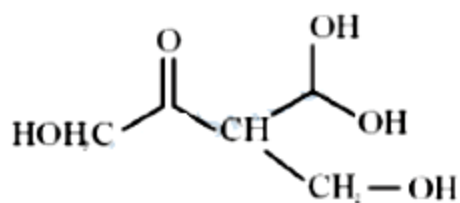
Compound A, $\text{C}_5\text{H}_{10}\text{O}_5$, given a tetraacetate with Ac_2O and oxidation of A with $\text{Br}_2 - \text{H}_2\text{O}$ gives an acid, $\text{C}_5\text{H}_{10}\text{O}_6$. Reduction of A with HI gives isopentane. The possible structure of A is :
[31-Jan-2023 Shift 2]

Options:

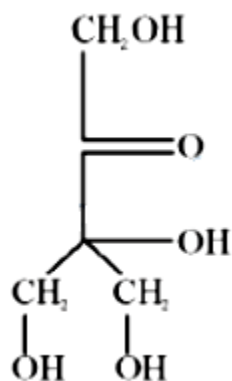
A.



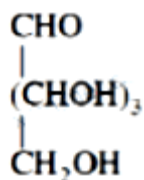
B.



C.



D.



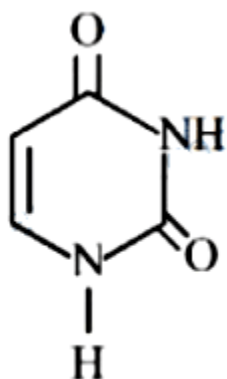
Answer: A

Solution:

(i) Formation of tetraacetate with Ac_2O means compound A has four $-\text{OH}$ linkage.
Reduction of A with HI gives Isopentane i.e. molecule contains five carbon atom.

Question57

Uracil is base present in RNA with the following structure. % of N in uracil is ____



Given :

**Molar mass N = 14gmol^{-1} ; O = 16gmol^{-1} ; C = 12gmol^{-1} ; H = 1gmol^{-1} ;
[24-Jan-2023 Shift 1]**

Answer: 25

Solution:

Mol. Wt of $\text{C}_4\text{N}_2\text{H}_4\text{O}_2 = 112$

$$\%N = \frac{28}{112} \times 100 = 25\%$$

Question58

**Total number of tripeptides possible by mixing of valine and proline is ____
[24-Jan-2023 Shift 2]**

Answer: 8

Solution:

No. of possible tripeptide :

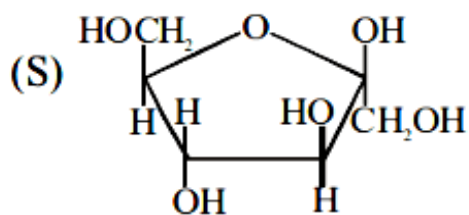
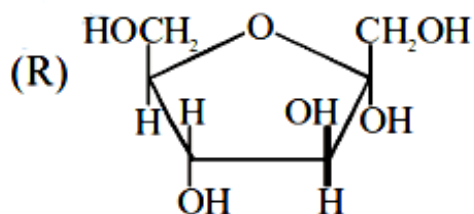
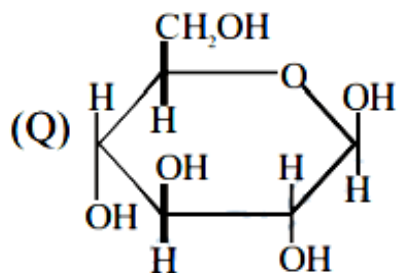
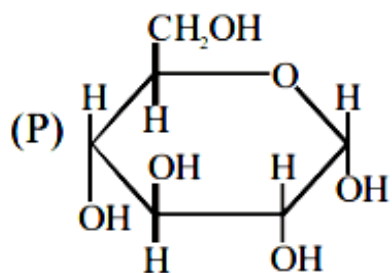
Val & Pro is 2^3

- (1) val - val - val
 - (2) pro - pro - pro
 - (3) val - pro - pro
 - (4) pro - val - pro
 - (5) val - val - pro
 - (6) val - pro - val
 - (7) pro - pro - val
 - (8) pro - val - val
-

Question 59

Match items of Row I with those of Row II.

Row I :



Row II :

(i) α -D-(-) Fructofuranose.

(ii) β -D-(-) Fructofuranose

(iii) α -D-(-) Glucopyranose.

(iv) β -D-(-) Glucopyranose

Correct match is

[25-Jan-2023 Shift 1]

Options:

A. P $\rightarrow$ iv, Q $\rightarrow$ iii, R $\rightarrow$ i, S $\rightarrow$ ii

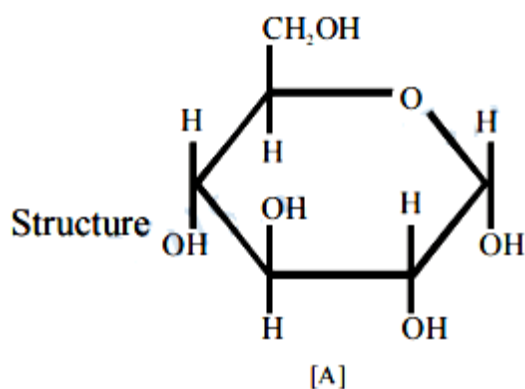
B. P $\rightarrow$ i, Q $\rightarrow$ ii, R $\rightarrow$ iii, S $\rightarrow$ iv

C. P $\rightarrow$ iii, Q $\rightarrow$ iv, R $\rightarrow$ ii, S $\rightarrow$ i

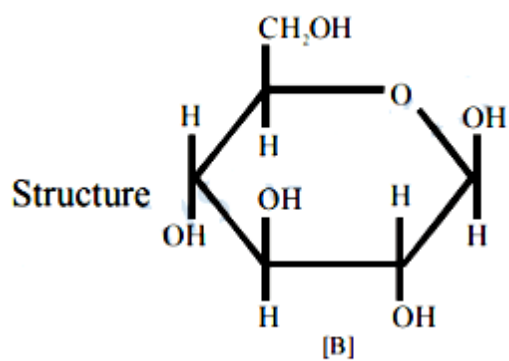
D. P $\rightarrow$ iii, Q $\rightarrow$ iv, R $\rightarrow$ i, S $\rightarrow$ ii

Answer: D

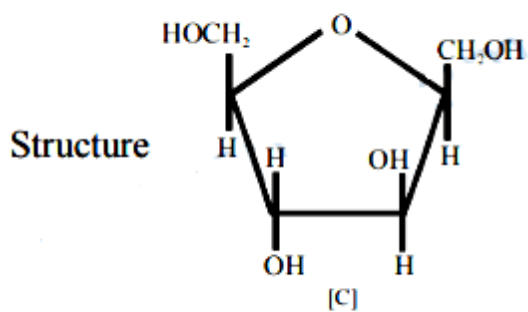
Solution:



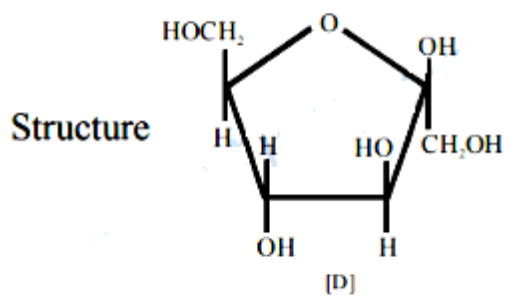
Represents α -D-(+) Glucopyranose



Represents beta-D- (+) Glucopyranose



Represents beta-D- (-) Fructofuranose



Represents beta-D- (-) Fructofuranose

(from the given options best answer is D)

Question60

Number of cyclic tripeptides formed with 2 amino acids A and B is:
[29-Jan-2023 Shift 1]

Options:

A. 2

B. 3

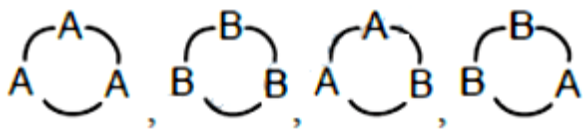
C. 5

D. 4

Answer: D

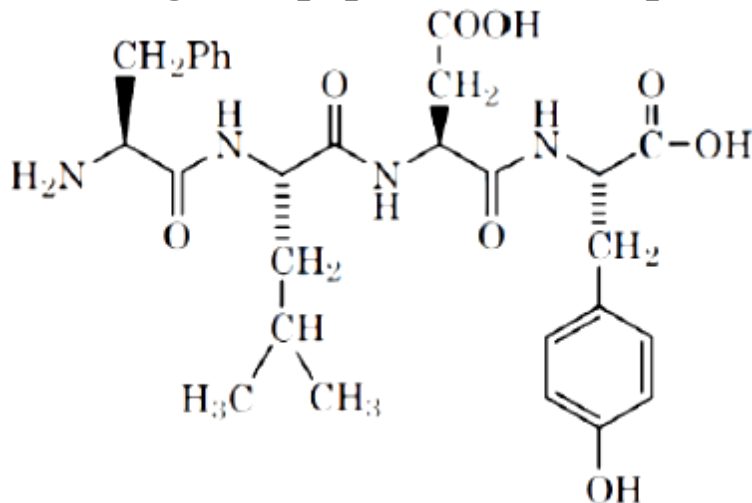
Solution:

Two amino acid are $\text{H}_2\text{N}-\underset{\text{R}_1}{\text{CH}}-\text{COOH}$ (A) $\text{H}_2\text{N}-\underset{\text{R}_2}{\text{CH}}-\text{COOH}$ (B) Tripeptide are formed from three amino acids



Question61

Following tetrapeptide can be represented as



(F, L, D, Y, I, Q, P are one letter codes for amino acids)
[29-Jan-2023 Shift 2]

Options:

A. FIQY

B. FLDY

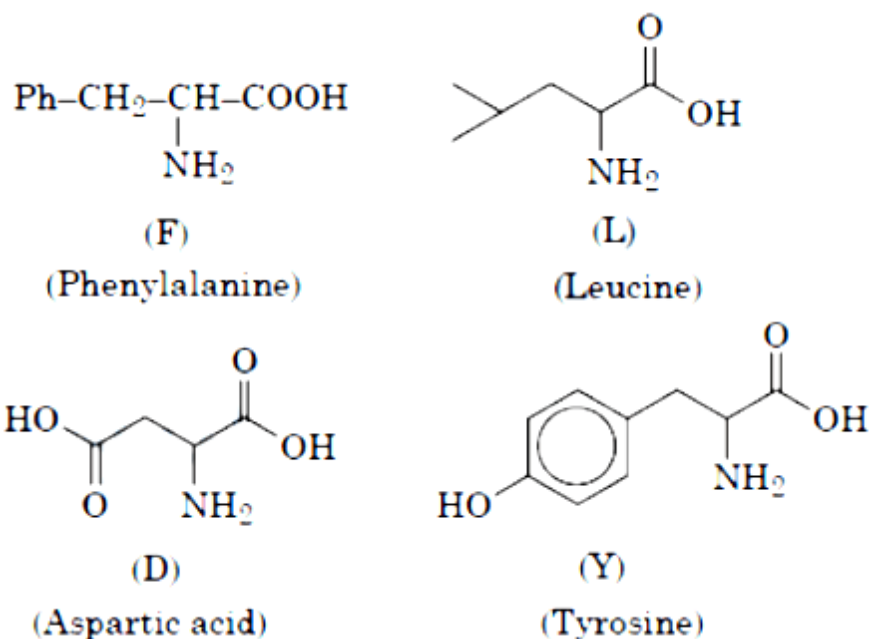
C. YQLF

D. PLDY

Answer: B

Solution:

Hydrolysis of the given tetrapeptide will give the following:



Question62

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Ketoses give Seliwanoff's test faster than Aldoses.

Reason (R) : Ketoses undergo β -elimination followed by formation of furfural.

In the light of the above statements, choose the correct answer from the options given below :

[30-Jan-2023 Shift 1]

Options:

A. (A) is false but (R) is true

B. Both (A) and (R) are true and (R) is the correct explanation of (A)

C. (A) is true but (R) is false

D. Both (A) and (R) are true but (R) is not the correct explanation of (A)

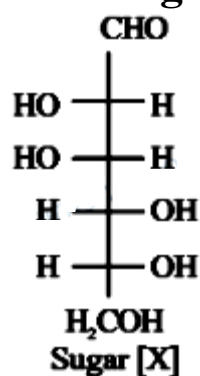
Answer: C

Solution:

Seliwanoff's test is a differentiating test for Ketose and aldose. This test relies on the principle that the keto hexose are more rapidly dehydrated to form 5 -hydroxy methyl furfural when heated in acidic medium which on condensation with resorcinol, Cherry red or brown red coloured complex is formed rapidly indicating a positive test.

Question63

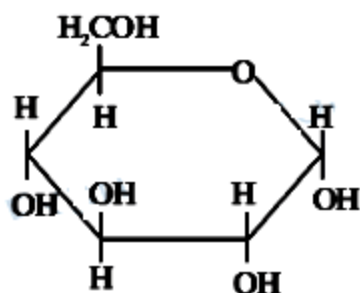
The correct representation in six membered pyranose form for the following sugar [X] is



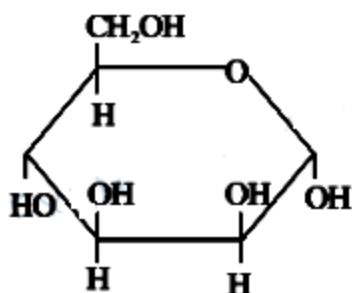
[1-Feb-2023 Shift 1]

Options:

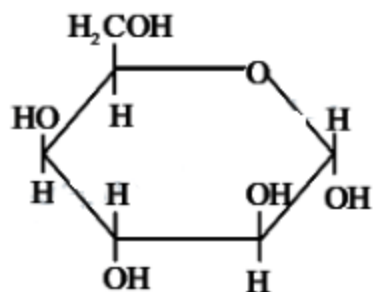
A.



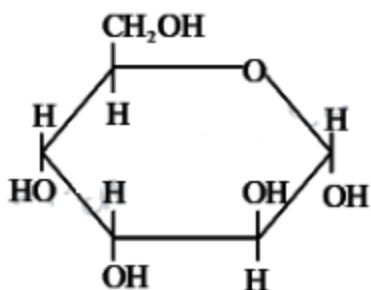
B.



C.



D.



Answer: B

Solution:

By Haworth structure of mannose.

Question64

Match List I and List II

List I Test	List II Functional group /Class of Compound
A. Molisch's Test	I. Peptide
B. Biuret Test	II. Carbohydrate
C. Carbylamine Test	III. Primary amine
D. Schiff s Test	IV. Aldehyde

Choose the correct answer from the options given below:
[1-Feb-2023 Shift 1]

Options:

A. (A) - I, (B) - II, (C) - III, (D) - IV

B. (A) - III, (B) - IV, (C) -I, (D) - II

C. (A) - II, (B) - I, (C) - III, (D) - IV

D. (A) - III, (B) - IV, (C) -II, (D) - I

Answer: C

Solution:

List I Test	List II Functional group /Class of Compound
A. Molisch's Test	II. Carbohydrate
B. Biuret Test	I. Peptide
C. Carbylamine Test	III. Primary amine
D. Schiff s Test	IV. Aldehyde

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : α -halocarboxylic acid on reaction with dil. NH_3 gives good yield of α -amino carboxylic acid whereas the yield of amines is very low when prepared from alkyl halides.

Reason (R) : Amino acids exist in zwitter ion form in aqueous medium.

In the light of the above statements, choose the correct answer from the options given below :

[1-Feb-2023 Shift 2]

Options:

- A. Both (A) and (R) are correct and (R) is the correct explanation of (A).
- B. Both (A) and (R) are correct but (R) is not the correct explanation of (A).
- C. (A) is correct but (R) is not correct.
- D. (A) is not correct but (R) is correct.

Answer: B

Question66

Match List I with List II

List I Vitamin	List II Deficiency disease
A) Vitamin A	I. Beri-Beri
B) Thiamine	II. Cheilosis
C) Ascorbic acid	III. Xerophthalmia
D) Riboflavin	IV. Scurvy

**Choose the correct answer from the options given below:
[6-Apr-2023 shift 1]**

Options:

A. A-IV, B-II, C-III, D-I

B. A-III, B-II, C-IV, D-I

C. A-IV, B-I, C-III, D-II

D. A-III, B-I, C-IV, D-II

Answer: D

Question 67

Match List I with List II

List I	List II
Natural Amino acid	One Letter Code
(A) Arginine	(I) D
(B) Aspartic acid	(II) N
(C) Asparagine	(III) A
(D) Alanine	(IV) R

**Choose the correct answer from the options given below :-
[6-Apr-2023 shift 2]**

Options:

A. (A)-IV, (B)-I, (C)-III, (D)-II

B. (A)-I, (B)-III, (C)-IV, (D)-II

C. (A)-III, (B)-I, (C)-II, (D)-IV

D. (A)-IV, (B)-I, (C)-II, (D)-III

Answer: D

Question68

Sulphur (S) containing amino acids from the following are:

(a) isoleucine

(b) cysteine

(c) lysine

(d) methionine

(e) glutamic acid

[8-Apr-2023 shift 1]

Options:

A. b, c, e

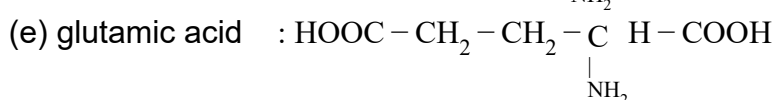
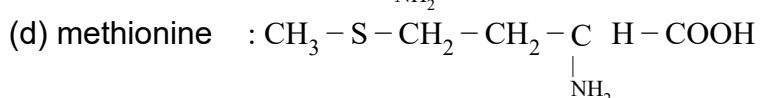
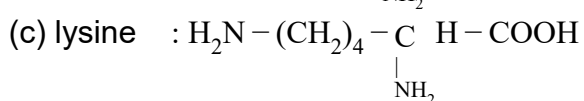
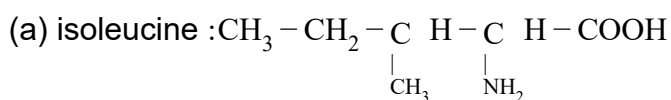
B. a, d

C. a, b, c

D. b, d

Answer: D

Solution:



Question69

Match list I with list II

LIST - I Natural amino acid	LIST - II One letter code
A. Glutamic acid	I. Q
B. Glutamine	II. W
C. Tyrosine	III. E
D. Tryptophan	IV. Y

Choose the correct answer from the options given below:
[8-Apr-2023 shift 2]

Options:

A. A-III, B-I, C-IV, D-II

B. A-IV, B-III, C-I, D-II

C. A-II, B-I, C-IV, D-III

D. A-III, B-IV, C-I, D-II

Answer: A

Solution:

A-III, B-I, C-IV, D-II

Fact

Question70

The one that does not stabilize 2° and 3° structures of proteins is
[10-Apr-2023 shift 1]

Options:

- A. H-bonding
- B. -S-S-linkage
- C. van der waals forces
- D. -O - O--linkage

Answer: D

Solution:

Fact

The main forces which stabilize the secondary and tertiary structure of proteins are

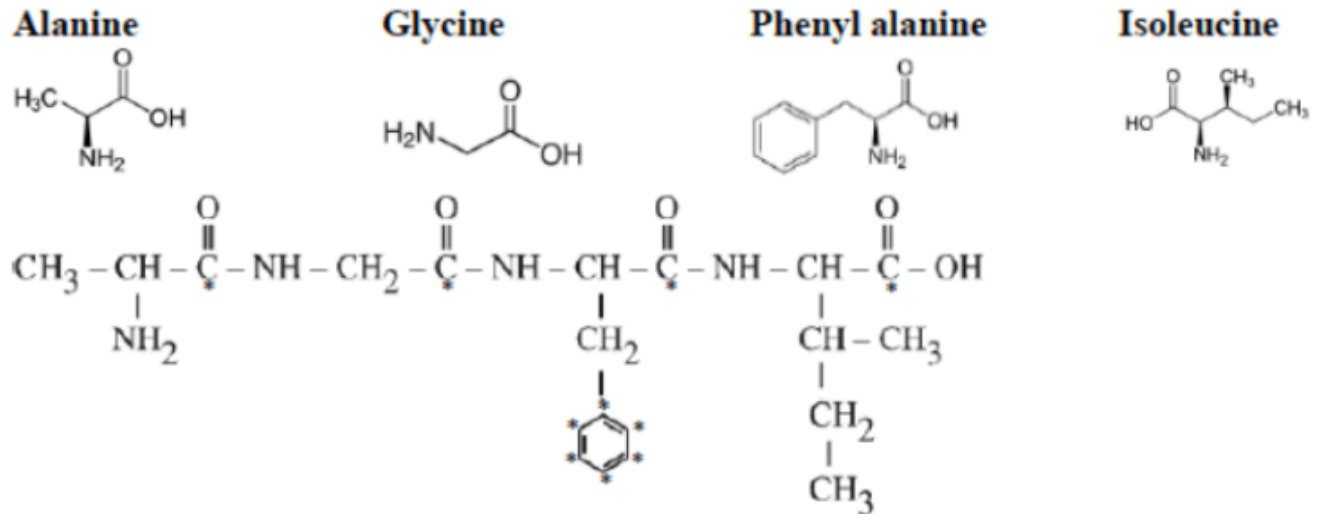
- Hydrogen bonds
 - S - S Linkages
 - vanderwaals force
 - electrostatic force of attraction
-

Question71

In an oligopeptide named Alanylglycylphenyl alanyl isoleucine, the number of sp^2 hybridised carbons is
[12-Apr-2023 shift 1]

Answer: 10

Solution:



Question72

The naturally occurring amino acid that contains only one basic functional group in its chemical structure is :
[13-Apr-2023 shift 2]

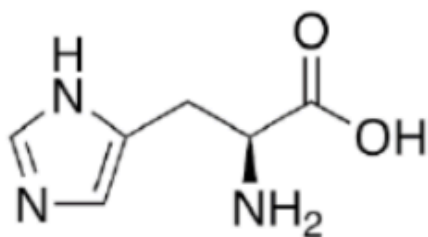
Options:

- A. histidine
- B. lysine
- C. asparagine
- D. arginine

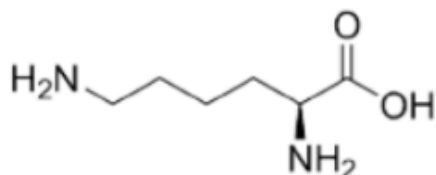
Answer: C

Solution:

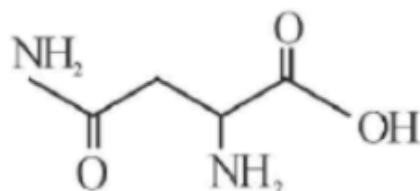
1. histidine



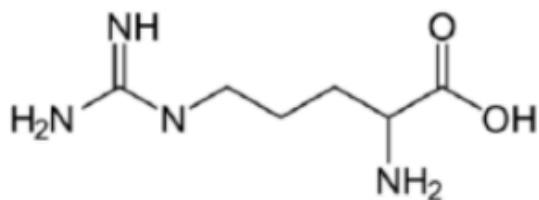
2. Lysine



3.



4. Arginine



Question73

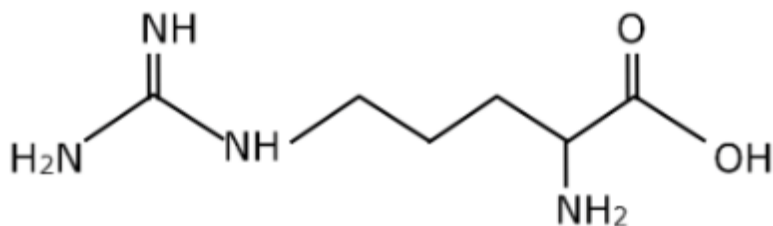
Which is not true for arginine?
[15-Apr-2023 shift 1]

Options:

- A. It has high solubility in benzene
- B. It is associated with more than one pK_a values.
- C. It is a crystalline solid.
- D. It has a fairly high melting point.

Answer: A

Solution:



Arginine exist is zwitterion, so solid nature and soluble in polar solvent.

(i) Polar, so not high soluble in benzene

(ii) It has 3 pKa values

(iii) True

(iv) High molecular mass to high M.P.

Question74

**Sugar moiety in DNA and RNA molecules respectively are
[29-Jun-2022-Shift-1]**

Options:

A. β -D-2-deoxyribose, β -D-deoxyribose.

B. β -D-2-deoxyribose, β -D-ribose

C. β -D-ribose, β -D-2-deoxyribose.

D. β -D-deoxyribose, β -D-2-deoxyribose.

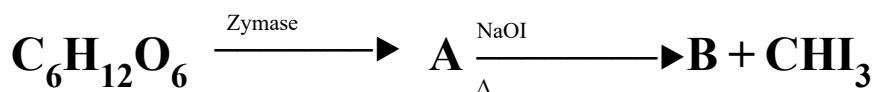
Answer: B

Solution:

DNA contains $\Rightarrow \beta$ -D-2-deoxyribose

RNA contains $\Rightarrow \beta$ -D-ribose

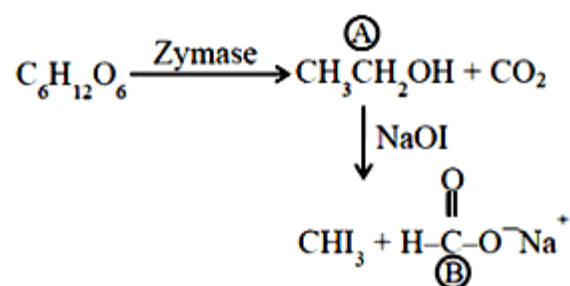
Question75



The number of carbon atoms present in the product B is _____
[29-Jun-2022-Shift-1]

Answer: 1

Solution:



no. of carbon atoms present in B is 1

Question 76

The structure of protein that is unaffected by heating is
[29-Jun-2022-Shift-2]

Options:

- A. secondary structure
- B. tertiary structure
- C. primary structure
- D. quaternary structure

Answer: C

Solution:

The primary structure of protein is unaffected by physical 'or' chemical changes.

Question77

Which one of the following is a water soluble vitamin, that is not excreted easily?

[26-Jun-2022-Shift-2]

Options:

A. Vitamin B₂

B. Vitamin B₁

C. Vitamin B₆

D. Vitamin B₁₂

Answer: B

Solution:

B-complex (vitamins B group) is a water-soluble vitamin. Also, vitamin C is a water-soluble vitamin. Vitamin A, D, E and K are fat soluble vitamins.

Question78

Given below are two statements.

Statement I : Maltose has two α -D-glucose units linked at C₁ and C₄ and is a reducing sugar.

Statement II : Maltose has two monosaccharides : α -D-glucose and β -D-glucose linked at C₁ and C₆ and it is a non-reducing sugar.

In the light of the above statements, choose the correct answer from the options given below :

[27-Jun-2022-Shift-2]

Options:

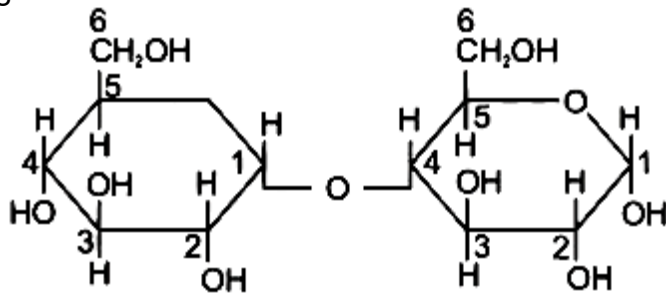
A. Both Statement I and Statement II are true.

- B. Both Statement I and Statement II are false.
- C. Statement I is true but Statement II is false.
- D. Statement I is false but Statement II is true.

Answer: A

Solution:

Maltose is composed of two α -D-glucose units in which C_1 of one glucose unit and C_4 of second glucose unit are linked.



Question79

**Stability of α -Helix structure of proteins depends upon
[28-Jun-2022-Shift-1]**

Options:

- A. dipolar interaction
- B. H-bonding interaction
- C. van der Waals forces
- D. π -stacking interaction

Answer: C

Solution:

Mostly H-bonding is responsible for the stability of α -helix form.

Question80

When sugar ' X ' is boiled with dilute H_2SO_4 in alcoholic solution, two isomers ' A ' and ' B ' are formed. ' A ' on oxidation with HNO_3 yields saccharic acid where as ' B ' is laevorotatory. The compound ' X ' is :

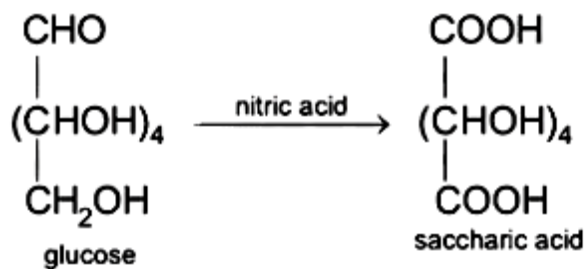
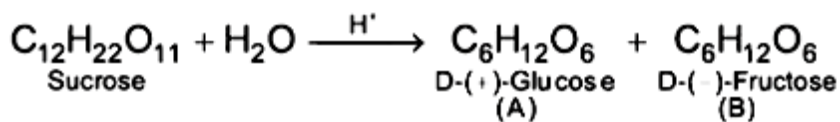
[28-Jun-2022-Shift-2]

Options:

- A. Maltose
B. Sucrose
C. Lactose
D. Starch

Answer: A

Solution:



Question81

2.5g of protein containing only glycine ($\text{C}_2\text{H}_5\text{NO}_2$) is dissolved in water to make 500 mL of solution. The osmotic pressure of this solution at 300K is found to be 5.03×10^{-3} bar. The total number of glycine units present in the protein is_____

(Given : $R = 0.083 \text{ L bar K}^{-1} \text{ mol}^{-1}$)
[28-Jun-2022-Shift-2]

Answer: 330

Solution:

Since,

$$\pi = icRT$$

$$5.03 \times 10^{-3} = \frac{2.5}{M} \times \frac{1000}{500} \times 0.083 \times 300$$

$$\text{Molar mass of protein} = 24751.5 \text{ g/mol}$$

$$\begin{aligned} \text{Number of glycine units in protein} &= \frac{24751.5}{75} \\ &= 330 \end{aligned}$$

Question82

A polysaccharide 'X' on boiling with dil H_2SO_4 at 393K under 2-3 atm pressure yields 'Y'. 'Y' on treatment with bromine water gives gluconic acid. 'X' contains β -glycosidic linkages only. Compound 'X' is :

[24-Jun-2022-Shift-1]

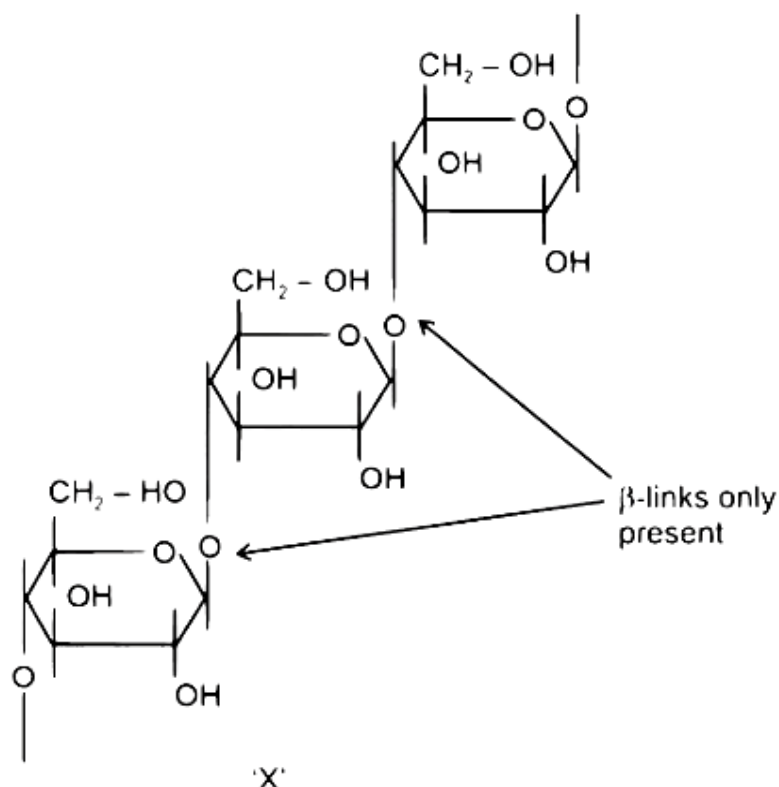
Options:

- A. starch
- B. cellulose
- C. amylose
- D. amylopectin

Answer: A

Solution:

Cellulose contains β -glycosidic linkages only. Structure of cellulose



On boiling with dil. H_2SO_4 at 393K under 2-3 atm, 'X' forms glucose, which gives gluconic acid on treatment with bromine water.

Question83

In *alanylglycylleucylalanylvaline*, the number of peptide linkages is

[24-Jun-2022-Shift-2]

Answer: 4

Solution:

The given pentapeptide is
ALA - GLY - LEU - ALA - VAL
It has 4 peptide linkages.

Question84

**How many of the given compounds will give a positive Biuret test ?
Glycine, Glycylalanine, Tripeptide, Biuret
[25-Jun-2022-Shift-2]**

Answer: 2

Solution:

Biuret test is given by all proteins and peptides having atleast two peptide linkages.
Hence positive test must be given by tripeptide and Biuret.

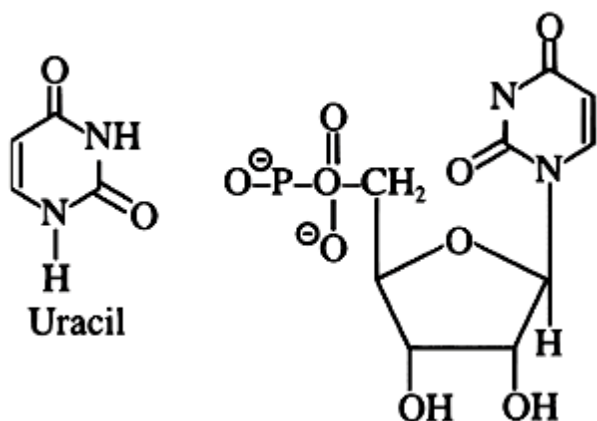
Question85

**The number of oxygens present in a nucleotide formed from a base, that is present only in RNA is ____
[26-Jun-2022-Shift-1]**

Answer: 9

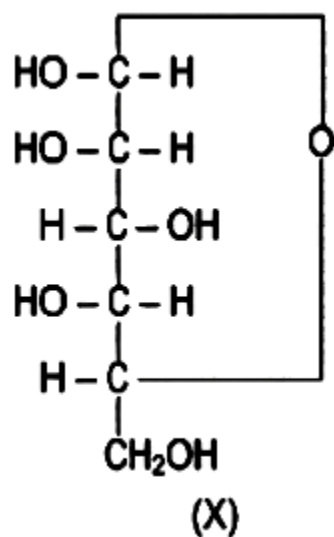
Solution:

Uracil is the base which only present in RNA.



Question86

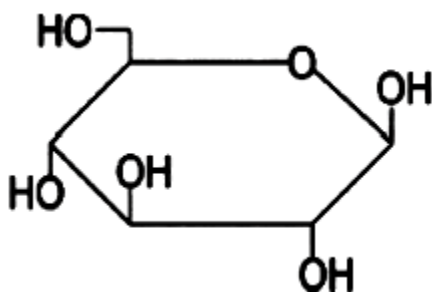
For the below given cyclic hemiacetal (X), the correct pyranose structure is :



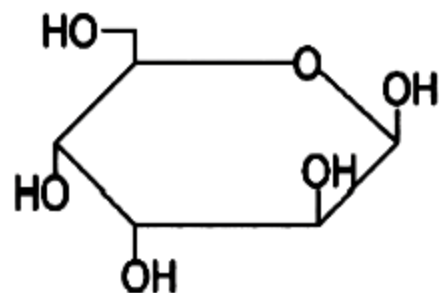
[28-Jul-2022-Shift-1]

Options:

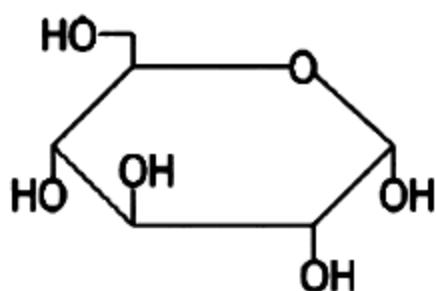
A.



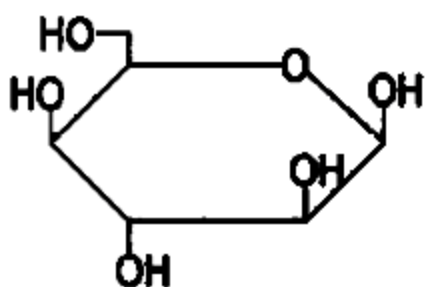
B.



C.

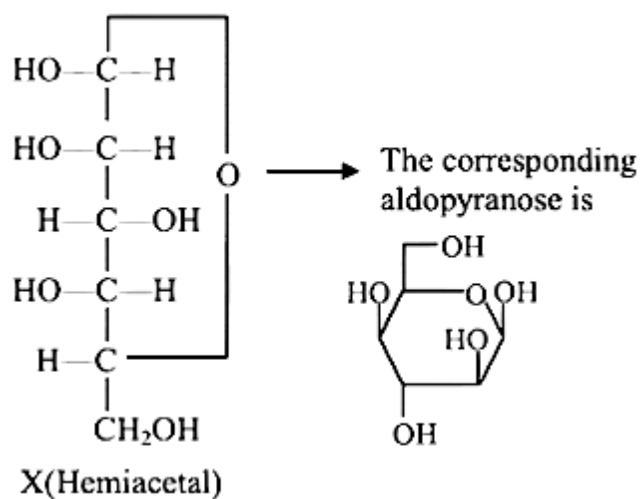


D.



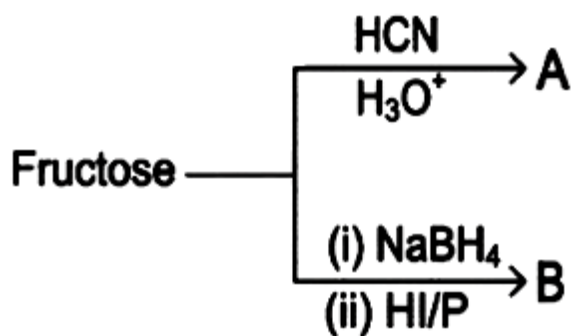
Answer: D

Solution:



Question87

The formulas of A and B for the following reaction sequence are



[28-Jul-2022-Shift-2]

Options:

A. $A = C_7H_{14}O_8$, $B = C_6H_{14}$ $A = C_7H_{14}O_8$, $B = C_6H_{14}$

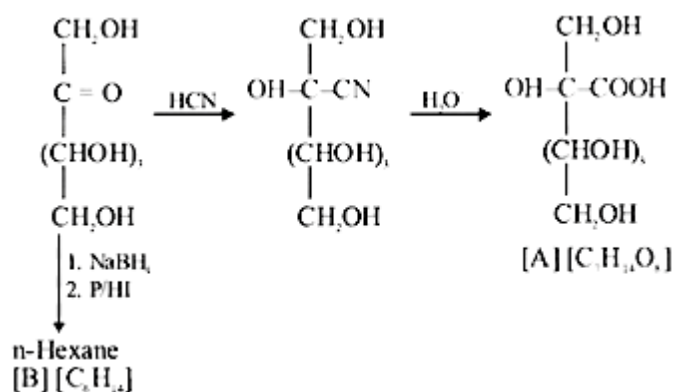
B. $A = C_7H_{13}O_7$, $B = C_7H_{14}$ $A = C_7H_{13}O_7$, $B = C_7H_{14}$

C. $A = C_7H_{12}O_8$, $B = C_6H_{14}$ $A = C_7H_{12}O_8$, $B = C_6H_{14}$

D. $A = C_7H_{14}O_8$, $B = C_6H_{14}O_6$ $A = C_7H_{14}O_8$, $B = C_6H_{14}O_6$

Answer: A

Solution:



Question88

Glycosidic linkage between C1 of α -glucose and C2 of β -fructose is found in

[25-Jul-2022-Shift-2]

Options:

- A. maltose
- B. sucrose
- C. lactose
- D. amylose

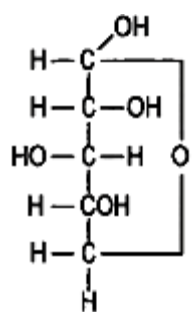
Answer: B

Question89

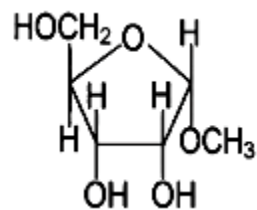
**Which one of the following is a reducing sugar?
[26-Jul-2022-Shift-1]**

Options:

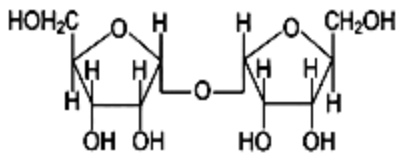
A.



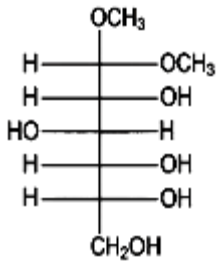
B.



C.



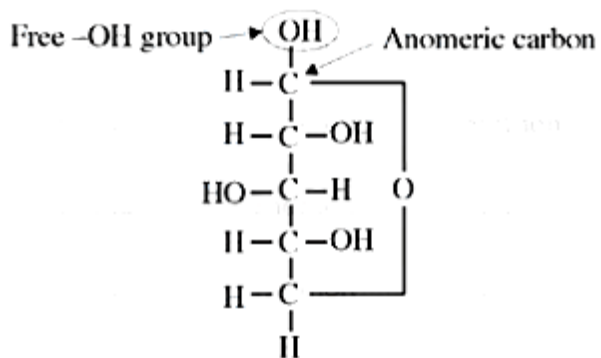
D.



Answer: A

Solution:

If any sugar is having free -OH group at anomeric carbon then it will be a reducing sugar



Question90

**Animal starch is the other name of
[26-Jul-2022-Shift-2]**

Options:

- A. amylose.
- B. maltose.
- C. glycogen.
- D. amylopectin.

Answer: C

Solution:

Animal starch is the other name of glycogen because its structure is similar to amylopectin.

Question91

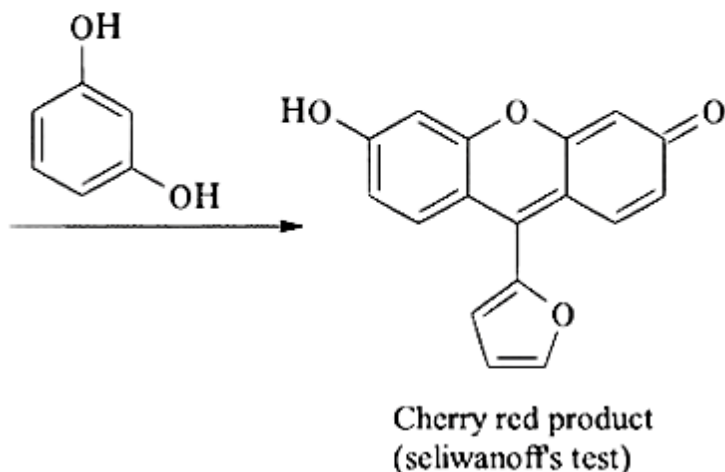
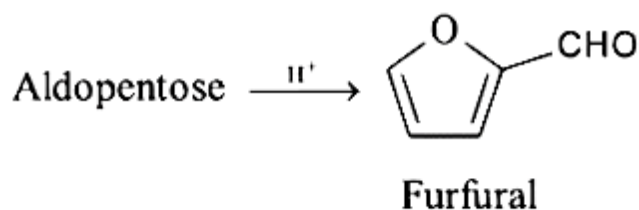
A sugar ' X ' dehydrates very slowly under acidic condition to give furfural which on further reaction with resorcinol gives the coloured product after sometime. Sugar ' X ' is
[27-Jul-2022-Shift-1]

Options:

- A. Aldopentose
- B. Aldotetrose
- C. Oxalic acid
- D. Ketotetrose

Answer: A

Solution:



Question92

Match List - I with Match List - II.

List I	List II
(A)Glucose + HI	(I)Gluconic acid
(B)Glucose + Br ₂ water	(II)Glucose pentacetate
(C)Glucose + acetic anhydride	(III)Saccharic acid
(D)Glucose +HNO ₃	(IV)Hexane

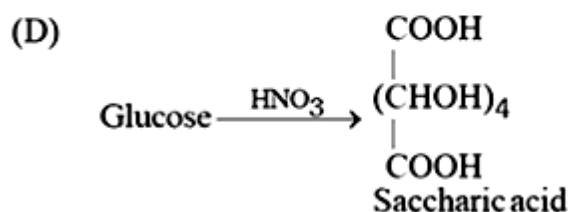
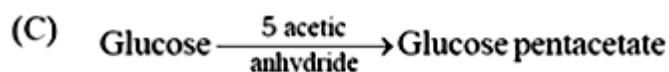
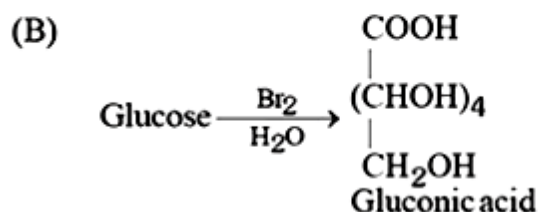
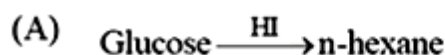
**Choose the correct answer from the options given below:
[27-Jul-2022-Shift-2]**

Options:

- A. (A)-(IV), (B)-(I), (C)-(II), (D)-(III)
- B. (A)-(IV), (B)-(III), (C)-(II), (D)-(I)
- C. (A)-(III), (B)-(I), (C)-(IV), (D)-(II)
- D. (A)-(I), (B)-(III), (C)-(IV), (D)-(II)

Answer: A

Solution:



Question93

In a linear tetrapeptide (Constituted with different amino acids), (number of amino acids) - (number of peptide bonds) is _____.
[29-Jul-2022-Shift-1]

Answer: 1

Solution:

In tetrapeptide

No. of amino acids = 4

No. of peptide bonds = 3

Hence, (1)

Question94

Given below are two statements. One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: Amylose is insoluble in water.

Reason R: Amylose is a long linear molecule with more than 200

glucose units.

In the light of the above statements, choose the correct answer from the options given below.

[29-Jul-2022-Shift-2]

Options:

- A. A Both A and R are correct and R is the correct explanation of A.
- B. Both A and R are correct but R is NOT the correct explanation of A.
- C. A is correct but R is not correct
- D. A is not correct but R is correct.

Answer: B

Solution:

Amylose is a linear polymer formed by combination of α -D glucose through 1, 4- glycosidic linkage.

It is water soluble.

So, assertion is incorrect.

Question95

Match List-I with List-II.

List-I	List-II
A. Sucrose	(i) β -D-galactose and β -D-glucose
B. Lactose	(ii) α -D-glucose and β -D-fructose
C. Maltose	(iii) α -D-glucose and α -D-glucose

Choose the correct answer from the options given below.

[26 Feb 2021 Shift 2]

Options:

- A. A $\rightarrow$ (i), B $\rightarrow$ (iii), C $\rightarrow$ (ii)

B. $A \rightarrow (ii)$, $B \rightarrow (i)$, $C \rightarrow (iii)$

C. $A \rightarrow (ii)$, $B \rightarrow (i)$, $C \rightarrow (iii)$

D. $A \rightarrow (iii)$, $B \rightarrow (ii)$, $C \rightarrow (i)$

Answer: C

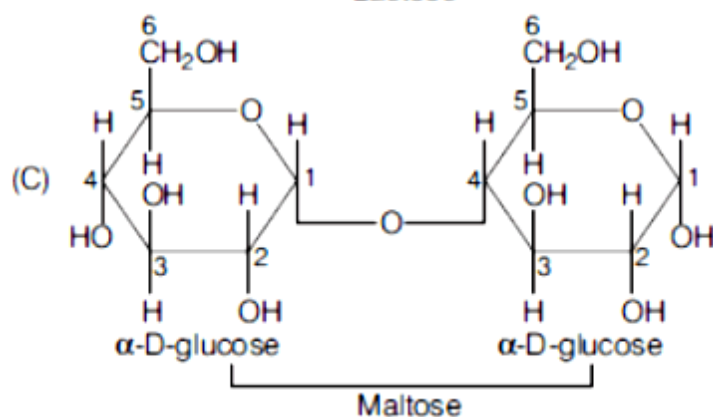
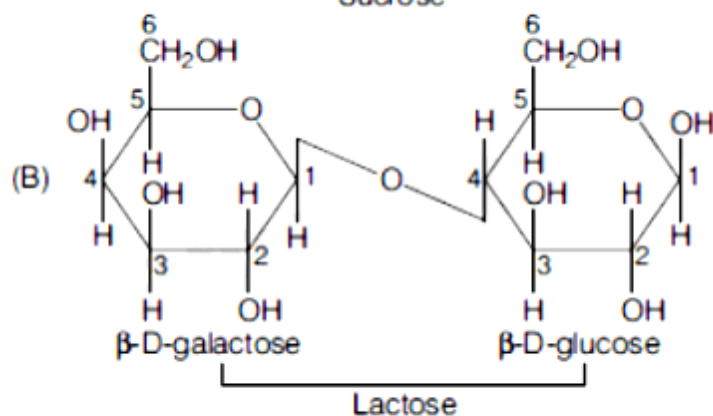
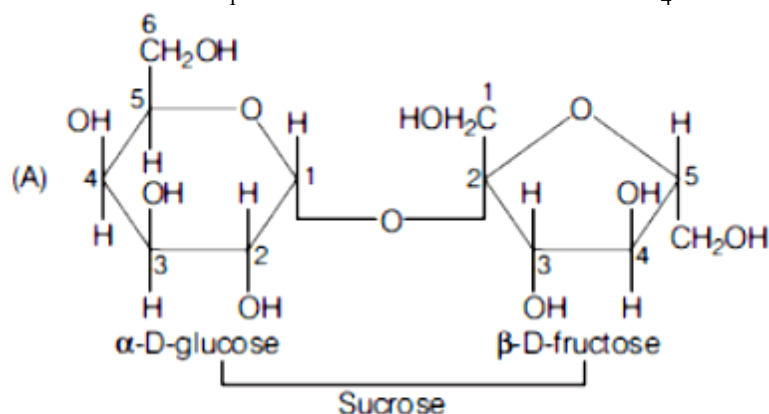
Solution:

Sucrose (cane sugar), lactose (milk sugar) and maltose are disaccharides in which two monosaccharides are held together by a glycosidic linkage.

(A) In sucrose, C_1 of α -D-glucose and C_2 of β -D-fructose are linked together (ii).

(B) In lactose, C_1 of β -D-galactose and C_4 of β -D-glucose are linked together (i).

(C) In maltose, C_1 of α -D-glucose is linked to C_4 of another α -D-glucose unit (iii).



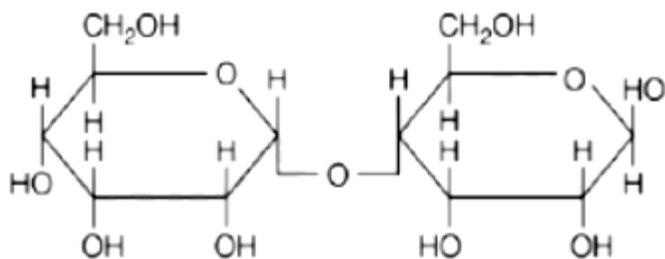
So, option (c) is the correct answer.

Question96

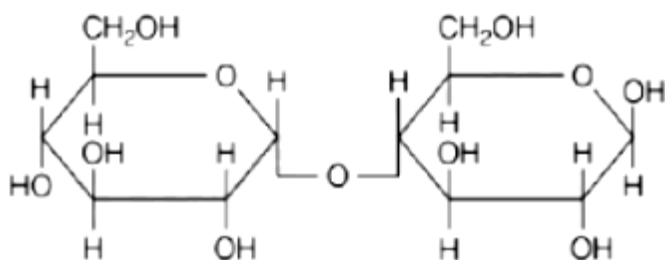
Which of the following is correct structure of α -anomer of maltose?
[25 Feb 2021 Shift 2]

Options:

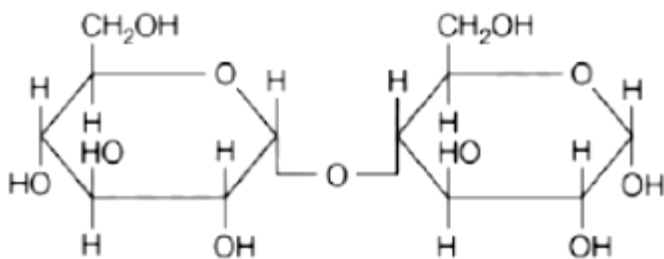
A.



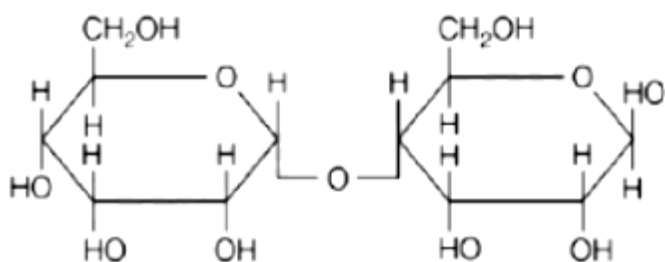
B.



C.



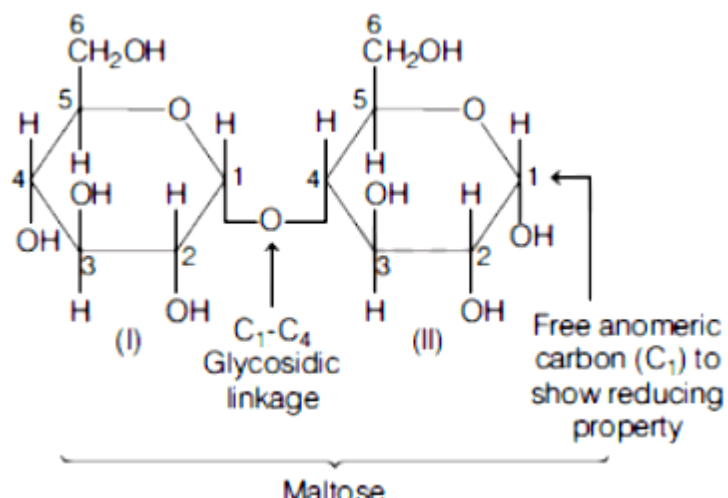
D.



Answer: C

Solution:

Maltose is a disaccharide which is made of two α -D-glucose units in which C_1 (anomeric carbon) of one glucose (I) is linked to C_4 of another glucose unit (II).



Question97

Which of the glycosidic linkage between galactose and glucose is present in lactose?

[25 Feb 2021 Shift 1]

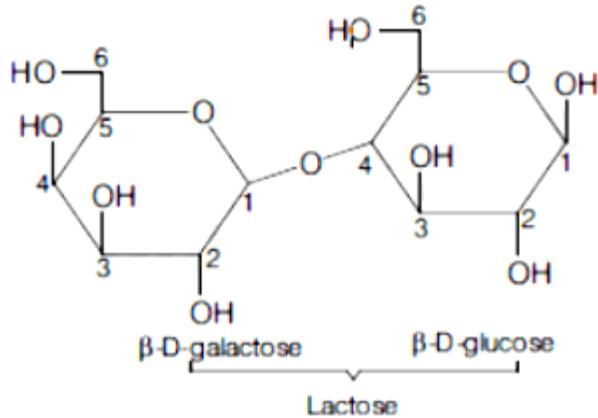
Options:

- A. C-1 of galactose and C-4 of glucose
- B. C-1 of galactose and C-6 of glucose
- C. C – 1 of glucose and C – 4 of galactose
- D. C-1 of glucose and C-6 of galactose

Answer: A

Solution:

Glycosidic linkage is a type of covalent bond that joins carbohydrate molecules to another group.



In lactose, glycosidic linkage is formed between C-1 of galactose and C – 4, glycosidic of glucose.

Question98

Which of the following vitamin is helpful in delaying the blood clotting ?

[26 Feb 2021 Shift 1]

Options:

- A. Vitamin C
- B. Vitamin B
- C. Vitamin E
- D. Vitamin K

Answer: D

Solution:

Deficiency of vitamin K increases blood clotting time. So, vitamin K is helpful in blood clotting. It is a fat-soluble vitamin.

Vitamin C is used to prevent and treat scurvy. It is a water soluble vitamin.

Vitamin B are also water soluble and play significant roles in cell metabolism and synthesis of RBC.

Vitamin E is a fat-soluble vitamin. Deficiency of it may cause increased fragility of RBCs, nerve problems and muscular weakness.

Vitamin E is a fat-soluble anti-oxidant which protects cell membranes.

Question99

Match List-I and List-II.

List-I	List-II
A. Valium	1. Antifertility drug
B. Morphine	2. Pernicious anaemia
C. Norethindrone	3. Analgesic
D. Vitamin B_{12}	4. Tranquiliser

Choose the correct answer from the option given below.

[24 Feb 2021 Shift 2]

Options:

A. (A-4), (B-3), (C-2), (D-1)

B. (A-4), (B-3), (C-1), (D-2)

C. (A-2), (B-4), (C-3), (D-1)

D. (A-1), (B-3), (C-4), (D-2)

Answer: B

Solution:

(A) Valium - (4) Tranquilizer

A tranquilizer drug became a standard drug for the treatment of anxiety and one of most commonly prescribed drugs of all time.

(B) Morphine - (3) Analgesic

Morphine is effective for both acute and chronic pain and often used before and after surgery.

(C) Norethindrone - (1) Antifertility drug

It is a form of progesterone, a female hormone important for regulating ovulation and menstruation.

(D) Vitamin B_{12} - (2) Pernicious anaemia

It is a nutrient that helps to keep our body blood cells and nerve cells healthy and help in making DNA.

Question 100

Out of the following, which type of interaction is responsible for the stabilisation of α -helix structure of proteins?

[24 Feb 2021 Shift 1]

Options:

- A. Ionic bonding
- B. Hydrogen bonding
- C. Covalent bonding
- D. vander Waals forces

Answer: B

Solution:

Hydrogen bonding is responsible for stabilisation of alpha-helix structure of proteins.

Question 101

A non-reducing sugar "A" hydrolyses to give two reducing monosaccharides. Sugar A is
[18 Mar 2021 Shift 1]

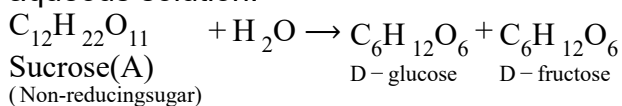
Options:

- A. fructose
B. galactose
C. glucose
D. sucrose

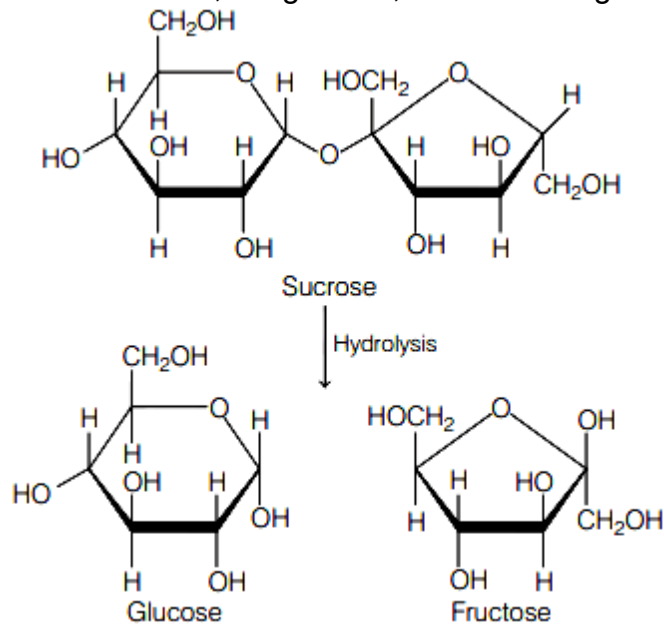
Answer: D

Solution:

A non-reducing sugar is one kind of carbohydrate that is not oxidised by a weak oxidising agent (an oxidising agent that oxidises aldehydes but not alcohols, such as the Tollen's reagent) in a basic aqueous solution.

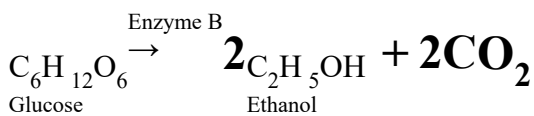
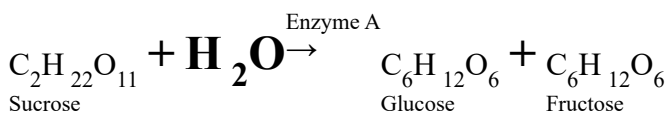


While all others, i.e. glucose, fructose and galactose are reducing sugar



Thus, sugar (A) is sucrose.

Question102



In the above reactions, the enzyme A and enzyme B respectively are [17 Mar 2021 Shift 2]

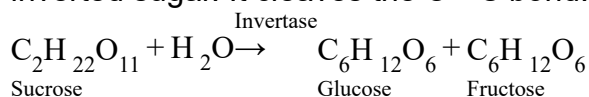
Options:

- A. amylase and invertase
- B. invertase and amylase
- C. invertase and zymase
- D. zymase and invertase

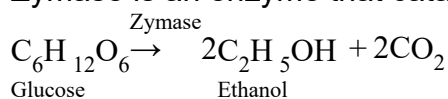
Answer: C

Solution:

Invertase and zymase are enzyme A and enzyme B respectively. Invertase is the enzyme that catalyses the hydrolysis of sucrose with a resulting mixture of fructose and glucose, which is called inverted sugar. It cleaves the O – C bond.



Zymase is an enzyme that catalyses the fermentation of sugar into ethanol and carbon dioxide.



Question103

**Fructose is an example of
[17 Mar 2021 Shift 2]**

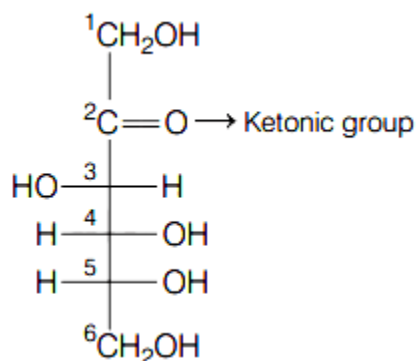
Options:

- A. pyranose
- B. ketohexose
- C. aldohexose
- D. heptose

Answer: B

Solution:

Fructose is an example of ketohexose is a class of sugars that contains six-carbon atoms and a ketonic group. Fructose has a ketonic group with six carbon atoms. It contains five hydroxyl group and one ketonic group. The structure of fructose is

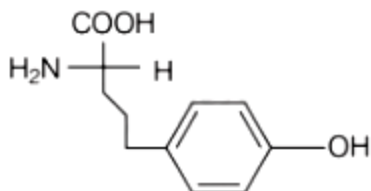


Question104

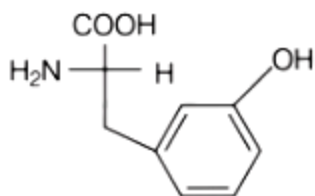
Which of the following is correct structure of tyrosine?
[17 Mar 2021 Shift 1]

Options:

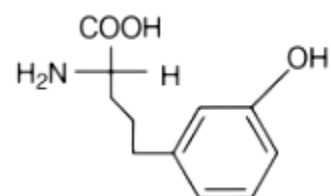
A.



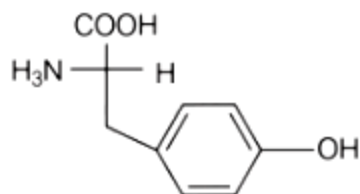
B.



C.



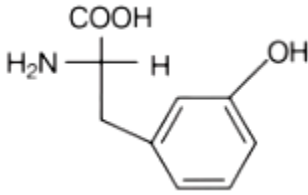
D.



Answer: B

Solution:

The structure of tyrosine amino acid is



It is an example of non-essential amino acid which contains aromatic (phenolic) ring.

Question105

**The secondary structure of protein is stabilised by
[16 Mar 2021 Shift 2]**

Options:

- A. peptide bond
- B. glycosidic bond
- C. hydrogen bonding
- D. van der Waals' forces

Answer: C

Solution:

In secondary structure of protein, hydrogen bond is formed

between $\begin{array}{c} \text{O} \\ \parallel \\ -\text{C}- \end{array}$ and $-\text{NH}-$ groups of the peptide bond.

Peptide linkage is an amide formed between $-\text{COOH}$ group and $-\text{NH}_2$ group while connecting two amino acids.

The two monosaccharides are joined together by an oxide linkage formed by the loss of a water molecule is called glycosidic linkage.

Question106

**Deficiency of vitamin K causes
[18 Mar 2021 Shift 2]**

Options:

- A. increase in blood clotting time

B. increase in fragility of RBC's

C. cheilosis

D. decrease in blood clotting time

Answer: A

Solution:

Vitamin K deficiency causes increase in blood clotting time because their blood does not have enough vitamin K to form a clot. The bleeding can occur anywhere i.e. inside or outside the body.
Note : Vitamin K related to blood factor.

Question107

Which among the following pair of vitamins is stored in our body relatively for longer duration?

[16 Mar 2021 Shift 1]

Options:

A. Thiamine and vitamin A

B. Vitamin A and vitamin D

C. Thiamine and ascorbic acid

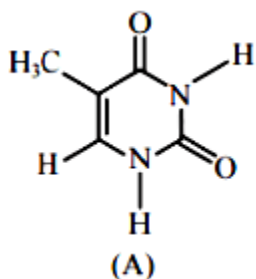
D. Ascorbic acid and vitamin D

Answer: B

Solution:

Vitamin A and vitamin D are stored in our body relatively for longer duration. These are water insoluble and soluble in fats and oils. Hence, these are called fat soluble vitamins.
Thiamine (vitamin B) and ascorbic acid (vitamin C) are water soluble vitamins and must be supplied regularly in diet. These vitamins are not stored in our body.

Question108



The compound 'A' is a complementary base of _____ in DNA stands.

[27 Jul 2021 Shift 1]

Options:

- A. Uracil
- B. Guanine
- C. Adenine
- D. Cytosine

Answer: C

Solution:

Given structure is Thymine and Thymine being paired with adenine

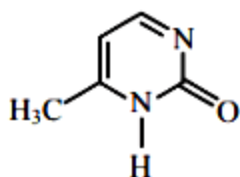
Question109

Which one of the following is correct structure for cytosine ?

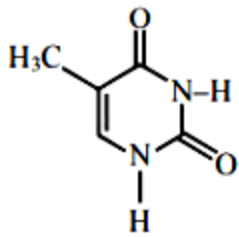
[25 Jul 2021 Shift 2]

Options:

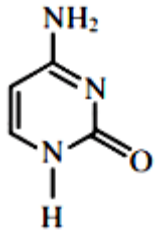
A.



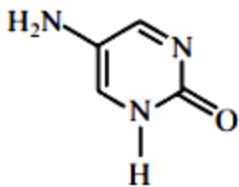
B.



C.



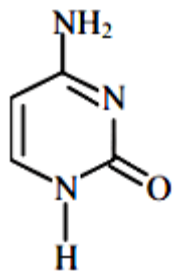
D.



Answer: C

Solution:

The correct structure of cytosine



Question110

**Thiamine and pyridoxine are also known respectively as :
[22 Jul 2021 Shift 2]**

Options:

A. Vitamin B₂ and Vitamin E

B. Vitamin E and Vitamin B₂

C. Vitamin B₆ and Vitamin B₂

D. Vitamin B₁ and Vitamin B₆

Answer: D

Solution:

Vitamine- B₁ is also known as Thiamine while vitamin B-6 is known as Pyridoxine

Question111

Compound A gives D-Galactose and D-Glucose on hydrolysis. The compound A is :

[27 Jul 2021 Shift 2]

Options:

A. Amylose

B. Sucrose

C. Maltose

D. Lactose

Answer: D

Solution:

Lactose : It is a disaccharide of β - D-Galactose and β - D-Glucose with C₁ of galactose and C₄ of glucose link.

Lactose : β -D-Galactose + β - D-Glucose

Question112

Identify the incorrect statement from the following [20 Jul 2021 Shift 1]

Options:

- A. Amylose is a branched chain polymer of glucose
- B. Starch is a polymer of α -D glucose
- C. α -Glycosidic linkage makes cellulose polymer
- D. Glycogen is called as animal starch

Answer: A

Solution:

Amylose is a linear chain polymer of α -D-glucose while amylopectine is branched chain polymer of α -D glucose

Question113

**Which one among the following chemical tests is used to distinguish monosaccharide from disaccharide ?
[27 Jul 2021 Shift 1]**

Options:

- A. Seliwanoff's test
- B. Iodine test
- C. Barfoed test
- D. Tollen's test

Answer: C

Solution:

Barford test is used for distinguish monosaccharide from disaccharide

Question114

Which one of the following compounds contain $\beta - C_1 - C_4$ glycosidic linkage ?

[31 Aug 2021 Shift 1]

Options:

A. Lactose

B. Sucrose

C. Maltose

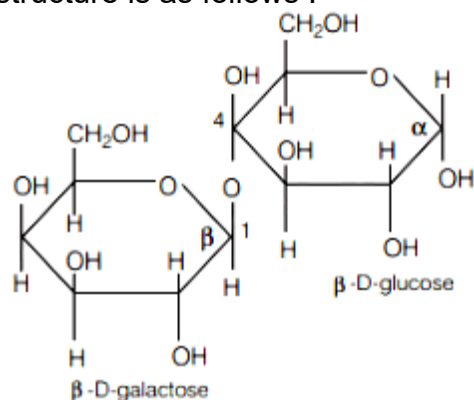
D. Amylose

Answer: A

Solution:

Lactose contains $\beta - C_1 - C_4$ glycosidic linkage.

The linkage is between C - 1 of galactose and C - 4 of glucose. It's structure is as follows :



Hence, correct option is (a).

Question115

Hydrolysis of sucrose gives
[27 Aug 2021 Shift 2]

Options:

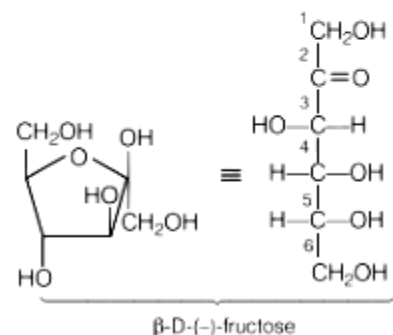
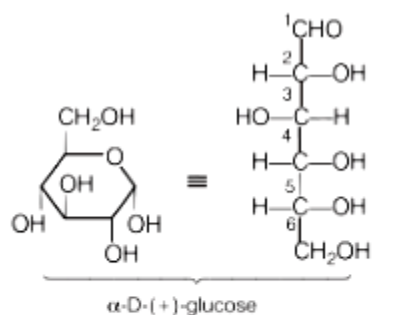
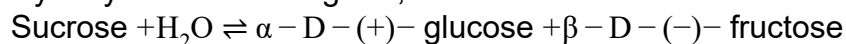
A. $\alpha - D - (-)$ -glucose and $\beta - D - (-)$ -fructose

- B. α - D - (+)-glucose and β - D - (+)-fructose
- C. α - D - (-)-glucose and β D - (+)-fructose
- D. α - D - (+)- glucose and β - D - (-)-fructose

Answer: D

Solution:

Hydrolysis of sucrose gives,



Therefore, option (d) is correct answer.

'D' represents the orientation of hydroxyl group at the chiral carbon that is farthest from the highest oxidised carbon (aldehyde group in case of glucose and keto group in fructose) when the - OH group is on the right side, 'L' is used when - OH group is on the left side. (+) and (-) represents the direction of rotation of plane polarised light (+) is clockwise rotation and (-) is anticlockwise rotation. α - is used when - OH group at anomeric carbon is below the ring in Haworth projection, β is used when - OH group at anomeric carbon is above the rings in Haworth projection.

Question 116

Given below are two statements : One is labelled as Assertion (A) and other is labelled as Reason (R).

Assertion (A) Sucrose is a disaccharide and a non-reducing sugar.

Reason (R) Sucrose involves glycosidic linkage between C_1 of β -glucose and C_2 of α - fructose. Choose the most appropriate answer

from the options given below.
[26 Aug 2021 Shift 2]

Options:

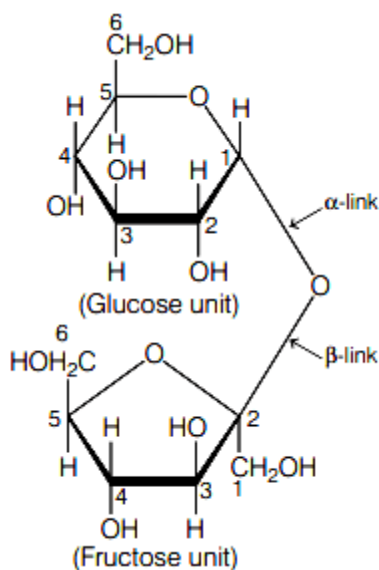
- A. Both (A) and (R) are true but (R) is not the correct explanation of (A).
- B. (A) is false but (R) is true.
- C. (A) is true but (R) is false.
- D. Both (A) and (R) are true and (R) is the correct explanation of (A).

Answer: C

Solution:

Assertion is true but Reason is false. Sucrose is formed by linkage of C_1 of α - D glucose and C_2 of β - D fructose. The glycosidic bond in sucrose is formed between reducing ends of both glucose and fructose.

Sucrose is a disaccharide and a non-reducing sugar as it contains no free —CHO group (Anomeric carbon of both monosaccharides are involved in formation of glycosidic bond). The structure is as follows



Question117

Which of the following is not an example of fibrous protein ?
[31 Aug 2021 Shift 2]

Options:

- A. Keratin
- B. Albumin
- C. Collagen
- D. Myosin

Answer: B

Solution:

Insulin and albumin are globular protein. Fibrous proteins are generally composed of long and narrow strands and have structural role while globular proteins generally have more compact and rounded shape and involved in metabolic functions.

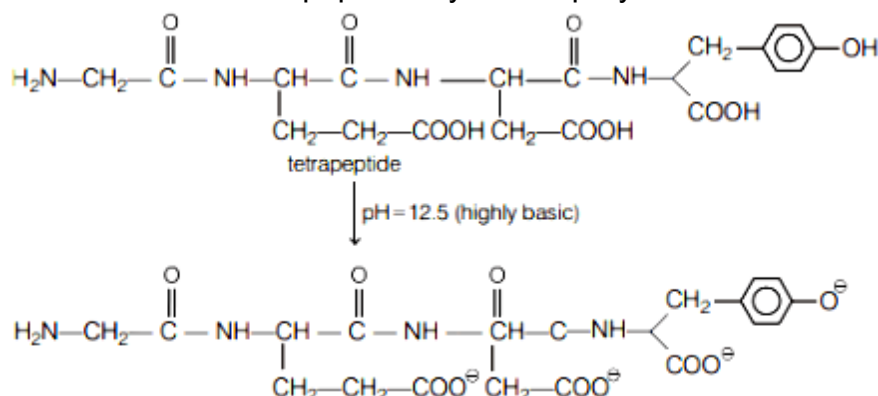
Question118

The total number of negative charge in the tetrapeptide, Gly - Glu - Asp - Tyr, at pH 12.5 will be (Integer answer)
[26 Aug 2021 Shift 1]

Answer: 4

Solution:

The structures of tetrapeptide Gly-Glu-Asp-Tyr is as follows



(All protons (H⁺) from carboxylic group and phenol are lost neutralised in basic medium)
 Total negative charge = 4.

Question119

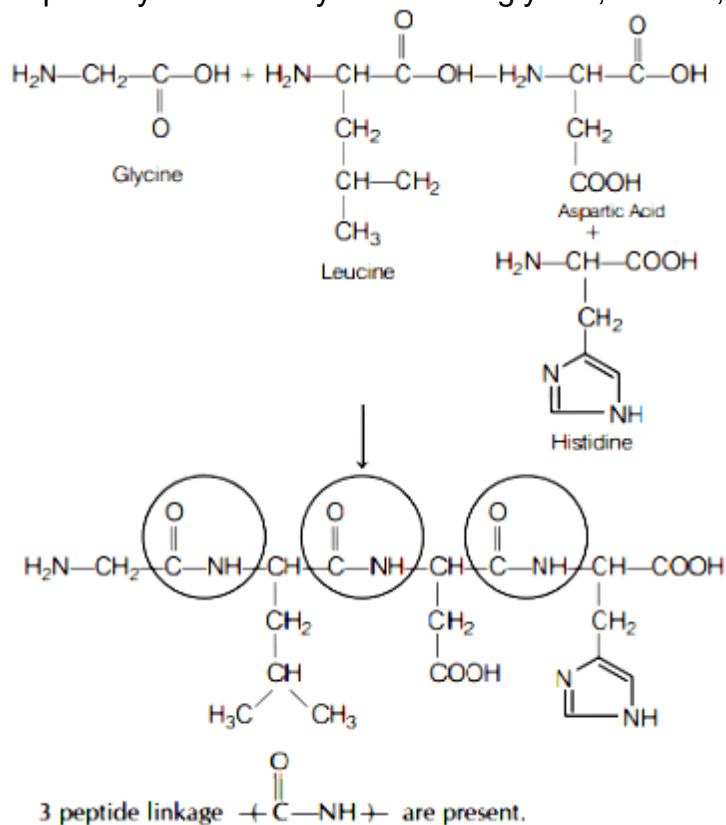
A peptide synthesised by the reactions of one molecule each of glycine, leucine, aspartic acid and histidine will have..... peptide linkages.

[1 Sep 2021 Shift 2]

Answer: 3

Solution:

Peptide synthesised by reaction of glycine, leucine, aspartic acid and histidine is as follows.



Hence, answer is 3.

Question120

Which of the following statement is not true for glucose?

[Jan. 08,2020 (II)]

Options:

- A. Glucose exists in two crystalline forms α and β
- B. Glucose gives Schiff's test for aldehyde
- C. Glucose reacts with hydroxylamine to form oxime The penta acetate of glucose does not
- D. react with hydroxylamine to give oxime

Answer: B

Solution:

Glucose exists in cyclic form in which aldehyde group is not free, therefore it does not give Schiff's test.

Question121

**Two monomers in maltose are:
[Jan. 08, 2020 (II)]**

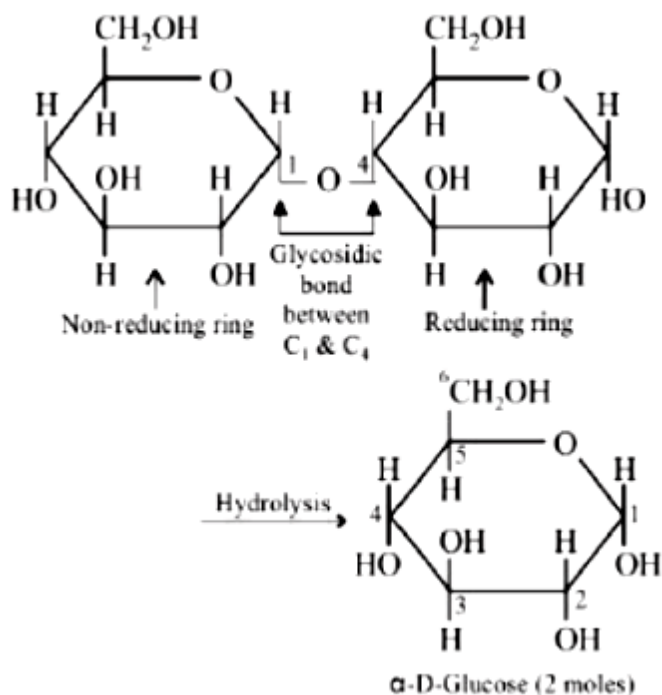
Options:

- A. α -D-glucose and β – D -glucose
- B. α -D-glucose and α -D-galactose
- C. α -D-glucose and α -D-fructose
- D. α – D-glucose and α -D-glucose

Answer: D

Solution:

Maltose on hydrolysis gives two moles of α -D-glucose.



Question122

Which of the following statements is correct?
[Jan. 07,2020 (II)]

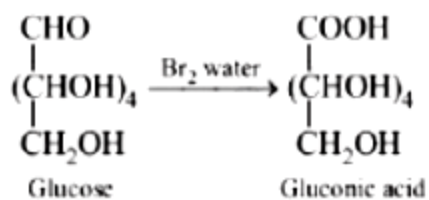
Options:

- A. Gluconic acid can form cyclic (acetal/hemiacetal) structure
- B. Gluconic acid is a dicarboxylic acid
- C. Gluconic acid is a partial oxidation product of glucose
- D. Gluconic acid is obtained by oxidation of glucose with HNO_3

Answer: C

Solution:

Gluconic acid is obtained by partial oxidation of glucose by mild oxidising agent e.g. Tollen's reagent, Fehling solution, Br_2 water.



Gluconic acid can not form hemiacetal or acetal.

Question123

The mass percentage of nitrogen in histamine is _____.
[NV, Jan. 09,2020(I)]

Answer: 37.84

Solution:

Molecular formula of histamine is $\text{C}_5\text{H}_9\text{N}_3$

Molecular mass of histamine

$$= 5 \times 12 + 9 \times 1 + 3 \times 14 = 111$$

Mass percentage of nitrogen in histamine

$$= \frac{42}{111} \times 100 = 37.84\%$$

Question124

Match the following:

(i) Riboflavin (a) Beriberi

(ii) Thiamine (b) Scurvy

(iii) Pyridoxine (c) Cheilosis

(iv) Ascorbic acid (d) Convulsions

[Jan. 07,2020(I)]

Options:

A. (i) – (a), (ii) – (d), (iii) – (c), (iv) – (b)

B. (i)-(c), (ii) –(d), (iii) – (a), (iv) – (b)

C. (i) – (c), (ii) – (a), (iii) – (d), (iv) – (b)

D. (i) – (d), (ii) – (b), (iii) – (a), (iv) – (c)

Answer: C

Solution:

Vitamins	Deficiency Diseases
Vitamin B_1 (thiamine)	Beribcri
Vitamin B_2 (riboflavin)	Cheilosis
Vitamin B_6 (pyridoxine)	Convulsions
Vitamin C (ascorbicacid)	Scurvy

Question125

Which one of the following statements is not true?

[Sep. 06,2020 (II)]

Options:

A. Lactose contains α -glycosidic linkage between C_1 of galactose and C_4 of glucose.

B. Lactose is a reducing sugar and it gives Fehling's test.

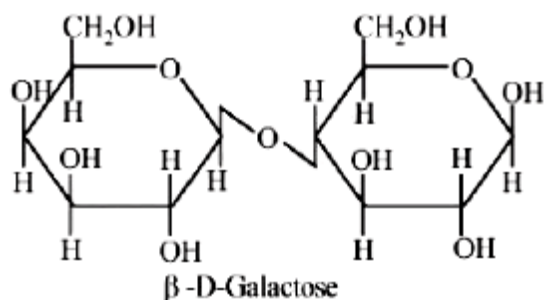
C. Lactose ($C_{11}H_{22}O_{11}$) is a disaccharide and it contains 8 hydroxyl groups.

D. On acid hydrolysis, lactose gives one molecule of D(+) -glucose and one molecule of D(+) -galactose

Answer: A

Solution:

Lactose contains β -glycosidic linkage between C_1 of galactose and C_4 of glucose.

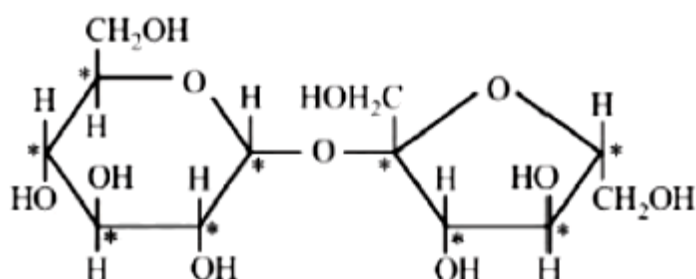


Question126

The number of chiral carbons present in sucrose is _____.
[NV, Sep. 05,2020 (II)]

Answer: 9

Solution:



No. of chiral centres = 9

Question127

What are the functional groups present in the structure of maltose?
[Sep. 04,2020(I)]

Options:

A. One ketal and hemiketal

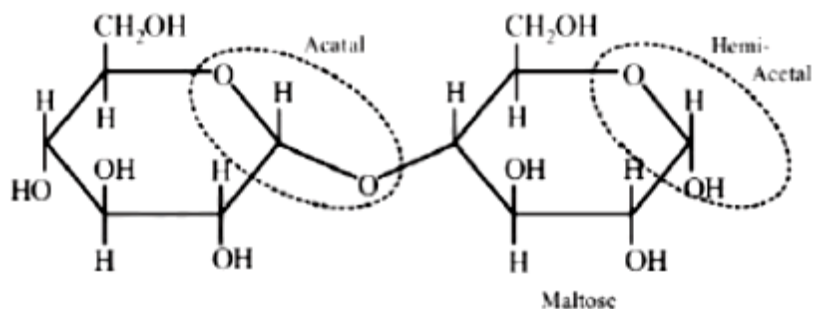
B. Two acetals

C. One acetal and one hemiacetal

D. One acetal and one ketal

Answer: C

Solution:



One acetal and one hemi acetal group is present in maltose.

Question128

Consider the following reactions :

(i) Glucose + ROH $\xrightarrow{\text{dry HCl}}$ Acetal $\xrightarrow{x \text{ eq. of } (\text{CH}_3\text{CO})_2\text{O}}$ acetyl derivative

(i) Glucose $\xrightarrow{\text{Ni/H}_2}$ A $\xrightarrow{y \text{ eq. of } (\text{CH}_3\text{CO})_2\text{O}}$ acetyl derivative

(i) Glucose + $\xrightarrow{z \text{ eq. of } (\text{CH}_3\text{CO})_2\text{O}}$ acetyl derivative

'x', 'y' and 'z' in these reactions are respectively.

[Sep. 02, 2020 (I)]

Options:

A. 5, 4 & 5

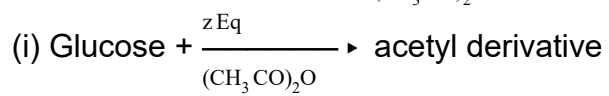
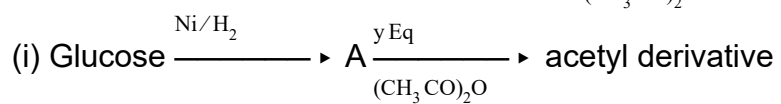
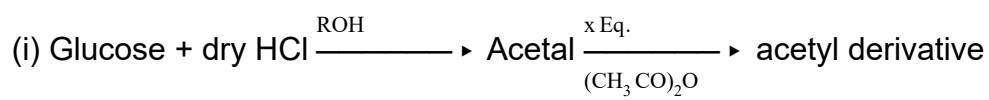
B. 4, 6 & 5

C. 4, 5 & 5

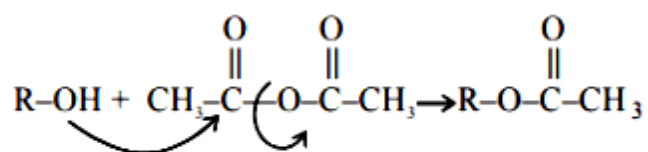
D. 5, 6 & 5

Answer: B

Solution:

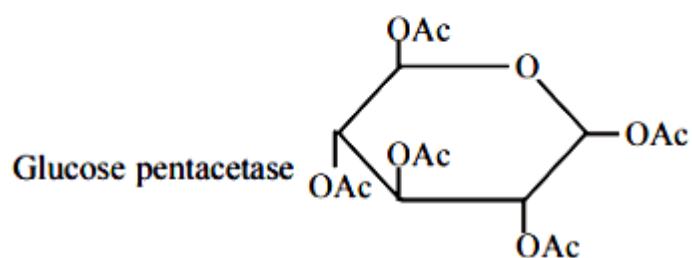
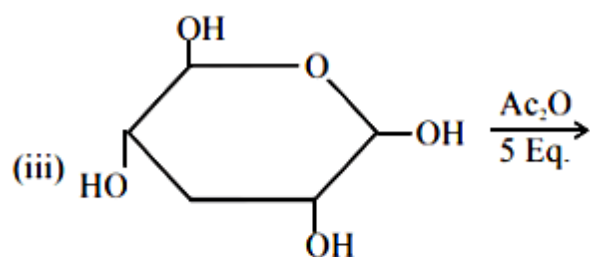
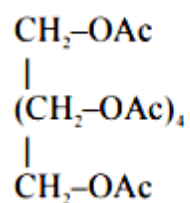
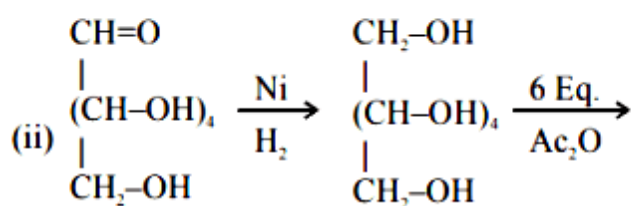
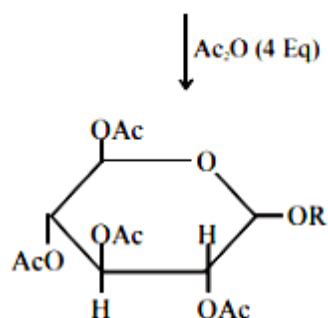
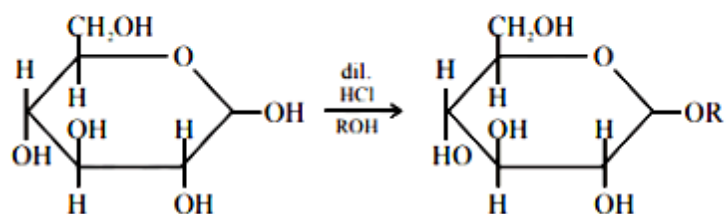


due to presence of -OH group in Glucose the reaction is



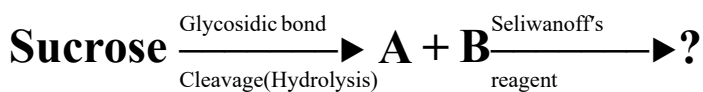
Acetyl derivative

so for (i)



Question129

The correct observation in the following reactions is:



[Sep. 02,2020 (II)]

Options:

- A. Formation of blue colour
- B. Gives no colour
- C. Formation of red colour
- D. Formation of violet colour

Answer: C

Solution:

Seliwanoff reagent $\rightarrow$ [Resorcinol + Conc. HCl] It is used to distinguish aldoses and ketoses. Ketoses show red colour whereas aldoses show light pink colour with Seliwanoff Reagent.

Question130

Which of the following is not an essential amino acid?

[Sep. 05,2020(I)]

Options:

- A. Tyrosine
- B. Leucine
- C. Valine
- D. Lysine

Answer: A

Solution:

Tyrosine is a non-essential amino acid.

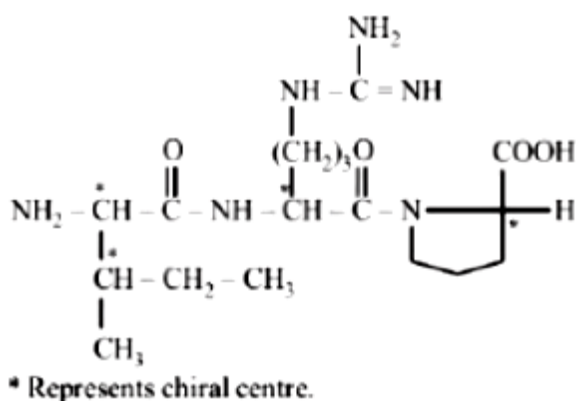
Question131

The number of chiral carbons (s) present in peptide, Ile Age-Pro, is _____.

[NV, Sep. 05, 2020(I)]

Answer: 4

Solution:



Question132

Which of the following will react with $\text{CHCl}_3 + \text{alc. KOH}$?

[Sep. 04, 2020 (I)]

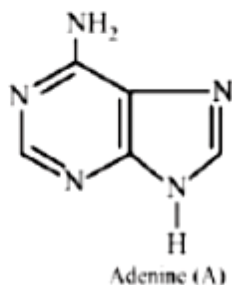
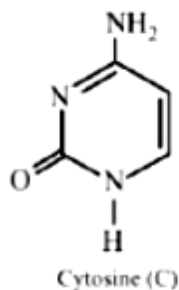
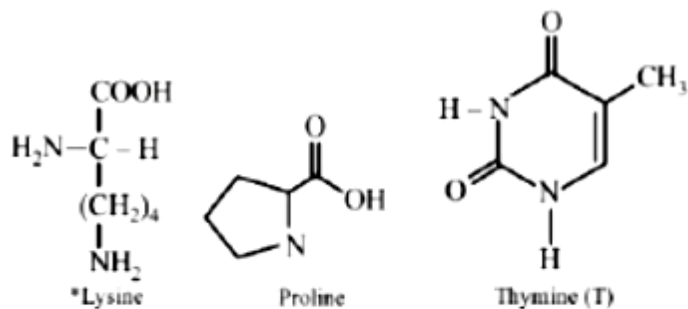
Options:

- A. Adenine and proline
- B. Thymine and proline
- C. Adenine and lysine
- D. Adenine and thymine

Answer: C

Solution:

Compounds having 1° amine give carbylamine reaction with CHCl_3 and alc. KOH .

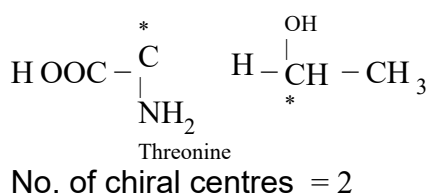


Question133

The number of chiral centres present in threonine is _____.
[NV, Sep.04,2020(II)]

Answer: 2

Solution:



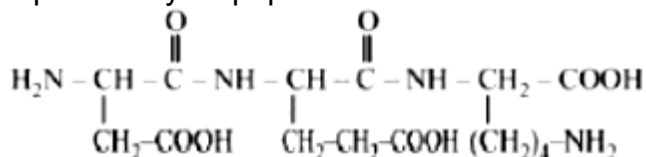
Question134

The number of ;C = O groups present in a tripeptide Asp - Glu - Lys is _____.
[NV,Sep.03,2020(II)]

Answer: 5

Solution:

Asp - Glu- Lys tripeptide is:



No. of CO group -5

Question135

Among the following compounds most basic amino acid is:
[Jan.12,2019(I)]

Options:

- A. Asparagine
- B. Lysine
- C. Scrine
- D. Histidine

Answer: B

Solution:

Lysine

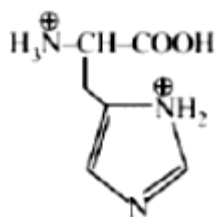
Question136

The correct structure of histidine in a strongly acidic solution (pH – 2) is :

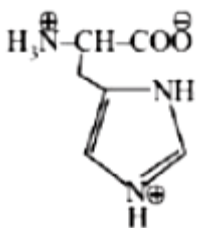
[Jan. 12, 2019 (II)]

Options:

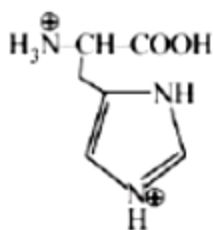
A.



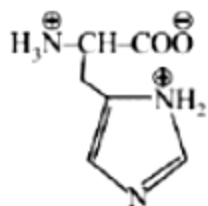
B.



C.

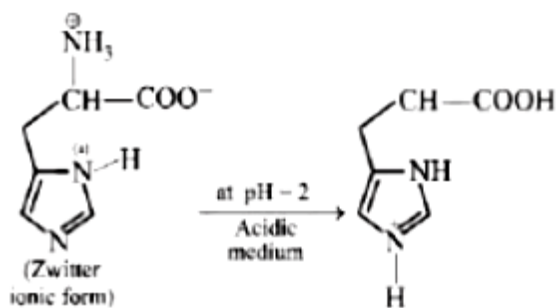


D.



Answer: C

Solution:



Note that 1p of electrons on N labelled as N^(a) are involved in delocalisation, hence not available for protonation.

Question137

The increasing order of pKa of the following amino acids in aqueous solution is:

Gly Asp Lys Arg

[Jan. 9,2019 (I)]

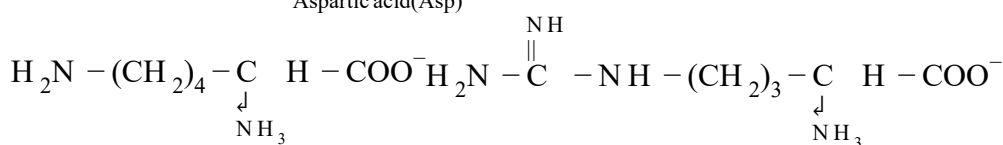
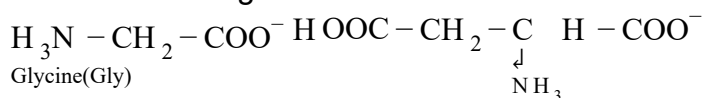
Options:

- A. Asp < Gly < Arg < Lys
- B. Gly < Asp < Arg < Lys
- C. Asp < Gly < Lys < Arg
- D. Arg < Lys < Gly < Asp

Answer: C

Solution:

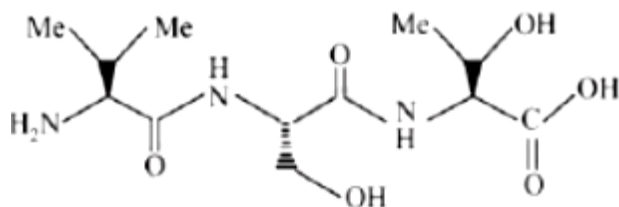
Structure of the given α -amino acids are:



Here, aspartic acid is acidic, glycine is neutral, while lysine and arginine are basic amino acids. Also, arginine is more basic due to stronger basic functional groups. $\therefore$ The order of pK_a value is directly proportional to the basic strength of amino acids, thus Arg > Lys > Gly > Asp.

Question138

The correct sequence of amino acids present in the tripeptide given below is:



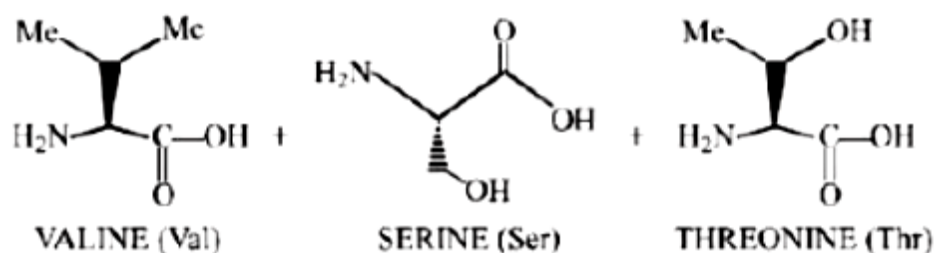
[Jan. 9,2019 (II)]

Options:

- A. Val-Ser-Thr
- B. Thr-Ser-Val
- C. L Leu –Ser – V al
- D. T hr – Ser – L_{cu}

Answer: A

Solution:



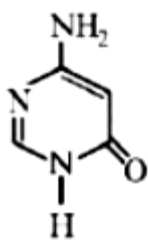
Question139

Among the following compounds, which one is found in RNA?

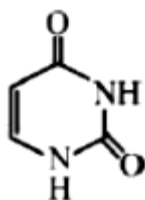
[Jan. 11, 2019 (I)]

Options:

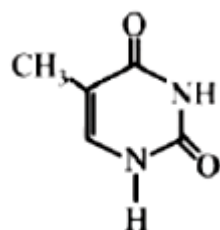
A.



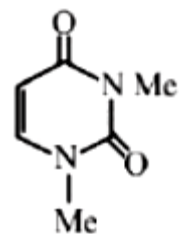
B.



C.



D.



Answer: B

Solution:

RNA contains adenine (A), guanine (g), cytosine (c) and uracil (u).

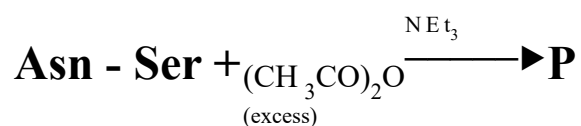
(1) is cytosine, found in RNA and DNA, both.

(2) is uracil, found in RNA only.

(3) is thymine, found in DNA only.

(4) is not a pyrimidine, i.e., not a constituents of RNA and DNA.

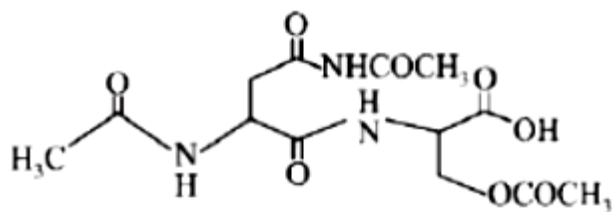
The correct structure of product 'P' in the following reaction is:



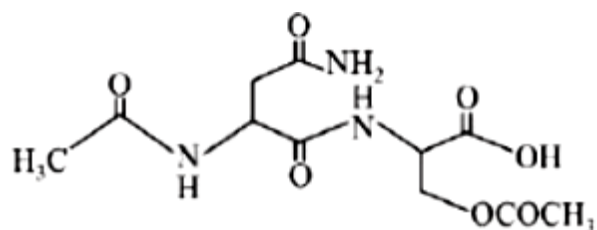
[Jan. 10, 2019(I)]

Options:

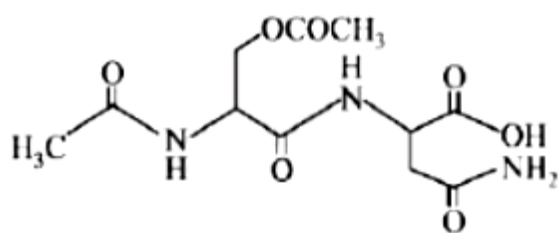
A.



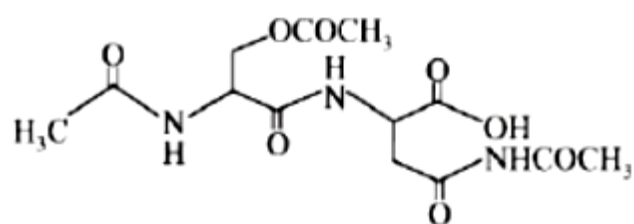
B.



C.

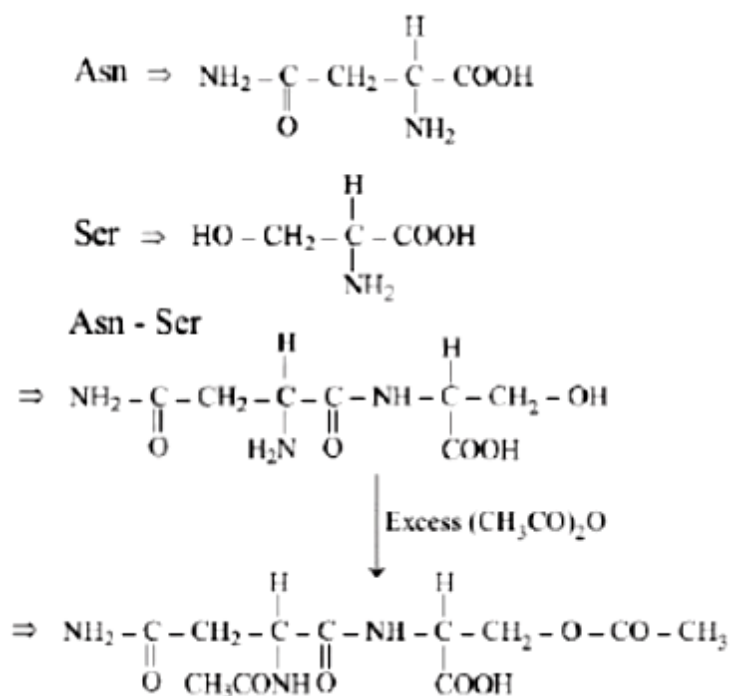


D.



Answer: B

Solution:



Question141

Glucose and Galactose are having identical configuration in all the positions except position.

[April 12, 2019 (I)]

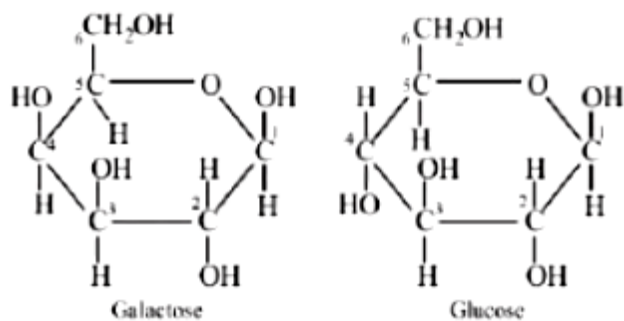
Options:

- A. C – 3
- B. C – 4
- C. C – 2
- D. C – 5

Answer: B

Solution:

Galactose and glucose are C_4 epimers.



Question142

Which of the given statements is **INCORRECT** about glycogen ?
[April 12, 2019 (II)]

Options:

- A. It is a straight chain polymer similar to amylose.
- B. Only α -linkages are present in the molecule.
- C. It is present in animal cells.
- D. It is present in some yeast and fungi.

Answer: A

Solution:

Structure of glycogen is similar to amylopectin. It is found in yeast and fungi and stored in animal body. It contains α -glycosidic linkages.

Question143

Amylopectin is composed of:
[April 10, 2019(I)]

Options:

- A. α - D -glucose , $C_1 - C_4$ and $C_1 - C_6$ linkages

B. β -D-glucose, $C_1 - C_4$ and $C_2 - C_6$ linkages

C. β -D-glucose, $C_1 - C_4$ and $C_1 - C_6$ linkages

D. α -D-glucose, $C_1 - C_4$ and $C_2 - C_6$ linkages

Answer: A

Solution:

Starch is a polymer of α -D-glucose. It has two components.

(i) Amylose, which has only $\alpha - 1, 4$ -glycosidic linkage and is a linear polymer.

(ii) Amylopectin, which has $\alpha - 1, 6$ -glycosidic linkage in addition to $\alpha - 1, 4$ -glycosidic linkage and is a crosslinked polymer.

Question144

Number of stereo centers present in linear and cyclic structures of glucose are respectively:

[April 10,2019 (II)]

Options:

A. 5 & 4

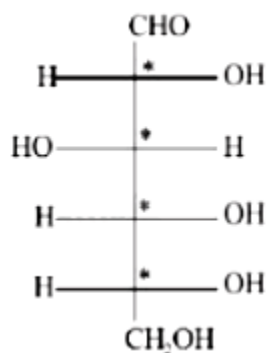
B. 4 & 4

C. 5 & 5

D. 4 & 5

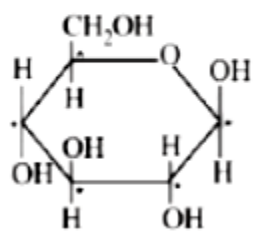
Answer: D

Solution:



4 stereogenic centres

Cyclic structure of glucose,



5 stereogenic centres

Question145

Which of the following statements is not true about sucrose?
[April 9, 2019(I)]

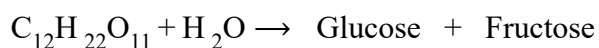
Options:

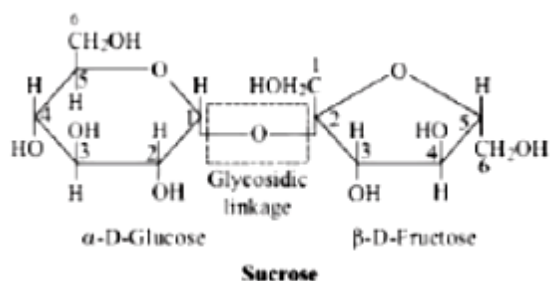
- A. It is a non reducing sugar
- B. The glycosidic linkage is present between C_1 of α glucose and C_1 , of β -fructose
- C. It is also named as invert sugar
- D. On hydrolysis, it produces glucose and fructose

Answer: B

Solution:

Sucrose contains glycosidic link between C_1 of α - D glucose and C_2 of β - D -fructose.





Question146

Which of the following statements is not true about sucrose?
[April 9, 2019 (I)]

Options:

- A. It is a non reducing sugar
- B. The glycosidic linkage is present between C₁ of α glucose and C₁ of β -fructose
- C. It is also named as invert sugar
- D. On hydrolysis, it produces glucose and fructose

Answer: B

Solution:

$$\text{Sucrose} \xrightarrow{\text{H}_2\text{O}} \alpha\text{-D-glucose} + \beta\text{-D-fructose}$$
 also named as invert sugar and it is a example of non-reducing sugar

The glycosidic linkage is present between C₁ of α -glucose and C₂ of β -fructose.

In sucrose, the components glucose and fructose are linked via an ether bond between C₁ on the glucosyl subunit and C2 on the fructosyl unit. The bond is called a glycosidic linkage.

Question147

Maltose on treatment with dilute HCl gives:
[April 8,2019(1)]

Options:

- A. D-Glucose and D-Fructose

- B. D-Fructose
- C. D-Galactose
- D. D-Glucose

Answer: D

Solution:

- (a) Fehling solution : CuSO_4 + Sod. - Pot. tartarate (Rochelle salt) gives red ppt. of Cu_2O with glucose and fructose both.
- (b) Barfoed reagent: 7% $(\text{CH}_3\text{COO})_2\text{Cu}$ + 1% CH_3COOH + 92% H_2O gives red ppt. of Cu_2O with both.
- (c) Benedict reagent: CuSO_4 + Sod. citrate + Na_2CO_3 also gives red ppt. of Cu_2O with both.
- (d) Seliwanoff's reagent: Resorcinol in conc. HCl gives red colour with both, but a ketose (fructose) reacts more quickly than an aldose (glucose).
-

Question148

The peptide that gives positive ceric ammonium nitrate and carbylamine tests is:
[April 9, 2019 (II)]

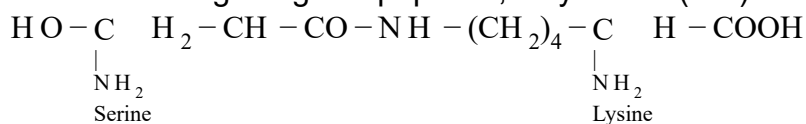
Options:

- A. Ser-Lys
- B. Gl n – Asp
- C. Lys - Asp
- D. A sp – Gin

Answer: A

Solution:

Ceric ammonium nitrate test is used for detecting alcohols, while carbylamine test is for primary amines. Among the given peptides, only serine (Ser) has alcoholic group.



Question149

Which of the following statements is not true about RNA?
[April 12, 2019 (I)]

Options:

- A. It controls the synthesis of protein.
- B. It has always double stranded helix structure.
- C. It usually does not replicate.
- D. It is present in the nucleus of the cell.

Answer: B

Solution:

RNA has a single helix structure, whereas, DNA has a double helix structure.

Question150

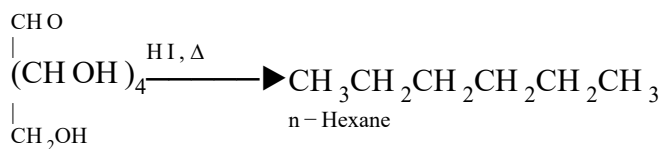
Glucose on prolonged heating with HI gives:
[2018]

Options:

- A. n -Hexane
- B. 1 - Hexene
- C. Hexanoic acid
- D. 6 -iodohexanal

Answer: A

Solution:



Question151

Among the following, the incorrect statement is:
[Online April 16, 2018]

Options:

- A. Cellulose and amylose have 1,4 -glycosidic linkage
- B. Lactose contains β -D-galactose and β -D-glucose
- C. Maltose and lactose have 1,4 -glycosidic linkage
- D. Sucrose and amylose have 1,2 -glycosidic linkage

Answer: D

Solution:

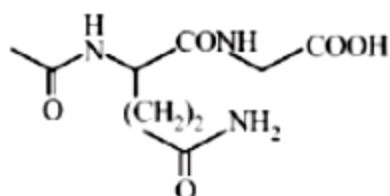
In amylose 1, 4 – α -glycosidic linkage is present.

Question152

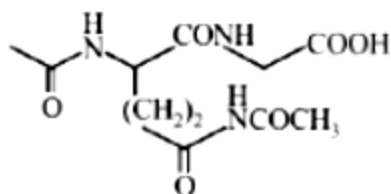
The dipeptide, Gln-Gly, on treatment with CH_3COCl followed by aqueous work up gives:
[Online April 15,2018(II)]

Options:

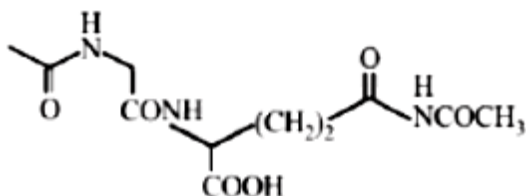
A.



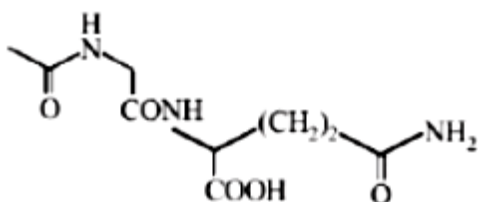
B.



C.



D.



Answer: A

Solution:

Amino group of glutamine is acetylated while amide group of glutamine is not acetylated.

Note : Acetylation of amide requires activation of amides and/or acyl donors, since the nitrogen atom of amides is less basic than that of the corresponding amines due to amide resonance.

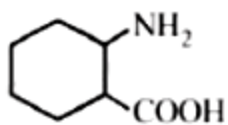
Question153

Which of the following will not exist in zwitter ionic form at pH = 7 ?

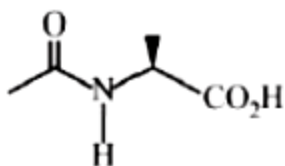
[Online April 15, 2018(I)]

Options:

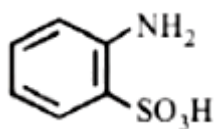
A.



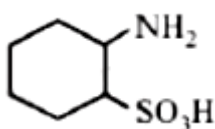
B.



C.



D.



Answer: B

Solution:

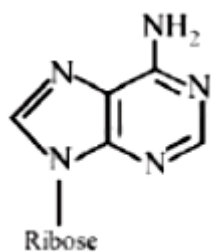
The N atom of amide is not basic so it will not exist in zwitter ionic form at pH 7.0.

Question154

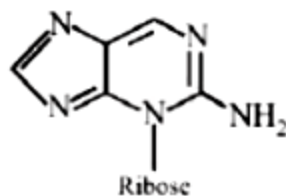
Which of the following is the correct structure of adenosine?
[Online April 15,2018(I)]

Options:

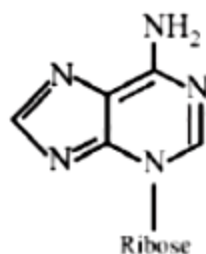
A.



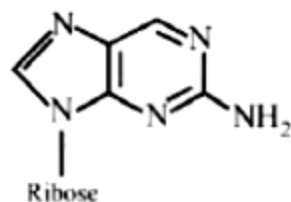
B.



C.



D.



Answer: A

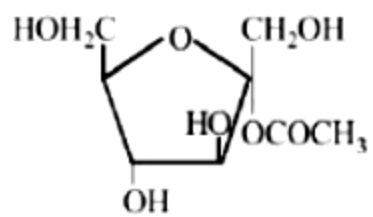
Question155

Which of the following compounds will behave as a reducing sugar in an aqueous KOH solution?

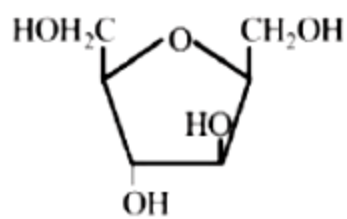
[2017]

Options:

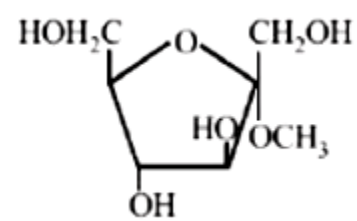
A.



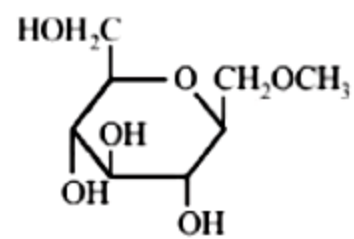
B.



C.

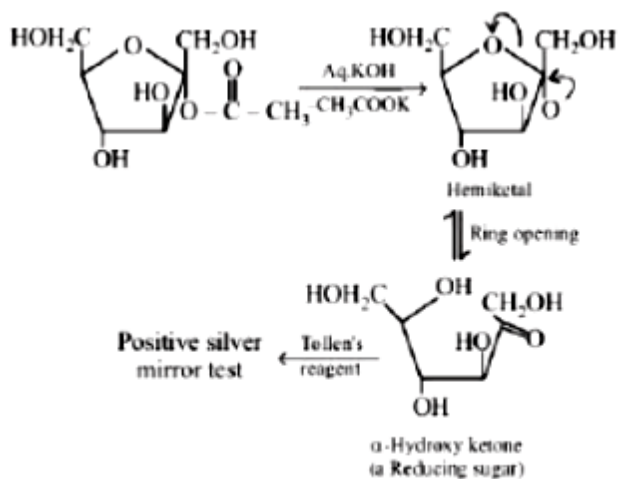


D.



Answer: A

Solution:



Question156

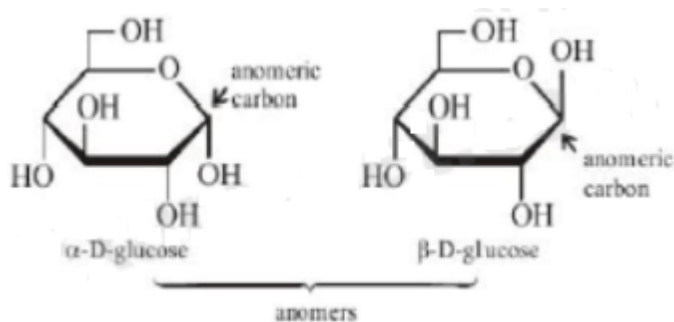
The incorrect statement among the following is:
[Online April 9, 2017]

Options:

- A. α -D-glucose and β -D-glucose are anomers.
- B. α -D-glucose and β -D-glucose are enantiomers.
- C. Cellulose is a straight chain polysaccharide made up of only β -D-glucose units.
- D. The penta acetate of glucose does not react with hydroxyl amine.

Answer: B

Solution:



Question157

Among the following, the essential amino acid is:
[Online April 8,2017]

Options:

- A. Alanine
- B. Valine
- C. Aspartic acid
- D. Serine

Answer: B

Solution:

Those amino acids that cannot be synthesized in our body and must be supplied in diet is called essential amino acid for ex. valine, histidine, isoleucine etc.

Question158

Observation of "Ruhemann's purple" is a confirmatory test for the presence of:
[Online April 10, 2016]

Options:

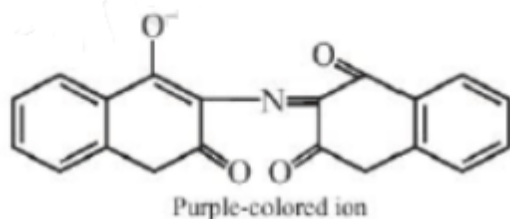
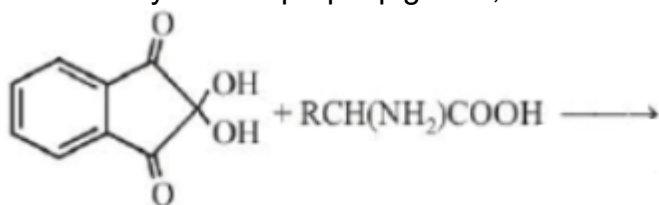
- A. Starch
- B. Reducing sugar
- C. Protein
- D. Cupricion

Answer: C

Solution:

Ninhydrin is often used to detect α - amino acids and also free amino and carboxylic acid groups on proteins and peptides. When about 0.5mL of a 0.1% solution of ninhydrin is boiled for one or two minutes with a few mL of dilute amino acid or protein solution, a blue color develops. Ninhydrin

degrades amino acids into aldehydes, ammonia, and CO_2 through a series of reactions to produce an intensely blue or purple pigment, sometimes called Ruhemann's purple.



Question159

Thiol group is present in:
[2016]

Options:

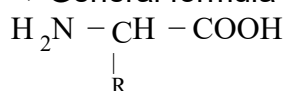
- A. Cysteine
- B. Methionine
- C. Cytosine
- D. Cystine

Answer: A

Solution:

Among 20 naturally occurring amino acids "cysteine" has ' $-\text{SH}$ ' or thiol functional group.

⇒ General formula of amino acid



⇒ Value of R = $-\text{CH}_2 - \text{SH}$ in cysteine.

Question160

Complete hydrolysis of starch gives: [Online April 10,2015]

Options:

- A. glucose only
- B. galactose and fructose in equimolar amounts
- C. glucose and galactose in equimolar amounts
- D. glucose and fructose in equimolar amounts

Answer: A

Solution:

Starch is a mixture of amylose & amylopectin polysaccharides and monomer is glucose. Thus on complete hydrolysis it gives only glucose.

Question161

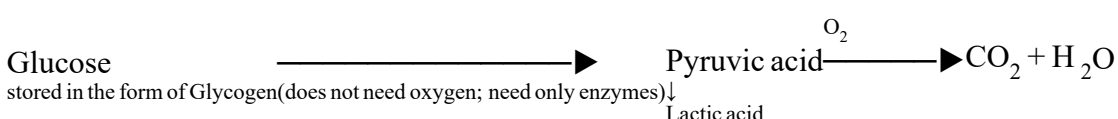
Accumulation of which of the following molecules in the muscles occurs as a result of vigorous exercise ?
[Online April 11, 2015]

Options:

- A. Glycogen
- B. Glucose
- C. Pyruvic acid
- D. L -lactic acid

Answer: D

Solution:



During vigorous exercise sufficient oxygen is not available to meet the energy demand, so energy is derived through conversion of pyruvic acid to lactic acid.

Question162

Which of the vitamins given below is water soluble?
[2015]

Options:

- A. Vitamin E
- B. Vitamin K
- C. Vitamin C
- D. Vitamin D

Answer: C

Solution:

Water soluble vitamins dissolve in water and are not stored by the body. The water soluble vitamins include the vitamin B-complex group and vitamin C.

Question163

Which of the following will not show mutarotation?
[Online April 12,2014]

Options:

- A. Maltose
- B. Lactose
- C. Glucose
- D. Sucrose

Answer: D

Solution:

Sucrose does not contain a free aldehydic or ketonic group, hence it does not show mutarotation.

Question164

**Which one of the following bases is not present in DNA?
[2014]**

Options:

- A. Quinoline
- B. Adenine
- C. Cytosine
- D. Thymine

Answer: A

Solution:

DNA contains ATGC bases Quinoline is not present in DNA or RNA.

Question165

**The reason for double helical structure of DNA is the operation of:
[Online April 19, 2014]**

Options:

- A. Electrostatic attractions
- B. van der Waals forces
- C. Dipole- Dipole interactions
- D. Hydrogen bonding

Answer: D

Solution:

The two polynucleotide chains of DNA molecules are twisted around a common axis but run in opposite directions to form a right handed helix. The two chains are joined together by specific hydrogen bonds. Adenine-Thymine (two hydrogen bonds) and cytosine-guanine (three hydrogen bonds)

Question 166

Among the following organic acids, the acid present in rancid butter is:

[Online April 19, 2014]

Options:

- A. Pyruvic acid
- B. Lactic acid
- C. Butyric acid
- D. Acetic acid

Answer: C

Solution:

Butyric acid, also known as butanoic acid is found in milk, and butter and is a product of anaerobic fermentation. It has an unpleasant smell and acid taste.

Question 167

**Synthesis of each molecule of glucose in photosynthesis involves :
[2013]**

Options:

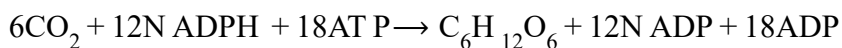
- A. 18 molecules of ATP
- B. 10 molecules of ATP

C. 8 molecules of ATP

D. 6 molecules of ATP

Answer: A

Solution:



Question168

**Natural glucose is termed D-glucose because :
[Online April 23, 2013]**

Options:

A. -OH on the second carbon is on the right side in Fischer projection

B. -OH on the sixth carbon is on the right side in Fischer projection.

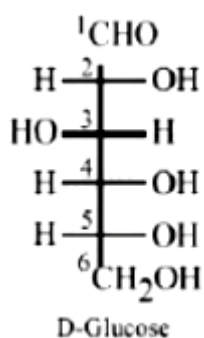
C. - OH on the fifth carbon is on the right side in Fischer projection.

D. It is dextrorotatory.

Answer: C

Solution:

Fischer gave the prefix "D" to compounds whose bottom chiral has its OH to the right. So natural glucose is called D-glucose or dextrose. Structure of D -Glucose :



Question169

Which of the following statement is not correct?
[Online April 25,2013]

Options:

- A. Amylopectin is a branched polymer of α - glucose.
- B. Cellulose is a linear polymer of β -glucose
- C. Glycogen is the food reserve of plants
- D. All proteins are polymers of α - amino acids.

Answer: C

Solution:

Glycogen is called animal starch and is found in all animal cells. It constitutes the reserve food material.

Question170

Glycosidic linkage is actually an :
[Online April 23,2013]

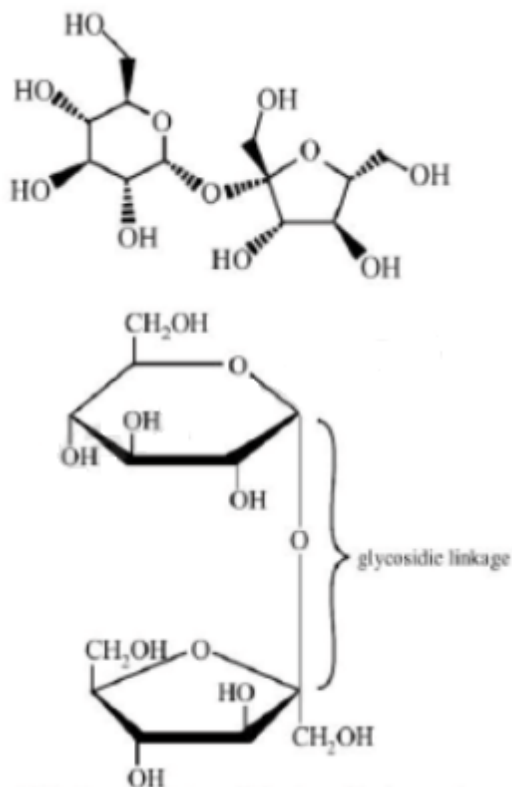
Options:

- A. Carbonyl bond
- B. Ether bond
- C. Ester bond
- D. Amide bond

Answer: B

Solution:

Glycosidic linkage is actually an ether bond as the linkage forming the rings in an oligosaccharide or polysaccharide is not just one bond, but the two bonds sharing an oxygen atom e.g. sucrose



Question171

Which of the following enzyme converts starch into maltose?
[Online April 9, 2013]

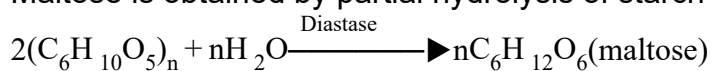
Options:

- A. Diastase
- B. Maltase
- C. Zymase
- D. Invertase

Answer: A

Solution:

Maltose is obtained by partial hydrolysis of starch by the enzyme diastase present in malt.

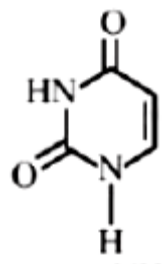


Question172

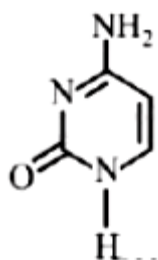
Which of the following structures represents thymine?
[Online April 22, 2013]

Options:

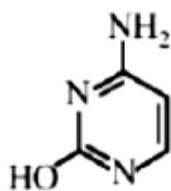
A.



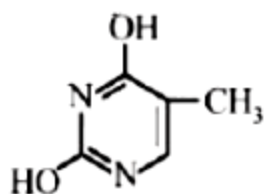
B.



C.



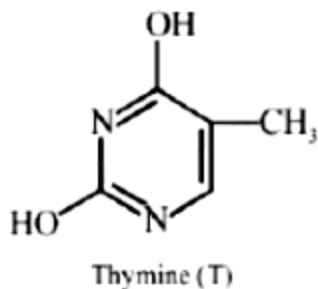
D.



Answer: D

Solution:

The correct structure of thymine is



Question173

Among the following vitamins the one whose deficiency causes rickets (bone deficiency) is
[Online April 25,2013]

Options:

- A. Vitamin A
- B. Vitamin B
- C. Vitamin D
- D. Vitamin C

Answer: C

Solution:

Deficiency of vitamin D causes rickets.

Question174

Which of the following compounds can be detected by Molisch's test?
[2012]

Options:

- A. Nitro compounds
- B. Sugars

C. Amines

D. Primary alcohols

Answer: B

Solution:

Molisch's test : This is a general test for carbohydrates. One or two drops of alcoholic solution of α - naphthol is added to 2mL glucose solution, 1mL of conc. H_2SO_4 solution is added carefully along the sides of the test-tube. The formation of a violet ring at the junction of two liquids confirms the presence of a carbohydrate or sugar.

Question175

**Amylopectin is a polymer of
[Online May 12, 2012]**

Options:

A. α - D - glucose

B. amino acid

C. β - D - glucose

D. amylase.

Answer: A

Solution:

Amylopectin is a polymer of α -D-glucose.

Question176

**Which of the following is a non-reducing sugar?
[Online May 19,2012]**

Options:

- A. Lactose
- B. Fructose
- C. Sucrose
- D. Maltose

Answer: C

Solution:

Sucrose is a non-reducing sugar as it does not reduce Fehling reagent and Tollen's reagent.

Question177

**Which one of the following statements is correct?
[2012]**

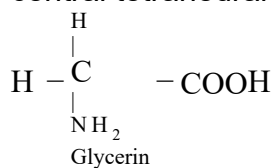
Options:

- A. All amino acids except lysine are optically active
- B. All amino acids are optically active
- C. All amino acids except glycine are optically active
- D. All aminoacids except glutamicacids are optically active

Answer: C

Solution:

With the exception of glycine all other amino acids have different functional groups (atom) on the central tetrahedral alpha carbon.



Question178

**All of the following statements apply to proteins except
[Online May 7,2012]**

Options:

- A. Proteins generally have no definite melting point
- B. Proteins contain the grouping –CON H
- C. Proteins have high molecular weight
- D. Proteins can only contain the elements C, H , O and N .

Answer: D

Solution:

Statement (d) is not correct. Some proteins also contain S, along with C, H , O and N .

Question179

**Which of the following statements is correct?
[Online May 26, 2012]**

Options:

- A. RNA controls the synthesis of proteins.
- B. The sugar present in DNA is D-(-)-ribose.
- C. RNA has double stranded α -helix structure.
- D. DNA mainly occurs in the cytoplasm of the cell.

Answer: A

Question180

Biuret test is not given by [2010]

Options:

- A. carbohydrates
- B. polypeptides
- C. urea
- D. proteins

Answer: A

Solution:

Biuret test produces violet colour on addition of dilute CuSO_4 to alkaline solution of a compound containing peptide linkage. Polypeptides, proteins and urea have $-\text{CO}-\text{NH}-$ (peptide) linkage,

$\begin{array}{c} \parallel \\ \text{O} \end{array}$

while carbohydrates have glycosidic linkages. So, test of carbohydrates should be different from that of other three.

Question181

The two functional groups present in a typical carbohydrate are: [2009]

Options:

- A. $-\text{CHO}$ and $-\text{COOH}$
- B. $>\text{C}=\text{O}$ and $-\text{OH}$
- C. $-\text{OH}$ and $-\text{CHO}$
- D. $-\text{OH}$ and $-\text{COOH}$

Answer: C

Solution:

Note: Glucose is considered as a typical carbohydrate which contains $-\text{CHO}$ and $-\text{OH}$ groups.

Question182

α – D – (+) -glucose and β – D – (+) -glucose are
[2008]

Options:

- A. conformers
- B. epimers
- C. anomers
- D. enantiomers

Answer: C

Question183

The secondary structure of a protein refers to
[2007]

Options:

- A. fixed configuration of the polypeptide backbone
- B. α - helical backbone
- C. hydrophobic interactions
- D. sequence of α -amino acids.

Answer: B

Solution:

The secondary structure of a protein refers to the shape in which a long peptide chain can exist. There are two different conformations of the peptide linkage present in protein, these are α -helix and β -conformation. The α helix always has a right handed arrangement. In β -conformation all peptide chains are stretched out to nearly maximum extension and then laid side by side and held together by intermolecular hydrogen bonds. The structure resembles the pleated folds of draping and therefore is known as β -pleated sheet.

Question 184

The term anomers of glucose refers to [2006]

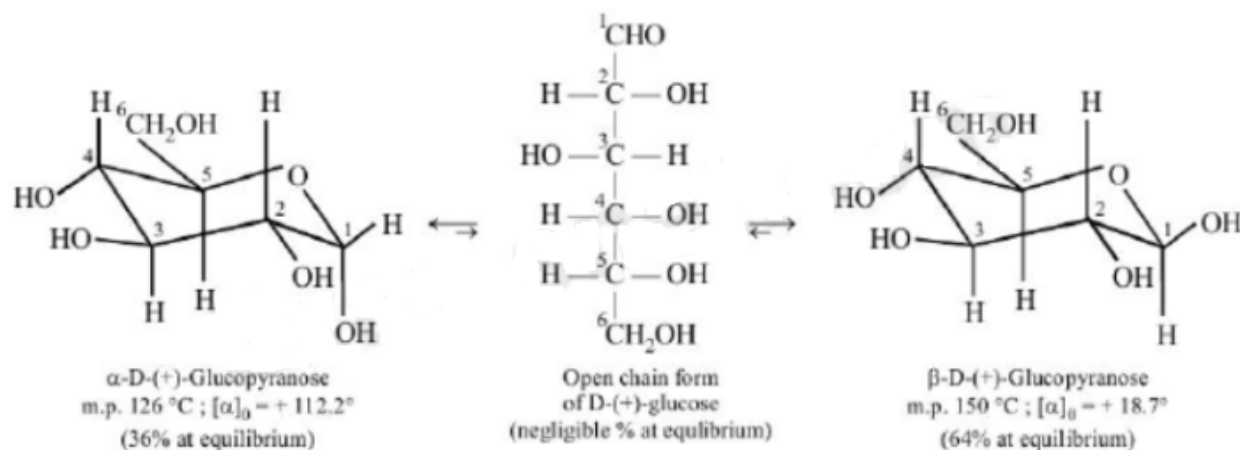
Options:

- A. enantiomers of glucose
- B. isomers of glucose that differ in configuration at carbon one (C – 1)
- C. isomers of glucose that differ in configurations at carbons one and four (C – 1 and C – 4)
- D. a mixture of (D)-glucose and (L)-glucose

Answer: B

Solution:

Cyclization of the open chain structure of D – (+) -glucose has created a new stereocenter at C_1 which explains the existence of two cyclic forms of D – (+) -glucose, namely α - and β - . These two cyclic forms are diastereomers, such diastereomers which differ only in the configuration of chiral carbon developed on hemiacetal formation (it is C_1 in glucose and C_2 in fructose) are called anomers and the hemiacetal carbon (C_1 or C_2) is called the anomeric carbon.



Question185

**The pyrimidine bases present in DNA are
[2006]**

Options:

- A. cytosine and thymine
- B. cytosine and uracil
- C. cytosine and adenine
- D. cytosine and guanine

Answer: A

Solution:

The pyrimidine bases present in DNA are cytosine and thymine.

Question186

**In both DNA and RNA, heterocyclic base and phosphate ester
linkages are at
[2005]**

Options:

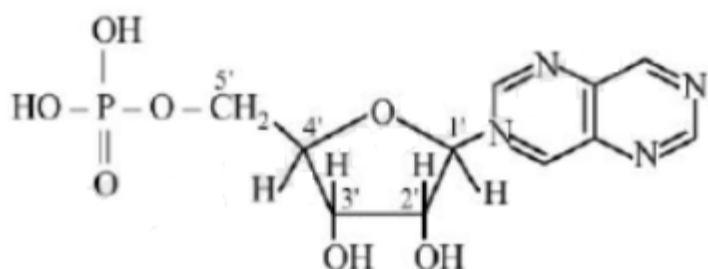
- A. C_5' and C_1' respectively of the sugar molecule
- B. C_1' and C_5' respectively of the sugar molecule
- C. C_2' and C_5° respectively of the sugar molecule

D. C_5' and C_2' respectively of the sugar molecule

Answer: B

Solution:

In DNA and RNA heterocyclic base and phosphate ester are at C_1' and C_5' respectively of the sugar molecule.



Question187

Which base is present in RNA but not in DNA?
[2004]

Options:

- A. Guanine
- B. Cytosine
- C. Uracil
- D. Thymine

Answer: C

Solution:

RNA contains cytosine and uracil as pyrimidine bases while DNA has cytosine and thymine. Both have the same purine bases i.e., guanine and adenine.

Question188

Insulin production and its action in human body are responsible for the level of diabetes. This compound belongs to which of the following categories?

[2004]

Options:

- A. An enzyme
- B. A hormone
- C. A co-enzyme
- D. An antibiotic

Answer: B

Solution:

Insulin is a biochemically active peptide hormone secreted by pancreas.

Question189

Complete hydrolysis of cellulose gives
[2003]

Options:

- A. D-ribose
- B. D-glucose

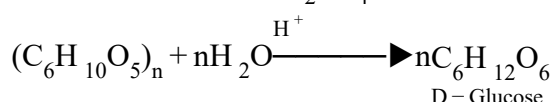
C. L-glucose

D. D-fructose

Answer: B

Solution:

Cellulose is a linear polymer of β -D-glucose in which C_1 of one glucose unit is connected to C_4 of the other through β -D glucosidic linkage. It does not undergo hydrolysis easily. However on heating with dilute H_2SO_4 under pressure, it undergoes hydrolysis to give only D-glucose.



Question190

The reason for double helical structure of DNA is operation of [2003]

Options:

A. dipole-dipole interaction

B. hydrogen bonding

C. electrostatic attractions

D. van der Waals' forces

Answer: B

Solution:

DNA consists of two polynucleotide chains, each chain forms a right handed spiral with ten bases in one turn of the spiral. The two chains coil to double helix and run in opposite direction held together by hydrogen bonding.

Question191

RNA is different from DNA because RNA contains [2002]

Options:

- A. ribose sugar and thymine
- B. ribose sugar and uracil
- C. deoxyribose sugar and thymine
- D. deoxyribose sugar and uracil.

Answer: B

Solution:

In RNA, the sugar is D-ribose and base is uracil, whereas in DNA, the sugar is D-2 deoxyribose and the nitrogenous base is thymine.
